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Workpackage G

Financial transactions data

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Executive summary

This report describes the final findings of "Workpackage G Financial Transactions Data". The main aim of WPG is to get an overview of the sources and the data infrastructure (metadata) of financial transactions data in the participating countries. The objective is to describe to what extent FTD are available as well as whether it is possible for NSIs to access them. Given the infrastructure, it is also a main aim of the WP to assess the statistical potential of these data sources. The statistical potential could be the quality improvement of official statistics as well as a new statistical product. In addition to the financial transaction part described above, WPG is also aiming at utilising financial transactions data to describe the sharing economy.

Part I concerns financial transactions data. After presenting the definition, different EU initiatives are presented before the main characteristics of payment system and payment instruments are presented as well as the legal aspects and the main actors moving in this market. The financial transactions data situation is described for the WPG countries: Bulgaria, Germany, Italy, Norway, Portugal and Slovenia. In Part I there is also an assessment of the official statistics that presumably can benefit of using financial transactions data, either as a new data source or as a source acting in addition to the current data source. This assessment was made prior to accessing data to provide a basis for selecting the empirical case studies performed within WPG. Promising statistics were the following: household budget survey, household expenditure of residents for tourism and balance of payments estimates, e-commerce turnover, retail trade index, early estimates of macroeconomic aggregates and indicators, business statistics indicators on flow between industries, and finally turnover of collaborative/sharing economy.

Part II contains some applications of FTD. First, case studies are presented from Italy, Norway and Portugal. The case studies show some of the potential of financial transactions data for official statistics. The Italian case studies show the gain of using time series of daily card transactions by aggregated merchant category code groups in forecasting and nowcasting of economic indicators. The Portuguese case study looks at how to improve the tourism statistics by addressing lodgings being too small to be covered by the current tourism survey. It also looks at how to measure transport by the recent alternatives to taxi. The Norwegian case study looks at to what extent debit card transactions at legal unit level can be used to estimate the monthly total retail sales by the ten industry classification groups used for the Retail sales index statistics.

As the final application, sharing economy is considered. First, the theoretical and then the operational definition are discussed, since there is no universal agreement on how to define sharing economy. Starting with a "pure" definition limited to sharing of temporarily available physical goods between persons, the operational definition finally adopted is the definition of collaborative or platform economy. The latter is both easier to measure as well as more relevant for economic statistics, enabling e.g. travel accommodation statistics covering more than hotels & regular guesthouses.

After the definition part of sharing economy, different approaches to assessing sharing economy is discussed. The selected approach is to use financial transactions data for finding indicators of sharing economy. First, a survey is presented, describing the availability of aggregated numbers on credit cards usage found in different WPG countries. This survey was the basis for a methodological study deriving sharing economy indicators. Finally, the possibility of calculating indicators and comparing indicators between WPG countries were investigated. It turns out that such a comparison is not possible with the available data. Further, the underlying problems of measuring sharing economy (here collaborative economy) as opposed to e-commerce is discussed, the former being a subset of e-commerce and more difficult to separate from the rest of e-commerce. Also, for e-commerce it is easier to integrate the financial transactions data with other sources such as administrative data and freely available statistics on internet activity on commercial pages.

After the conclusions, this report ends with the recommendations for countries who want to approach the use of FTD in statistical production and also recommendations for the direction of further developments at EU and the national institutional level that should facilitate data access and processing. WPG members advocate the inclusion of FTD among the priority areas identified by Eurostat in exploring potential options of public-private co-operation and in the definition of the European Data Spaces designed by the European Commission in order to enhance the access to privately-held data, via industrial data platforms.

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PART I - Financial transactions data (FTD)

1. Introduction

Financial transactions data (FTD) potentially contain all financial transactions in a country and can potentially enrich official statistics in various fields such as economic statistics and business structure statistics. In a society where new payment solutions emerge as an option to cash and where the internet enables easier trading, FTD are a natural data source for statistics monitoring the economic activity.

This report is the final and cumulative deliverable of the ESSnet Big Data work package "WPG Financial transactions data". The main aim of the report is to give an overview of FTD in each of the participating countries in WPG: Bulgaria, Germany, Italy, Norway, Portugal and Slovenia. Further, the statistical potential of the FTD sources is an important part of WPG, illustrated by case studies. An aim has also been to study the application of FTD for measuring the sharing economy.

The first part of this report covers an FTD inventory. Section 2 is devoted to recent European Statistical System (ESS) initiatives related to FTD apart from the Essnet Big Data. Then in Section 3, we give a general description on types of FTD, processes and actors (Section 3.1), the legal situation and cooperation with data providers (Section 3.2), the metadata on FTD (Section 3.3), and finally statistics that seem promising for benefitting from FTD. Section 4 provides insight into the special situation in each of the WPG countries Bulgaria, Germany, Italy, Norway, Portugal and Slovenia. Details for each country are presented in Appendix 2-Appendix 7.

In the second part, different applications of using FTD are presented. In Section 5, general considerations are made on the case studies being presented in Section 6-Section 8 for the Italian, Norwegian and the Portuguese case studies respectively. Then, in Section 9-Section 10, the use of FTD on sharing economy is investigated: Section 9 is devoted to defining sharing economy in a suitable way for measuring the impact of sharing economy on economic statistics, given that there is no commonly internationally accepted definition. Then in Section 10.1, first the different sharing economy definitions are discussed before the three approaches to measuring sharing economy are discussed. The approach of measuring the sharing economy subset of FTD is chosen and operationalised in Section 10.2, where also a survey of the availability of relevant data in the WPG countries is analysed with the aim of developing indicators of the sharing economy. The possibility of deriving and comparing sharing economy indicators in practice are discussed in Section 10.3, before it is discussed in Section 10.4 whether sharing economy-related transactions are measurable. Finally, perspectives for future developments on sharing economy indicators are described in Section 10.5.

In the third part, after the conclusions in Section 11, recommendations are given for future use of FTD in official statistics.

1.1 Terminology

A financial transaction is an agreement between a buyer and a seller to exchange goods, services or financial instruments. The two actors are denoted *peers*. The process described here builds up a transaction: 1) a transaction of goods etc. from a *provider* to a *user,* represented by process 1 in Figure 1.1, and 2) a financial transaction going in the opposite direction, represented by process 2 in Figure 1.1.[1]

[1] The payer and the receiver of the process 2, is the user and the provider in process 1, respectively.

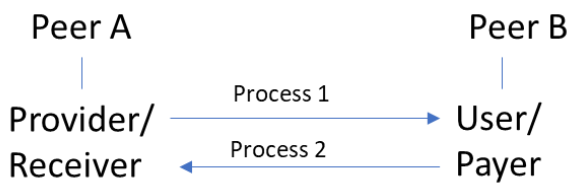

Figure 1.1. The two transactions in a sharing economy transaction.

There are several ways to subdivide and categorise FTD, one of them is to cross-classify into six groups by type of transactions and by type of payer-/payee pair. Transaction types are debit or credit card transactions and "giro"-transactions (bank-to-bank credit or direct debit transfers). Payer-/payee-pair types are consumer-to-consumer (C2C), business-to-business (B2B) or consumer-to-business (B2C)[2]. A consumer in our context can be a person or a household.

As noted above, a peer could be either a business or not a business. In this report, a person is defined as a business whenever the tax authorities define the person as a business. When the peer is not a business, we denote the peer a person, which covers both private persons, households and any other entity not being a business according to the definition above. In the literature it is also common to use the term 'consumer' on the non-business peer, but by using 'person' instead of 'consumer', we can separate the role of being a non-business peer from the role of being consumer of the goods provided in the sharing economy transaction.

By *financial transactions data,* we will refer to data on any monetary transaction (between persons, businesses, or between a person and a business) involving the transfer of money between a payer and a payee. Such transactions are also sometimes denoted "payment transactions". Related to financial transactions data are *purchase transaction data*, which consists of receipts and other data about the contents of the purchase but does not contain information about the payer.

A *sharing economy platform* is a business being mediator between the payer and payee in a *sharing economy transaction* where typically under-used goods are shared for a limited time. This will be more precisely defined in Section 9. The sharing economy platform collects/generates data on the sharing transaction. At least some of these *sharing economy platform data are* stored in and/or owned by the sharing economy platform.

1.1.1. Abbreviations

We will use the following abbreviations in the rest of this document:

ADL	Autoregressive Distributed Lag model
API	Application programming interface
AR model	Autoregressive model
ATM	Automated Teller Machine
B2B	Transaction between two businesses
B2P or B2C	Transaction between a business and person (independently of the direction, i.e. who is the payer and payee)
BI-COMP	Bank of Italy compensation system
DSOP	Digital Cooperation between Governmental and Public sector (in Norway)
ESCB	European System of Central Banks
ESS	European Statistical System
FTD	Financial transactions data

[2] For «consumer-to-business», the consumer could be either the payer or the payee.

IFT	Internet financial transactions
MAE	Mean absolute errors
MAPE	Mean Absolute Percentage Error
MCC	Merchant category code
MIDAS	MIxed DAta Sampling model
NACE	Nomenclature statistique des Activités économiques dans la Communauté Européenne (standard industry classification)
NSI	National statistical institute
NSP	National Statistical Plan
NSS	National statistical system
P2P or C2C	Transaction between two persons (sometimes having the role of “consumer”)
POS	Point of Sale
RMSE	Root mean squared errors
SCT	SEPA Credit Transfers
SDD	SEPA Direct Debits
SEP	Sharing economy platforms
SEPA	Single Euro Payments Area
SEPA	STEP2-T retail payment system for and settlement. It is run by
SIA	Società interbancaria per l'automazione
TARGET2 or T2	Trans-European Automated Real-Time Gross Settlement Express Transfer System

2. FTD – a category of privately-held data

The issue of accessing and using FTD for official statistics is also under investigation by the EU, as well as by the ESS[3], exploring potential public uses of "privately-held data". Within the framework of the European Data Strategy[4] and the provisions for data protection given by the UE GDPR Regulation[5], the aim of fostering "business to government" data sharing has become a priority at EU level. Some general principles and practices have been laid down by the *High-Level Expert Group on Business-to-Government (B2G) data sharing*, an independent expert group set up by the European Commission in November 2018 that published a detailed report[6] about how the public and private sector could effectively co-operate to use available data for their independent needs. Several recommendations and classifications might be also useful to improve the co-operation between public institutions and FTD owners. For instance, it is not always needed to share FTD micro-data, but data providers could commit themselves to run specific routines, agreed with their public counterparts, in order to produce aggregated data and to deliver the requested information without disclosure of confidential information (not necessarily only about individuals but industries or territories, as well).

At ESS level, an initiative was taken by the Steering Group of the joint network of EU National Statistical Institute (NSI)'s Directors for methodology and Directors for Information Technology (DIME / ITDG) by establishing an ad-hoc Expert Group with the aim of exploring potential uses of privately-held data[7]. The Expert Group was given as a task the identification of potential areas to explore for the acquisition by NSIs of privately-held data for the purposes of EU official statistics. Actually, the Group mostly collected information on on-going research and testing activities (including those of the ESSNet BigData) and developed a check list of activities to be taken by the ESS and its members in the near future[8]. Moreover, a list of principles to serve as foundations of future public-private partnerships was set out and, more on a practical level, an evaluation

[3] European Statistical System.

[4] See: https://ec.europa.eu/info/strategy/priorities-2019-2024/europe-fit-digital-age/european-data-strategy_en and Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. *A European strategy for data*, Brussels, 19.2.2020, COM(2020) 66 final.

[5] The General Data Protection Regulation (EU) 2016/679 (GDPR) is a regulation in EU law on data protection and privacy in the European Union (EU) and the European Economic Area (EEA). It also addresses the transfer of personal data outside the EU and EEA areas. The GDPR's primary aim is to give control to individuals over their personal data and to simplify the regulatory environment for international business by unifying the regulation within the EU. Source: Wikipedia and official EU sources.

[6] *Towards a European strategy on business-to-government data sharing for the public interest*. Final report prepared by the High-Level Expert Group on Business-to-Government Data Sharing, Luxembourg: Publications Office of the European Union, 2020

[7] The Expert Group was established in July 2019 by answering for a need emerged in the May 2019 meeting of the ESS Committee; Since then, the activity of the Expert Group has been closely monitored by both the ESSC and the Partnership Group (see the agenda of the 74th meeting of the Partnership Group in March 2020 and, in particular, the document *PG 2020/74/4.1.a* on "Actions enabling the use of privately held data for official statistics").

[8] From PG 74th meeting's Doc *PG 2020/74/4.1.a*: "Actions: 1. The ESS (NSIs and Eurostat) should engage in activities building the data stewardship functions and the related network of data stewards to position itself as a leading player on the side of the public sector, and to influence governance structures supporting the B2G data sharing. 2. Add new or extend the already identified and described case studies of the position paper mentioned earlier (ESS, 2017) in support of a future impact assessment, providing compelling evidence that access by the ESS is needed, focusing on comparison of risks of not using new data sources with benefits for the society, the economy and the environment of using them to produce official statistics. Such case studies should build as far as possible on existing data collaborations (e.g. in the area of MNO or scanner data). 3. Investigate how the creation of common European data spaces could become a platform for enabling the use of the data for European statistics. 4. Elaborate strategies for legal solutions and start drafting legislative text to be included in a possible future legal initiative. 5. Continue publishing (experimental) statistics based on new data sources to strengthen the case for sustainable access. 6. Continue the creation of information products in collaboration with data holders. 7. Conduct project using privacy preserving methodologies in using privately held data, building on the initiatives carried out in the context of the GWG and HLG-MOS."

tool to be used in order to assess the potential of private data sources to be effectively used for official statistics purposes. The tool is illuminating for the future perspectives on FTD, and it is based on seven dimensions used to classify each single potential source of data according to the following scheme, where we quote from Group on privately held data (2020), using different formatting and numbering:

“

1. **Data assignability**: A classification dimension is related to the “data assignability” of big data and of the relevant sources to persons/economic unit. In this respect, data can be:
    a. Assignable, meaning that data are directly or indirectly associated to a person/economic unit. Examples are data from wearables or smart meters, mobile network operators’ data, etc.
    b. Non-assignable. An example of data that are not associated to a person or an economic unit is data related to environmental sensors.

    This classification dimension is relevant to legal issues.
2. **Concession**: A classification dimension is related to the relationship of the big data holder (company) with the government. Several services such as water supply, electricity supply and the use of mobile telephone networks may be operated as a concession. In this respect a big data holder can be:
    a. A private company with public concession, e.g. Mobile Network Operators, suppliers of gas and electricity, etc.
    b. A private company (without any public concession), e.g. Twitter
    c. A public organisation/subject to restrictions for data access

    This classification dimension is relevant to practical issues such as business models and to legal issues.
3. **Data provision**: A classification dimension is related to direct provisioning of data with respect to the presence of intermediaries.
    a. Direct provisioning, e.g. Mobile Network Operators data
    b. Intermediary-based provisioning, e.g. Smart meters data

    This classification dimension is relevant to practical issues such as business models and to legal issues.
4. **Data access**: A classification dimension is of technical nature and maybe related to big data access
    a. On the premises of the data provider, i.e. the data provider makes available results of elaborations. This can be the case for Mobile Network Operators’ data.
    b. Via API: data are accessed and processed either on the premises of the NSI or by intermediaries. This can be the case of Twitter data.
    c. Via a direct data collection: data are collected via apps or wearable either by the NSI or by intermediaries.

    This classification dimension is relevant to technical issues.
5. **Trade confidentiality**: A classification dimension is related to trade secrets or confidential information that could have an impact on companies’ competitiveness. This classification dimension is relevant to practical and legal issues.
6. **Data structure**: A classification dimension is related to the structure of the data that is relevant for the usability of the data.
    a. Structured data such as road traffic data
    b. Semi-structured data, such as scraped data from web pages
    c. Unstructured data such as those of social media

7. **Global data holders**: An additional dimension is related to the national, European and international nature of the approach to access the data source. For instance, for MNOs data, the providers can be national or multinational. In the case of multinational companies, the approach to access data could be national by considering for instance business models of the Telco companies that are different across countries.

“

A number of different potential sources of private data was screened by the Expert Group against these criteria to check for the feasibility of using them as inputs of official statistics. The results for FTD are given in Table 1.1 and confirm some critical issues related to the accessibility of FTD.

Table 1.1. Classification made by the Expert group of FTD against the seven dimensions (Group on privately held data (2020)).

Data source	**Data assignability**	**Concession**	**Data provision**	**Data access**	**Trade confidentiality**	**Data Structure**	**Global data holders**
Financial transaction data (credit cards, debit cards, online payment systems)	Assignable	Private company	Direct or Intermediaries	Via APIs	Yes	Structured	National / European/ International

Since FTD are assignable, i.e. associated with units, FTD have a much higher potential for statistics than many other Big Data, c.f. the case studies in Section 6-Section 8, and at the same time: being "private property" highlights a major challenge, as the WPG group has experienced in the work of getting data. Data provision is a key issue, because NSIs should aim at getting in touch with data providers without intermediaries, which could substantially improve the level of co-operation and the quality of data transmitted. The identification of access channels only via APIs is not correct: FTD can be accessed in multiple ways provided that a collaboration with the data owner could be established. Trade confidentiality is an additional critical issue since this requires more aggregated data than would otherwise be possible. Finally, the availability of highly structured FTD at both national and international level has to be seen as an asset, in contrast to the situation for many other types of Big Data being highly unstructured.

3. Financial transactions data (FTD) (from the finance industry)

In this section we will look at the common properties of FTD for the WPG-countries. In the rest of this document, we use terms that might not be widely known outside the banking sector. Therefore, many terms related to financial transactions are described in Appendix 1.

3.1 The different financial transactions data and where to be found

The transactions among the economic actors need to rely on a well-designed payment system, that is on a complete set of payment instruments, intermediaries, rules, procedures, processes and interbank fund transfer systems aimed at facilitating the circulation of money in a country or currency area. In this sense, a payment system comprises three main elements:

1. the payment instruments (i.e. payment cards such as debit, prepaid, credit; credit transfers and direct debits), which are a means of authorising and submitting a payment;
2. the processing, which involves the payment instruction being exchanged between the banks (and accounts) concerned;
3. a means of settlement for the relevant banks (i.e. the central bank money).

Starting with the perspective of the payer and payee of the FTD, we will first look at the payment instruments and payment platforms. Then we will look at the process behind a financial transaction and the actors within different parts of the process.

3.1.1 Payment instruments and platforms

The payment instruments are:

- Cash, i.e. banknotes and coins.
- Debit cards: Cards enabling cardholders to have their purchases (or cash withdrawals) directly and immediately charged to their bank or payment accounts (except for e-money accounts), whether held with the card issuer or not.
- Delayed debit cards: Cards issued with a contractual agreement granting a credit line, but with an obligation to settle the debt incurred at the end of a pre-defined period without charging the cardholder any interest rate (commonly referred to as “charge cards”).
- Credit cards: Cards enabling cardholders to make purchases and/or to withdraw cash up to a pre-arranged ceiling. The credit granted may be settled in full by the end of a specified period or may be settled in part, with the balance taken as extended credit on which interest is usually charged.
- Prepaid cards - a card on which a monetary value can be loaded in advance and stored either on the card itself or on a dedicated account on a computer. Those funds can then be used by the holder to make purchases. Prepaid card is a typical example of e-money.
- Credit transfers (“Giro”)- payment instruments based on payment orders or possibly sequences of payment orders made for the purpose of placing funds at the disposal of the payee
- Direct debits (“Giro”[9]) - payment instruments based on preauthorised debits, possibly recurrent, of the payer’s account by the payee

[9] i.e. direct transfer from a bank account to another bank account.

In addition, cheque[10] is another payment instrument, but less used today and we will not consider it.

Payment platforms are:

- POS (point of sale) terminal - devices typically used at a retail location to capture payment information electronically and – in some cases – on paper vouchers
- ATM (*Automated teller machines)* - terminals that allow authorised users, typically by using a card, to access a range of services such as cash withdrawals, balance enquiries, transfers of funds and/or acceptance of deposits. Note that not all ATMs need to have cash withdrawal functionality.
- Web - electronic transactions conducted via the internet or other computer-mediated (online communication) networks.
- Mobile payment solution - A solution used to initiate payments for which the payments data and the payment instructions are transmitted and/or confirmed via mobile communication and data transmission technology through a mobile device. This category includes digital wallets and other mobile payment solutions used to initiate P2P and/or P2B transactions.
- Traditional mail/telephone - Traditional mail was the traditional platform for paying giro, and about three decades ago, telephone also appeared as a platform for giro. Also, telephone is used for some kind of card transactions (so called "mail order/telephone order").

Traditional mail/telephone is a platform less used today and its use is decreasing. It will be disregarded for the rest of the document.

The payment instruments and platforms can be combined. Table 3.1 shows all the possible combinations: of course, cash payments are not possible on the web and on a mobile payment solution.

Table 3.1. The possible combinations of payment instruments and payment platforms. 'x' indicating a possible combination.

Payment instrument	POS terminal	ATM	Web	Mobile Payment Solution
Cash	x	x		
Prepaid cards	x	x	x	x
Delayed debit Cards and credit cards	x	x	x	x
Debit cards	x	x	x	x
Direct debits ("Giro") [1)]			x	x
Credit transfers ("Giro") [1)]	x	x	x	x

[1)] As a standardised system for transfers between bank accounts, "Giro" can also be considered as a separate payment platform

3.1.2 Processes during the financial transaction

After a payment is initiated by the payer, until the payee has received the data and all accounts have been settled, we can distinguish two types of processes: 1) clearance and settlement, and 2) transaction processing.

[10] Negotiable payment instruments based on written orders from one part (the drawer) to another (the drawee, normally a bank) requiring the drawee to pay a specific amount from a specified transactional account held in the drawer's name with that institution to the drawee, or a third part specified by the drawer.

3.1.2.1 Clearance and settlement systems

The retail payment system market generally uses the clearing and settlement mechanism model, in which one or more operators perform *clearing* (i.e. transmission, matching, confirmation of payments and calculation of a final settlement position) and *settlement* (fulfilling the obligations that arose during clearing). For a clearing and settlement mechanism to run smoothly there must be rules governing its operation and participants' access and exit criteria, as well as specific functionalities and technical standards that interconnect all the participants and the other systems. Clearance means the calculation of the process of transmitting, reconciling and, in some cases, confirming transfer orders prior to settlement, potentially including the netting of orders and the establishment of final positions for settlement. Settlement is the actual transfer of money between the banks/financial institutions to finance the retail payments/micro transactions. The settlement can be done "net" or "gross": a) net settlements: a net position is calculated as a result of the clearance process described above. The net position for transactions happening during the time interval in question is transferred from one entity (the one with the largest sum of payments) to the other (the one with the smallest amount of payments); b) Gross settlements: The amount that shall finance a single micro transaction is transferred from the payer's bank / financial institution to the payee's bank / financial institution. Gross settlements may take place for every single micro transaction or just for some transactions, like transactions exceeding a specific large amount.

The Bank of Italy, along with Deutsche Bundesbank and Banque de France, contributed to develop the payment system TARGET2 which is used within all of EU, and settles the most part of the wholesale transfers, on a gross-basis in real-time. As for the retail payments, the payment system is not fully integrated at the European level yet. However, since 2014 the Single Euro Payments Area (SEPA) has strongly fostered the standardisation and interoperability among different national clearing and settlement retail systems, especially for credit transfers and direct debits instruments.

A retail payment system operating across many countries of SEPA is the STEP2-T retail payment system for clearance and settlement. It is run by the private company EBA CLEARING via a pan-European platform, for the centralised settlement of SEPA payments by participating European banks. As a cornerstone of the SEPA processing infrastructure, STEP2-T provides full reach for SEPA Credit Transfers (SCT) and SEPA Direct Debits (SDD) to over 4,800 payment service providers across all SEPA. Payment transactions and retail collections in STEP2-T are forwarded in the form of files to the central system of EBA through the service provider SIA, where they are processed for clearing and transferred for regulation on the TARGET2 platform. In this system, the processing solutions for the interbank management of collection and payment instruments SEPA Credit Transfer (SCT) and Direct Debit (SDD) are offered by SIA and provided in compliance with the SEPA standards in force.

Banks belonging to different banking groups can also exchange retail payments through TARGET2 payment system. TARGET2 is based on a single shared platform and, from a legal point of view, it is structured as a set of national systems. TARGET2 settles payments on accounts opened by participating banks at their national central bank: therefore, the presence of central bank money makes the system safer compared to using commercial bank money, which is subject to the risk of default by the bank where the settlement account is held. This characteristic enhances the robustness of the platform to financial crises, thereby helping to mitigate systemic risk. TARGET2 is used to settle payments relating to monetary policy operations, interbank payments, operations on behalf of customers of participating banks and ancillary system transactions (i.e. customer payment systems, securities settlement systems, central counterparties and money markets)

For the non-EU country Norway, the clearance and settlement are performed within Norwegian organisations, and a large part of the clearance is done through a net clearance system implying an aggregation of single transactions to the net transfer between the banks during a certain time interval. Details of this clearance and settlement system is described in Section 4 and in Appendix 5.

3.1.2.2 Transfer processing

While clearing and settlement are performed to timely secure and provide liquidity to finance retail payments, every single payment is made with the intention of timely transporting money from the payer to the payee. The value for all statistics produced to measure consumption lies in the information connected to retail payments.

When gross settlements take place, clearing and settlement providers may be useful as a source for information about every payment made from a bank account. Otherwise, payment service providers /actors registering retail payments should be used as a source. Payments with giro, cheques and debit cards take place by debiting a bank account. To find information about payments with these instruments, banks or the service providers processing bank data, will be a natural data source. For information about payments with delayed debit cards, credit cards and prepaid cards/prepaid accounts, the actual issuer of card/account should be used as a data source.

3.1.3 The actors -- Payment service providers and owners of FTD

The transactions among the economic actors need to rely on a well-designed payment system. It also relies on institutions that provide payment accounts, instruments and services to customers and on organisations that operate payment, clearing and settlement services.

The payments between financial institutions are named wholesale payments while those made by banks or other payment service provider on behalf and between non-financial institutions (households, firms, government) are named retail payments.

Data referring to payment card transactions (debit/prepaid or credit), credit transfers and direct debits will usually be owned by banks. In the case where the transaction is processed between two accounts in the same bank, this bank is the owner. When two banks are involved, they own the data together. Thus, for many transactions there is a shared ownership. To handle such transactions between different banks, the financial institutions have established privately owned service providers with the role of operating the transactions. These service providers can take care of the technical role as bank data centrals, and the acquiring as well as the clearance and settlement role. Different entities can have the same role, where each of them is covering different parts of the financial market, e.g. different banks.]

Central banks have a keen interest in the smooth functioning of the national payment system and the way it affects the economy since modern economies are dependent on the safe and efficient flow of transactions. They are involved in payment, clearing and settlement in many different ways: as operators and providers of settlement services in central bank money; as participants in or users of such systems; as oversight/supervision authorities; and as promoters of efficiency in the payment system as a whole.

The central bank usually has a role in settlement, but it may also have a role in the clearance of retail payments. If we look at the role of the central bank in the financial transaction market more broadly, it is somewhat different across countries (as we will see in Sections 7 – 12), but there are also some similarities. The availability of payment data at the central bank depends on the different role of the central bank in the payment system:

- Supervision/oversight: All central banks have a role in supervising/auditing the financial market, and this necessitates the acquisition of certain data from the market, that might or might not be relevant for the NSI;
- Clearance/settlement role: The national banks have all a common role in contributing to the development of an effective financial market. Still, the central banks have different roles in clearance and settlement of the transactions.

- Official statistics production: one difference concerning the role of the national banks, is that national central banks in EU have the responsibility of all the official statistics on the financial market and thus for that purpose they produce payment data in-house. On the contrary, the NSIs have this responsibility only in some other countries. As an example, in Norway some of the financial market statistics is made by Statistics Norway.

In Section 3.2.1, we will look at the differences in the role of the national central banks from a legal point of view.

Payment data referring to debit cards, credit cards and prepaid cards are owned by every single bank or financial institution that has issued the actual card. The ownership of data connected to the receipt of amounts from these cards will usually be a payee's bank, but this depends on how/to which account the amount is transferred. International payment cards, that can be both debit cards, delayed debit cards and credit cards, are issued by banks under the brand managed by international card schemes. The international card schemes define standards and rules for cards that can be used worldwide such as VISA, MasterCard, Diners, American Express etc. assigning licenses to local banks or financial corporations as principal members for issuing card or acquiring merchants. As the licenses also cover the duty to process transactions, with such licenses the banks or national card companies (via their card processors) may be valuable as a data source for card payments information. International card companies also assign roles to local banks or other financial corporations as acquirers of merchants (POS). Since acquirers (through the processor) processes card payments on behalf of a merchant, they may be useful as a data source for information connected to user locations. They will also be able to give information about the geo-localisation of the usage of cards (at national level or abroad)

Payment card transactions and credit transfers may be addressed through mobile devices (mobile payments), and the operational framework can be handled by banks, card operators as well as separate mobile payment operators. In case of "closed loop" payment models, payments between payer and payee are settled in a separated e-money account (prepaid account) such as in the case of PayPal. In case of "closed loop" models, a card (or direct debit) transactions initiated at the bank account can be done for pre-funding the e-money account.

In addition to the owners of the FTD, other service providers mentioned above involved in the actual processing of payments have such data, sometimes also national central banks. In some cases, they may have data on behalf of many banks/financial institutions. However, the service providers typically have only the FTD relevant for their role in managing the financial transactions.

The access to FTD might not necessarily have to go through their owners. Both the legal authorisation as well as the technical access can sometimes go through the entities mentioned above instead of accessing through the owners, but this depends on each country's legislation. Since there are many banks in each country, this will often mean more efficiency and less work both for the NSI and for the banks to get direct debits and debit card transactions through a service provider. For international card data, this depends on the number of actual issuers of cards v/s entities with licences as issuers and/or acquirers in the market.

Above, we have the following list of entities having/maintaining FTD in addition to banks:

- National central banks
- interbank clearing/settlement entities
- private payment service providers for banks
- credit / debit / prepaid cards operators
- mobile payment operators
- tax / control authorities' data

Table 3.2 summarises all the FTD data considered within the WPG by each participating country. The table presents on the left-hand side the already accessed data and on the right the data that exists in the country and might be accessed in the future. For each country, the data accessed are broken down by type of maintainer (e.g. National central banks, interbank clearing/settlement entities, credit or debits cards operators, tax authorities or other national control authorities). The table shows the complexity of the framework and the actors having card data in each country. Among the already accessed sources, the interbank clearing/settlement entities are slightly prevalent (see the cases of Italy, Norway and Portugal), but also the credit card providers are very relevant data providers for our project (Italy and Norway, and possibly in Bulgaria and Slovenia).

Table 3.2. Framework of the considered FTD data (already accessed or not) in each WPG participating country so far, by type of maintainer.

Country	Types of (accessed) data by type of maintainer				Possible data (not already accessed) by type of maintainer		
	NATIONAL CENTRAL BANK DATA	INTERBANK CLEARING/SETTLEMENT OF THE SYSTEM OF PAYMENTS DATA	CREDIT / DEBIT CARDS OPERATORS DATA	TAX / CONTROL AUTHORITIES DATA	NATIONAL CENTRAL BANK DATA	CREDIT / DEBIT CARDS OPERATORS DATA	TAX / CONTROL AUTHORITIES DATA
BULGARIA					**Bulgarian National Central bank** data **-** data from the RING system (established in 1996, it performs real time gross settlement of all payments in the national currency within the territory of Bulgaria) and from another settlement system for interbank payments in euro.	Data from the Company that maintains card payments technical infrastructure	
GERMANY	Total number and value of transaction/payment by type of card and terminal.						
ITALY	**Bank of Italy – Financial Intelligence Unit (FIU) Anti-money laundering aggregate reports (SARA) database** of transactions over 15K euro**.**	**Bank of Italy data from the inter-exchange and settlement systems BI-COMP and TARGET2 retail of the System of payments**	.			**SIA (Società interbancaria per l'automazione) data; Sia is** a processing entity both for issuing and acquiring side (debit and credit card) **Nexi S.p.A data -** it is a PayTech not only a processing entity but also a credit card company, as well as a primary bank issuing debit card. Information collected both on the issuing and on the acquiring side	
PORTUGAL		**Interbank settlement of the System of payments - SIBS data.** It is connected to the main issuers of payments in Europe, in all operations and in issuers, with several units and several networks - Visa, MasterCard, American Express, UnionPay). Data refer to Data of ATM Express and MULTIBANCO networks, as SIBS send to Statistics Portugal monthly data (only) at NUTS IV level of cash withdrawals and POS payments and they are published by Statistics Portugal by municipality**.**		**Tax and Customs Authority (AT) data** - E-Invoice data E-Invoice is the invoice confirmation system issued under a taxpayer number (communicated by the entity that provides the service or sells the good). The following information will be available to Statistics Portugal: info: VAT Number (taxpayer number),VAT of the buyer, Month of transaction, Country (of the buyer), Taxable value (by VAT Number)	**Bank of Portugal data** - it is manages information on the SEPA (Single Euro Payments Area) financial transfers such as: name of the payer and/or I**BAN of the payer's** payment account; amount to be transferred; IBAN of the payee's payment account		
SLOVENIA						**BANKART data** - it is a company founded by the 22 Slovenian banks processing of ATM operations, card business, payment systems and services (SEPA transactions in both SEPA and the SIMP-PS payment system and the E-invoice system), ATM and POS terminals**.**	**Fiscal cash registers data** - Financial Administration of the Republic of Slovenia monitors all issued invoices, which are paid in cash. This includes all payments except credit transfers, meaning also payments with credit or debit cards are include. adopted measures against grey economy and tax evasion**.**
NORWAY		**Interbank clearing system for the Norwegian Krone (NOK) - NICS data.** NICS manages bank account transfer covering both P2P, P2B,B2P and B2B	**Private provider debit card data**. Data are collected on the acquiring side.				**DSOP data -** Digital cooperation between public and private sectors. Source of B2B,P2P,B2P transactions, the new system will cover both debit card data and bank account to bank account transfers.

3.1.4 Dimensions of FTD

We have so far identified three dimensions regarding FTD:

- Dimension 1: Entities having/maintaining FTD
- Dimension 2: Payment instrument
- Dimension 3: Payment platform.

In Table 3.3, we have represented Table 3.2 in a different way by seeing what the status is of getting FTD for the combinations of dimension 1 and dimension 2.

Table 3.3. The existence of FTD in WPG countries, by payment instrument and by type of entity maintaining the data.

	Type of entity maintaining data			
Payment instrument	Central bank	Clearing/settlement	Card operators	Authorities
Cash				SI*
Delayed debit and credit cards	DE	IT*	IT*, NO***, SI	NO**
Debit cards	BG, DE, IT*	IT*	IT*, NO, SI	NO**
"Giro"	BG, IT*, PT	NO*, SI, PT		NO**, PT?

* test data are already accessed or will soon probably be available for the NSI (covering part of the population)
** will appear as a data base outside NSI in a couple of years.
*** Such data have been identified also in Norway, held by the Issuers.

Let us consider expanding Table 3.3 by also including payment platform, the third dimension. The question then is: is data from a payment instrument handled differently by the actors of Table 3.3, depending on whether payment was done (at POS terminal or on the Web)? In Norway, as an example, one of the largest providers of payment systems data has information on all the use of card data in POS terminals accepting the national card scheme. To receive information about use of cards in an ATM or over the internet, we have to go to the owner of the ATM (for ATM transactions) or card issuer or the entity with the issuing license (for both ATM transactions and transactions over the internet).

As we have seen both in Table 3.1 and Table 3.3, there are some pairwise combinations of the three dimensions that will not exist. We can see which combinations of all three dimensions not being possible in Table 3.4. All or some of the cells of Table 3.4 will be needed to cover as much as possible of the statistical population for certain statistics to be produced. Notice that for a given row (payment instrument), a data maintainer might contain all POS-data and all web-data, and in that case only the corresponding columns are relevant for these statistics. This also means that the relevant cells of Table 3.4 for covering as much as possible of the statistical populations, will sometimes be different depending on the country.

Table 3.4. Types of FTD by payment instrument, by type of entity maintaining the data, and by payment platform. Grey cells indicating impossible combinations

	Type of entity maintaining data							
	Central bank		Clearing/ settlement		Card operators		Authorities	
Payment instrument	POS/ ATM	Web	POS/ ATM	Web	POS/ ATM	Web	POS/ ATM	Web
Cash								
Credit cards								
Debit cards								
"Giro"								

3.2 Getting FTD from an actor having the data

To get data from the different actors (cf. Section 3.1.3), two essential activities to implement are: 1) meeting national legal requirements, and 2) setting cooperation agreements.

3.2.1 Legal requirement for getting FTD

In considering the legal aspects allowing or facilitating the access to the FTD we must consider simultaneously two dimensions, the prerogatives that the national statistical law (in the provided forms) guarantees to the national statistical institutes (NSIs) for statistical production purposes and the prerogatives of autonomy in the field of the supervision and the control of payment system that the law provides to the central banks in the different countries.

Among the countries participating in the WPG on FTD of the ESSnet Project BIG DATA II, there are similarities regarding the legal requirement ruling the general data access set by the national statistical acts, but major differences in the norms setting the relationships between the NSI and the National central banks. This is due to fact that four participating countries belong to the European system of central banks (ESCB) (Germany, Italy, Portugal and Slovenia) whereas Bulgaria and Norway do not belong to the Euro area and consequently present different rules.

In line with art. 105 of the Treaty establishing the European Community and in line with art. 3 and 22 of the Statute of the European System of Central Banks and the European Central Bank, individual national central banks of the Eurosystem may provide processing facilities for retail payments in euro for credit institutions, either via participation in private retail payment systems or acting as operators of their own retail payment systems, in order to contribute to the safety and efficiency of payment systems in the euro area. Depending on the specific national circumstances, they may also facilitate access to payment systems for all credit institutions.

Regarding the general principles ruling the data production in the ESCB, the Article 5 of the Protocol on the Statute of the European System of Central Banks and of the European Central Bank *(Official Journal of the European Union C 202/230, 7.6.2016)* sets for the ESCB, assisted by the national central banks, the tasks of data collection of the necessary statistical information for its aims. The national central banks carry out the collection, and the ESCB contribute to the harmonisation of the rules and practices governing the collection, compilation and distribution of statistics. The ESCB attaches great importance to the quality of its statistics. It therefore takes into consideration internationally agreed quality standards, such as those formulated in the IMF's Special Data Dissemination Standard and Data Quality Assessment Framework, which are in turn rooted in the UN's Fundamental Principles of Official Statistics.

Nevertheless, without prejudice to the cited Protocol, the ESCB collaborates with the European Statistical System (ESS), which comprises Eurostat, the National statistical institutes and other national statistical authorities, and takes account of the principles laid down in the European Statistics Code of Practice for the National and Community Statistical Authorities.
The Article 21 of EC Regulation n. 223/2009 and the Article 8 of the EC Regulation n. 2533/98 establish that the exchange of confidential data between ESS and ESCB subjects is authorised if necessary for the efficient development, production and dissemination of European statistics or improvement of their quality and that this need has been justified.

An outline of the main legal features at national level may highlights the national specificities.

3.2.1.1 Bulgaria

Statistical Act stipulates "The National Statistical Institute shall collaborate with the Bulgarian National Bank, including by means of exchange of individual data for statistical purposes, in developing, producing and

disseminating official national or European statistical information or the statistical information of the European Central Bank." Art.4 (5)

Further on Art.20 (6) says "Central and local government authorities, other central government institutions, the Bulgarian National Bank and the bodies keeping registers and information systems as stipulated by a law, shall have the obligation, upon written request and free of charge, to provide the National Statistical Institute and statistical authorities with the individual data collected by them and/or statistical information that is necessary for conducting the statistical surveys included in the National Statistical Programme".

3.2.1.2 Germany

The activities of the Federal Statistical Office in Germany are based on the Act on Statistics for Federal Purposes (Federal Statistics Act - BStatG) in the version promulgated on 20 October 2016 (Last amended by Article 10 paragraph (5) of the Act of 30 October 2017 and published in the Federal Law Gazette I, p. 3618). The act is originally published in German language. The provisions cited here are based on an English working draft. But only the German version is authentic.

According to the Federal Statistics Act every statistical survey has to be based on a legal regulation. Apart from the case of a European regulation this can be an act adopted by the legislative or another form of legal regulation enacted by the government (Stipulated in Section 5 of the Federal Statistics Act). Further provisions (Sections 5a, 6 and 7 of the Federal Statistics Act) give attention to:

- The use of administrative data
- Measures for the Preparation and Production of Federal Statistics and
- Surveys for Special Purposes

Altogether, the activities of the Federal Statistical Office are not determined by itself but by the government and the ministries respectively.

In Section 3 of the Federal Statistics Act concerning the duties of the Federal Statistical Office the relationship to the national central bank (Deutsche Bundesbank) is described: "[The Federal Statistical Office] shall cooperate closely with the Deutsche Bundesbank to minimise the effort involved in data collection and to ensure the quality and coherence of the statistics compiled."

The tasks of the national central bank are stipulated in the Act on the Deutsche Bundesbank (Bundesbankgesetz – BBankG). Section 18 of this act stipulates that the Deutsche Bundesbank is authorised to arrange and to conduct statistical surveys in the field of banking and finance if this is necessary to perform its tasks.

In addition to these provisions stipulated by law, the Federal Statistical Office and the Deutsche Bundesbank have in 2014 agreed to a common Memorandum of Understanding. In this Memorandum the cooperation between both institutions is concretised. Especially, it is stated that both partners provide each other with statistical data concerning the fields of activities that are subject of their cooperation.

3.2.1.3 Italy

The statistical legislation regarding the NSS is mainly described in the Italian Legislative Decree no. 322 of the 6 September 1989 (and subsequent amendments) and in the Annex A3 to the Privacy Code (the Code of Ethics and good practices for the NSS). According to that legislation, Istat is in charge of proposing a National Statistical Plan (NSP) in which it establishes statistical projects representing a "substantial public interest". The NSP is submitted to the Committee for Safeguarding Statistical Information for advice and then approved and implemented by government's decree. The NSP establishes, among others, the purposes, the appropriate safeguards, the information notice and for which specific projects participation of natural and

legal persons must be deemed as compulsory. Public and private entities can consequently be requested by the NSS to provide individual data in the context of such a mandatory project.

The processing of personal data must also respect the provisions of the Code, Annex A3 and the Statistical Legislation. All projects whose participation is not explicitly defined as mandatory in the NSP are deemed as voluntary and are submitted to the application of the usual rules of the Code and Annex A3 if personal data are processed. Regarding the secondary use of personal data for statistical purposes, it is always possible to use for statistical purposes personal data that third parties have collected for whatever purpose. Nevertheless, Istat has the obligation to inform private and public subjects, which hold or disseminate data through the various multimedia channels, about the specific uses of the processed data.

In carrying out its institutional functions, the Bank of Italy collects data and produces statistics on money, credit, balance of payments and other economic indicators. According to the 1993 Banking Law (art. 146), the legal provisions of the Treaty on European Union (art. 127.2) and the Statute of the ESCB (art. 22), Bank of Italy as oversight authority[11] of payment systems collects payment transactions data from payment systems, bank and non-banks operators and produce (aggregated) statistics on payments.

Moreover, pursuant to the Italian Anti-Money Laundering Law (Legislative Decree no. 231/2007), banks and other financial intermediaries have to report periodically to the Financial Intelligence Unit established at the Bank of Italy (FIU) payment transactions over certain thresholds, after aggregating them by branch, customer sector and type of transaction.

Payment data as well as other data collected within the supervision duties of the Bank (i.e. payment system oversight, bank supervision, anti-money laundering controls) are covered by secrecy and confidentiality rules. However, some data may be available for official publication in aggregated way.

3.2.1.4 Norway[12]

Norway is neither a member of EU nor of the ESCB. The national central bank has a similar role as those in EU when it comes to supervising and auditing the financial market, but the role is different when it comes to official statistics production and dissemination. The Central bank of Norway disseminates the statistics on the financial market. However, contrary to the ESCB system, the financial market statistics of Norway have been produced by Statistics Norway since the Central bank unit that used to produce these statistics, moved to Statistics Norway in 2007. This also means that the legal authorisation for getting the FTD data was changed. The central bank get data from the Central bank Act §27 when needed to perform its main tasks but also for production of Official statistics.

The Statistical Act §2-2 grants Statistics Norway access to any register that Statistics Norway needs for production of official statistics, unless access is prohibited by another law. This right is granted whether the register is governmentally owned or privately owned. When Statistics Norway needs data stored in a register for statistical production, a decision is made which requires the register owner or another governmental or private organisation/business having the data to provide the register. There is an exception when the organisation is just a data handler. Anyhow, a good cooperation process with the data provider (whether being owner or not) is important also in a situation with so strong legislation.

[11] With this regard, according to the banking law and the Eurosystem framework Bank of Italy within the role of payment system overseer and promotes the smooth functioning of the payments system through its direct management of the main circuits and by exercising oversight powers of guidance, regulation and control. This activity, coupled with market supervision, is intended more generally to contribute to the stability of the financial system and foster effective monetary policy.

[12] The description is based on the legal acts operating in 2019.

The paragraph above applies to all registers. The question is then whether access by Statistics Norway is prohibited by other acts. For FTD, the "Financial Institutions Act" and "Central Bank Act" are relevant.

The Financial Institutions Act, §16-2 (1) is about obligation of secrecy and implies that restrictions are not preventing access authorised by another legal act. As seen above, the Statistical Act authorises such access if these is a justified need due to official statistics production.

Obligation of secrecy is covered by the Central Bank Act §12 and prevents access by others with some exceptions. One of the exceptions is that the Public Administration Act §13b no. 4 applies. Together with the Statistical Act §2-2, this authorises Statistics Norway access to Central bank data (given a justified need due to official statistics production).

3.2.1.5 Portugal

The National Statistical System (NSS) includes the Statistical Council, the State body that superintends and coordinates the system; the National Statistical Institute (Statistics Portugal), the central body responsible for the production and dissemination of official statistics, that ensures the supervision and the technical and scientific coordination of the NSS; the Bank of Portugal that, as part of its mission, is responsible for the collection and compilation of monetary, financial, foreign exchange and balance of payments statistics; the Regional Services of Statistics of the Autonomous Regions of Açores and Madeira that act as delegations of the Statistics Portugal in relation to nationwide statistics as statistical authorities in what concerns regional statistics; and other entities producing official statistics by delegation of Statistics Portugal. Statistics Portugal, Bank of Portugal, Regional Services of Statistics of the Autonomous Regions of Açores and Madeira and entities producing official statistics by delegation of Statistics Portugal are considered statistical authorities, having responsibility for the production of official statistics, and empowered to require (mandatory and gratuitously) to all departments or agencies, individuals and legal entities, information necessary for the production of official statistics.

The Bank of Portugal collects and compiles monetary, financial, foreign exchange and balance of payments statistics, particularly within the scope of its cooperation with the European Central Bank (ECB) (Article 13 of the Organic Law). The Law on the National Statistical System (NSS) (Law No 22/2008 of 13 May 2008) acknowledges the Bank of Portugal as statistical authority without prejudice to the guarantees of independence deriving from its participation in the European System of Central Banks (ESCB). According to SEPA (Single Euro Payments Area), the Bank of Portugal is able to manage several information on the financial transfers, such as the name of the payer and/or IBAN of the payer's payment account; the amount to be transferred; the IBAN of the payee's payment account.

3.2.1.6 Slovenia

According to the National Statistical Act (ZDSta), reporting units that are holders of official and other administrative databases are by the statistical survey program defined as data providers (Article 4). According to Article 35 of the National Statistics Act, these reporting units must provide the Statistical Office and the authorised performers with complete and correct data free of charge, in a timely and in a prescribed manner. This duty lasts as long as they perform the observed activity or until the reporting obligations cease. When compiling statistics for which no compulsory reporting is specified, reporting is based on the agreement of the reporting unit (at its sole discretion and on a voluntary basis). According to Article 5 of the National Statistics Act, the reporting unit must be acquainted with the basic characteristics of the statistical survey, with a definite period of statistical processing, as well as with the right to decline cooperation for any reason and at any stage of the statistical survey.

The Banking Act regulates the conditions for the establishment, operation and regular termination of credit institutions established in the Republic of Slovenia and the competent authorities, measures and powers to

exercise supervision over their business. Article 125 of the Act defines that confidential information under this Act is all information, facts and circumstances regarding the individual client at the disposal of the bank. By the Act bank must protect the confidential information referred to in the Article 125, irrespective of the manner in which it was obtained. Article 126 defines cases in which the bank can share data with external users. It is precisely defined to whom the bank can provide information: The Bank of Slovenia, the European Central Bank, the supervisory authorities, the Commission for the Prevention of Corruption, the Commission of Inquiry (in the case of a parliamentary inquiry), the Court of Auditors of the Republic of Slovenia and members of the information exchange system customer ratings. Statistics Office is not defined as one of those users and this represents the biggest obstacle in obtaining the data, both from banks and from private operators owned by the banks.

3.2.2 Cooperation process for getting FTD

A good cooperation process for getting FTD is a crucial activity. It can be regarded as a process following different phases:

- Identifying a need for data
- Collecting overall infrastructure information on the financial market and its actors, both in the micro transaction processing stage, in the clearance stage and in the settlement stage of the transaction processes.
- Identifying the default potential data provider
- Setting clearly the cooperation agreement terms
- Maintaining a good cooperation relation with the potential data provider.

The National central Bank should probably be the first institution that should be addressed to receive a thorough insight into the infrastructure of FTD in a country, in the sense of

- which are, if any, the other governmental institutions involved in the financial market, either by monitoring or regulating the financial market, or playing other roles.
- which are the financial institutions/companies that are involved in financial transactions, both banks, payment card companies and other data providers.
- which are the organisations involved either in monitoring the financial market, being "spokespersons" for the actors in the financial market etc.

The National central Bank accesses FTD as a part of their auditing role and may or may not be able to provide data for the NSI (cf. Section 3.2.1 on legal requirements at national level).

The private institutions having a role in payment processing, clearance or settlement of the transactions (c.f. Section 3.1.1) are possible data providers. Getting bank-to-bank transfers or card data from the owner means contacting a lot of banks and credit card companies, and this approach is not suggested if there are other possibilities. Alternatives could therefore be processing companies and acquirers, as well as payment card issuing companies.

Sometimes there are several potential data providers, but only one of them is the "default" provider, given its more comprehensive coverage of the population, its infrastructure being more well-adapted to data delivery, or its role fits better with the data provider role than is the case for the other potential data providers. In that case, this default provider should be addressed first, since other potential providers could be negative to provide data when they are not themselves the default data provider. Since data delivery requires quite some resources, it should be based on a clear need of getting the data from *this* potential data provider.

A good cooperation process with a potential data provider is an advantage for the following several reasons:

- The potential data provider has extensive knowledge of its data and metadata, whereas the NSI might not exactly know what to ask for, at least for a first test data delivery. A successful outcome of a data delivery is depending on the data provider to use additional resources and together with the NSI derive what data are needed by the NSI and what metadata should be provided. Inevitably, this process is depending on a constructive and positive cooperative relation between the potential data provider and the NSI.
- Since FTD are generally considered sensitive due to the object (money) being transferred, it is important that the data provider/data owner is confident that the NSI will treat their data with extreme caution and can guarantee a confidential treatment. When data about individuals are collected, data must be treated according to the GDPR rules. Note that providers of card data have a sharp focus on that card data must be treated according to PCI security standards https://www.pcisecuritystandards.org/).
    - A cornerstone in the process of guaranteeing confidential treatment, is to explain the NSI's role to the data provider/data owner. In contrast to most other governmental institutions, the NSI has a sole statistical purpose for its use of the data and will not look at the data for any identified person/local unit/enterprise. The latter will however be a purpose for most of other governmental institutions as a part of their control/surveillance/monitoring activity. It is of importance to emphasise and convince data providers that their data will be treated confidentially, totally safe and for statistical purposes only.
- Although the legal basis for data acquisition is present, the data provider can choose to cause a lot of extra work for the NSI as well as delay through a legal dispute process which not only costs a lot of time and money for the NSI, but also could be negative for the NSI's reputation.

Thus, the first recommendation is formulating a clear need for data as well as an assurance that the data will be adequately secured, but also use of other means for improving the cooperation relation is recommended as the participation in investment costs in developing a reporting framework. Some resources are often required by the data provider to perform a successful data delivery, but the NSI in some countries cannot pay for getting data. However, the NSI can contribute to the cost reduction helping the data provider by letting one of their own employees work inside the data provider's office with the data preparation for a period of time. This is only possible if such an arrangement can be fitted into the confidentially rules of the potential data provider, and at least requires the NSI employee to sign a confidentiality declaration concerning his/her work inside the potential data provider's office, and perhaps also to be employed at the potential data provider for a limited time period, where the salary is paid by the NSI. Notice that the expenses for the NSI not only contributes to getting the data, but also to improve knowledge of metadata inside the NSI.

3.3 Metadata for FTD

3.3.1 Population and coverage

The FTD has as base unit the single financial transaction between the payer and the payee of money. The total of all possible FTD is the set of all base units where the transaction is done electronically. Our population of interest is however often the "all payments"-population, which consists of all possible FTD together with all payments made in cash, i.e. by coins or banknotes.

Notice that a single credit card transaction is a transaction between the card holder financed by a credit card company, representing the issuer, and the payee. Thus, since a bank account is not debited, and the banks therefore are not directly involved in the single transactions, credit card data typically must be received from other sources than when debit card data and other payment instruments for bank to bank transactions (giro) are used.

The coverage of data covering only payment instruments is over time susceptible to changes in the consumer behaviour. It is possible in some countries to receive bonus points for e.g. flights when using certain credit cards. Such incentives could lead to an increased proportion of credit cards usage. The increase of mobile payments may reduce volumes of transactions processed by banks and card companies and increase the volume of transactions processed by mobile payment operators.

3.3.2 Variables and units: relevance

Just as for an ordinary administrative register, FTD are collected for administrative purposes, not for statistical purposes. Below are some challenges concerning metadata, some of them are common to any administrative register whereas others are particular for FTD as well as for other big data resembling administrative registers.

- Neither the units nor the variables are necessarily adapted to the statistical purpose. As an example, for debit card data, the main administrative purpose is to secure that the correct amount of money is transferred between the correct accounts, accompanied by some explanatory details needed for the payer and the payee. In addition, the bank(s) needs the data on a single financial transaction to keep track of the remaining sum of money in the two accounts in question. These needs will influence the units and the variables, and hopefully the units and variables will be adapted to the statistical purpose as well, but as an example the system might not be concerned with whether the bank account is directly connected to a legal unit or a local unit. Somewhere deep in the database system, there will most likely be a link to a legal unit, but this information might not be easily available for the NSI.
- The infrastructure of the FTD data might not be immediately compatible with the statistical need of representing all the transactions (within a domain) as a dataset (table) containing all the relevant variables existing in the data provider's system. The main requirement on the infrastructure is to provide each account owner with a table of the transactions related to this account during a period of time, in addition to keeping track of the total amount of money in the account at any time. Furthermore, since FTD contains a huge amount of data, there could be technical/practical constraints on how data can be stored and otherwise handled by the banks and the data provider. Construction of a relevant data set for the NSI could thus be hard and time-consuming work for the data provider, and some variables could be more difficult to find than others.
- The variable-descriptions received by the NSI could be:
    - Totally or partially missing if
        - the data provider has only poor variable-descriptions, or the metadata system and thus the variable-descriptions are fragmented. For a data provider working only with a limited set of variables being well-known to them, the need for good variable descriptions might not have been acknowledged in the data provider organisation. Instead, everyone could just have the descriptions in their mind, and there is no perceived need of having the written documentation easily available. In such situations, the data provider will have to use much time for producing a list of variable-descriptions.
        - the NSI has taken the variable-description for granted without explaining the NSI's need for such a description when the delivery agreement has been made.
        - the NSI receives additional variables to those specified in the delivery agreement, which was not thought of in the delivery agreement, but which are regarded as

necessary additional variables by technical staff at the data provider's organisation, and that might – or might not - turn out to be useful for utilising the main variables.[13]

- Incomplete in the sense of using concepts that are imprecise for a statistical purpose but could very well be sufficient and even precise for the administrative purpose.

3.3.3 More about units

The single transaction can be regarded as a base unit. For a certain official statistics production process, the transaction might however not be the *statistical* unit. Instead, the statistical unit could be e.g. payer (in consumer expenditure statistics). Payer is a compound unit with respect to the base unit above, i.e. an aggregation of base units. Using a compound unit, leads to a coarser granularity. The typical unit in the data set received by the NSI will typically be a compound unit. As an example, the unit could be the set of retailers in a region receiving payment(s) related to a certain industry classification code during one day. Then, the transaction amount received by this compound unit is the sum of all transactions received within this region and industry classification code for this day.

In the retail sales example above, the focus for defining the compound unit is the businesses receiving payment. On the other hand, with the consumer perspective, e.g. the consumer expenditure survey, a compound unit based on the same industry classification code and day, could be defined as the set of persons in this region who has spent money this day related to the industry classification code in question. Then, the corresponding transaction amount sent by this compound unit (to different retailers inside or outside this region) is the sum of all transactions sent from persons within this region and industry classification code during this day.

3.4 Promising official statistics based on FTD

Based on FTD, there could be potential for improving current official statistics or for developing new official statistics.

An improvement of current official statistics can be done in three different ways:

- Replace the current source: This option has the advantage of reducing the response burden connected with the current data collection that usually is based on industries or persons filling in questionnaires.
  - An example is the retail sales index. Today, shops need to report their total monthly sales to the NSI. FTD aggregated to shop, could potentially replace the current statistics once both credit and debit card data are included in FTD, and in addition the cash payment amount is reduced. In Norway, the cash payment rate has already decreased to about 20%, and it will decrease further in the future and eventually disappear someday. In the future, also giro payments need to be added to the FTD, since internet payment is increasing (web shops where one can order and pay before picking up in the physical shop or receiving by mail delivery).
- Improve estimation based on the current data collection:
  - Still using the retail sales statistics example, FTD will give totals for the whole population of shops, thus the survey-based estimation can be adjusted/calibrated to the totals known through the FTD. If the debit sales total is known in both the sample and the population, then we could use this in a rate estimator, as an example. Such an adjustment can reduce the sample variance as well as non-response and measurement error.
- Quality assessment of the current official statistics:
  - In the retail sales statistics example, there could be reasons not to change the estimator, e.g. due to uncertainty about the FTD coverage or other considerations. Thus, in this setting we

[13] Obvoiusly, these additional variables would not have been received by the NSI unless the data provider has classified it as non-sensitive.

regard the FTD source as having insufficient quality for being included in the official statistics estimator. However, FTD could still be used to assess the quality of the current official statistics. One way of doing this is to produce an alternative estimator as in the bullet point above, and although the alternative estimator is considered worse than the traditional estimator, it can still tell something about the quality of the current estimator.

Before accessing FTD and testing their potential for improving a certain official statistics production process closer, it is difficult to indicate which official statistic product could benefit the most from the use of FTD and, consequently, on which product is most promising to invest. Thus, it is possible to make such an assessment considering only the information coming from the available metadata at this stage. Every WPG participating country is supported in doing this assessment by the results of an internal survey conducted within the NSIs. The results of these surveys are synthetised in national case studies reported in the appendices. According to these results, promising official statistics that could benefit from the use of FTD are:

- *household expenditure of residents for tourism and balance of payments estimates* – new data can be used for comparison with official statistics in order to improve the quality of current statistics, as indicated by Italy;
- *e-commerce turnover* – private credit card operator data can be compared to the current e-commerce turnover index (traditionally calculated with surveys) and can help in getting some insights even by *merchant category code* (MCC) groups, as reported by Italy;
- *retail trade index* – FTD are a potential alternative source to be used either in the estimation of the index or in the quality assessment of the current statistics, either in form of card use data as reported by Norway, or by using the fiscal cash register as reported by Slovenia accuracy;
- *early estimates of macroeconomic aggregates and indicators* – FTD aggregated (monthly or daily) series can be used for nowcasting/forecasting purposes. The availability of private credit card operators FTD will allow refining the forecasting, for instance, of private consumption and retail trade index at the national level or for the early estimates of turnover in Services sector, as proposed by Italy;
- *business statistics indicators* – new indicators on monetary flows among companies can describe the economic relations between industries, as indicated by Norway;
- *turnover of Collaborative economy or Sharing economy* - Portugal indicated that the new FTD sources can help in the production of some indicators measuring these new economic activities dimensions.

4. Overview of FTD situation in the WPG countries

The FTD situation varies between countries. Below is a short overview of the situation in the different WPG countries. More details for each of the countries are presented in Appendix 2 to Appendix 7.

4.1 Bulgaria

Types of FTD

Bulgarian National Bank publishes monetary survey on the monthly base. Monetary survey provides data about foreign assets, money in circulation, deposits and monetary aggregates M1, M2 and M3.

In addition to monetary survey, BNB publishes data about POS payments and online payments. Data are aggregates and do not make difference between legal persons payments and natural persons payments.

Also, BNB publishes more detailed data about deposits of households and Non for-Profit Institution Servicing Households and deposits of non-financial enterprises grouped by amount of deposits. BNB publishes data about credits by amount and industry as well.

Process for getting FTD

Bulgarian National bank, being administrator of settlement system, is a potential provider of granular data on consumers to business payments and business to business payments. Another potential provider is the private company Borica JSC. Borica is a technology company that provides payment services. The company developed and maintained payment platform which is used by its clients.

Commercial banks are another potential data provider. Top five banks hold about 60 percent of overall bank assets in Bulgaria. It is reasonable to focus on the biggest banks when negotiate data access.

Metadata for FTD

Bulgarian National Bank publishes monetary survey each month. Data about deposits grouped by number and value and credits grouped by value and economic activity are published quarterly.

Also, Bulgarian national bank publishes number and value of POS payments and online payments annually. These are aggregated data about consumer to business and business to business data. Consolidation of B2B and C2B payments is a shortage of the data for we cannot separate final demand related payments from intermediate consumption related payments.

Granular data access is subject to negotiation.

Promising statistics

FTD are much more valuable when combined with national account data and business statistic data. For example, mortgage loans when combined with data about houses built and building permission are much more informative about housing market than considered alone. Consumption loans are more informative when combined with private consumption data and income data than considered separately.

The examples suggest FTD could be very useful when combined with existing data. FTD refer to the money flow within the economy. Business statistics and national accounts refers to the flow of goods and services within the economy. Having data about both good/service flows and money flows could enrich economic statistics.

4.2 Germany

Types of FTD

The Deutsche Bundesbank publishes data about the number and volume of card transactions and payments to POS terminals. The available data are however only at the aggregate level and do not inform about the nature of purchase done per transaction or payment.

Process for getting FTD

The Deutsche Bundesbank is part of the TARGET2 system: About half of the payments in terms of volume and over one-third in terms of value are submitted via the Deutsche Bundesbank through the Target2 payment system, with the German system called TARGET2-Bbk.

Metadata for FTD

In addition to the data passing through the TARGET2 payment system, the Deutsche Bundesbank currently receives monthly data on credit and debit card turnover in relation to travel (balance of payments) including payments cards transactions of German residents travelling abroad as well as payments cards transactions of foreign travellers in Germany. However, more detailed information, as for example the MCC, is not known. The current data therefore does not allow for empirical analyses on estimating shared economy transactions.

Promising statistics

Given the current data available at the NSI, it is not possible for now to identify potential official statistics that would benefit from FTD.

4.3 Italy

Types of FTD

The Bank of Italy directly manages the information on the interbank retail payment systems, through BI-COMP clearance and TARGET2 settlement systems. The EBA-clearance system operated by SIA and NEXI (the main card processors in Italy) is in addition to the national central bank system. The private payment data processing entities manage the data on behalf of banks.

Process for getting FTD

The Italian Institute of Statistics has undertaken a collaboration agreement with the Bank of Italy for the exchange of aggregated statistical information. SIA and NEXI were contacted by the Bank of Italy for an experimental project for sharing aggregated daily information. Data collection from card processors (*SIA and NEXI)* requires an intense experimental phase before proceeding with their exploitation for statistical production purposes, because problems related to either the harmonisation of different IT platforms or the set-up of data lakes arise.

Metadata for FTD

FTD managed by Bank of Italy are timely, with a good broken down by payment instruments but not granular in terms of sectoral and geographical breakdowns, because they come from predefined interbank procedures at the settlement level. We could have access to private operators' daily data with information on the location of the merchant and the issuer, both from the acquiring side and the issuing side (useful for

measuring the cross-border flows). Other information allows the distinction between e-commerce and proximity payments (useful to measure the retail trade turnover) and the Merchant Category Code (MCC).

Promising statistics

Promising statistics are considered: 1) *household expenditure of residents for tourism and balance of payments estimates* for comparison with official statistics; 2) *Nowcasting/forecasting of macroeconomic aggregates* - BI-COMP and TARGET2 and private credit card operators series used for refining the forecasting of private consumption, retail trade index, or service sector turnover; 3) *e-commerce turnover by MCCs groups* - private credit card operators used to make comparison with the current e-commerce turnover index and to get some insights by MCCs groups.

4.4 Norway

Types of FTD

- In Norway, there is a combination of a net and a gross clearance and settlement system. The net system for each period (of some hours during a day) contains the aggregated net transactions of transactions between bank A and bank B
- Norway has, in the same way as countries like Denmark and Canada, a national debit card scheme that covers most of the debit card transactions.
- The Register of Cross border Currency Transactions and Currency Exchange contains all transactions crossing the national border.
- The DSOP Control system is a system for cooperation between some governmental organisations and the private financial institutions, making the current governmental audit requests more automatic. The system will start during 2020 and is based on an API interface.

Process for getting FTD

- Clearance and settlement data are not a relevant source for Statistics Norway due to the net clearance system.
- Norway has received micro-transactions of national debit card scheme data for one day in 2016.
- The DSOP Control system are being addressed by Statistics Norway for cooperation on future data delivery to Norway. Some keywords are: getting and discussing metadata, preparing for the API solution, and getting synthetic test data.

Metadata for FTD

- The national debit card scheme data covers more than 75 percent of the debit card payments on point of sale terminals.
- The Register of Cross border Currency Transactions and Currency Exchange: for smaller transactions, the receiver is aggregated to country and month.
- DSOP Control will cover all debit card- and giro-transactions. Only samples, e.g. random sample, of persons and businesses will be available for Statistics Norway due to the server capacity limitations within the DSOP Control system.

Promising statistics

- Retail sales index

- Economic relations between businesses by industry classification code.

4.5 Portugal

Types of FTD

In Portugal FTD can be accessed in different ways, each of them covering slightly different domains. The universe of Financial transactions data includes electronic and non-electronic transactions. Among electronic transactions we may distinguish from the following main types of FTD: MBWAY and MULTIBANCO (electronic payment and money transfer processes managed by SIBS, a payment services provider); Electronic transactions managed by Credit Card companies); peer–to-peer electronic transfers between banks; and data from the Tax Authority.

Data owners and data providers

The data owners consist of the private companies, and banks, and the Bank of Portugal. The access to data from private companies and banks is difficult since these companies are not obligated by law to report microdata to Statistics Portugal. The Bank of Portugal collects this information for the transactions data in aggregated way, and SIBS (who manages MBWAY and MULTIBNCO), manages almost all electronic transactions (over 60% or all transactions are made electronically and almost all are monitored by SIBS). The Tax Authority is an important data provider, as it collects the data for all the transactions in Portugal.

Process for getting FTD

The different transactions, B2B, B2C through bank cards, direct debits and transfers, payments, withdrawals and purchases are all monitored by the Bank of Portugal and the Tax Authority. After an agreement between Statistics Portugal and The Tax Authority, this will be the data provider for the project.

Promising statistics

Either as an additional source for improving the quality of statistics, or as a source that can replace the existing source we aim at focusing on turnover in specific sectors of the Portuguese sharing Economy, such as accommodation and transportation.

4.6 Slovenia

Types of FTD

- Central bank of Slovenia collects quarterly aggregated data on payment cards, cash withdrawals and deposits, electronic banking, credit transfers and direct debits.
- BANKART, private company owned by Slovenian banks, ensures processing of transactions with various banking payment instruments. As a result, they own data on ATM operations, card payments, payment systems (SEPA, SIMP-PS and E-invoice) and on POS terminals of majority of Slovenian banks.
- With fiscal cash register Financial Administration of the Republic of Slovenia monitors all issued invoices, which are paid in cash, credit and debit cards, and do not exceed value of EUR 5,000 for consumers (B2C) and EUR 420 for businesses (B2B).

Process for getting FTD

As Bank of Slovenia collects only aggregated quarterly data, this data is not relevant for Statistical office of Slovenia. We were last year in talks with both Financial Administration of Slovenia and BANKART regarding

obtaining the data on financial transactions, but have realised that based on legislation we could only obtain data from Financial Administration of Slovenia. We have then started negotiations with them in the beginning of 2020 and have talked about getting test data set on fiscal cash registers, which would be anonymised (tax numbers would be hidden). We would then aggregate this data and compare them with aggregated data on retail trade index. If the data would be useful to estimate the retail trade index, we would then include this sets of data in our Annual Programme of Statistical Surveys. Unfortunately the covid-19 crisis happen and we couldn't get the data so far, but are planning on doing so. The only thing that we could have done is include this sets of data in our Annual Programme of Statistical Surveys, which is the legislative basis for gaining such data.

Metadata

Obtaining data on fiscal cash registers would give us access to individual cash or card payments in terms of tax number and location of the issuer, time and date of issue of invoice and the value of the invoice, broken down by tax rates. Missing is the information on the payer, unless in case of B2B sale.

Promising statistics

The most promising statistics are statistics on retail trade. Currently we calculate indices on retail trade based on data from survey and administrative data on value added tax, which becomes available to us 45 days after reference period. As the data on cash registers would be available 10 days after reference period at the latest, we could get more accurate results at time of first publishing of provisional data, which is 30 days after reference period.

PART II -- Applications

Part II.1 – Case studies

5. Case studies

5.1. Purpose of the case studies

The purpose of the case studies is to illustrate how financial transactions data can contribute into official statistics, either as a source for new statistics, replacing current sources, adjusting the current estimation, or at least contribute to quality evaluation of the current statistics.

A new source such as financial transactions data might have an information power that enables the production of new statistical production that was earlier not possible. The Italian case study is such an example in the field of forecasting models of macroeconomic variables and will be presented in Section 6.

When the aim is not to produce new statistics, there exist current official statistics that are often based on a survey. All surveys generate response burden for those selected to participate in the survey. Many businesses perceive the surveys as a heavy burden, and they experience that they use a lot of time on answering surveys from the NSI. Financial transactions data (FTD) are not based on anyone reporting answers but are instead automatically collected. If FTD could replace a survey, this would be an improvement for the response burden and perhaps also for the NSI's production costs. The Norwegian case study, cf. Section 7, investigates whether such an approach is possible for the retail sales statistics.

Sometimes, the FTD at hand are not sufficiently complete in coverage or otherwise not of sufficient quality to replace the current source. Then there are still two possibilities. Firstly, the FTD can be used as auxiliary information for adjusting the current estimation method. As an example, consider statistics based on the variable of interest Y, say total sales. The typical estimator is to calculate the sum of Y in the sample and then weight up to an estimate for the population. The idea of the *rate estimator* is to base the weight on an auxiliary variable correlated with the total sales, e.g. number of employed taken from a register. The rate of the total of employed X for the population and x for the sample, is used as weight. If FTD cover the whole population, as is the case both for the invoice data of Portugal and for debit cards data of Norway (c.f. Section 8 and Section 7), the FTD-based total sales can in our example be used instead of the number of employed. This will be an improvement if the FTD-based total sales is more correlated to total sales than employment is correlated to total sales.

The second of the possibilities mentioned above can be rather similar to the use of auxiliary information described above, but if for some reason it is concluded that the adjustment will not improve the statistics, the FTD still is an alternative source that may say something about the quality of the current statistics using a modelling approach. An example of the opposite approach, i.e. of quality evaluating a statistical register based on a survey, is given in Fosen & Zhang (2011).

5.2 Integration process

Before using the FTD for the statistical production, integration of different sources is usually necessary. The FTD source itself often lacks some relevant variables that can be obtained from registers within the NSI provided that the FTD source can be linked to the registers. In both the Portuguese and the Norwegian case study, such linkage is done.

To compare the FTD source with the current source in order to assess its potential to replace the current source, it is usually desirable to link the two sources at the micro-level. This is often not straightforward, especially if there is no unique ID that can be used as linkage variable and some kind of record linkage is required instead. Even in the presence of the ID for a linkage variable, this might only allow linkage at an

undesired level, c.f. the Norwegian case in Section 7 where linkage is only possible at the legal unit level, whereas the correct linkage level would be the local unit level.

5.3 Quality framework

When assessing the potential of an FTD source for official statistics, it is useful to consider two phases in the data process. Phase 1 is the processing of the FTD data and other data one by one before they are integrated, whereas phase 2 is the rest of the process until a statistical dataset (statistical register) has been made and the statistics can be produced by simple or weighted enumeration based on the statistical dataset. Our setting has many similarities to the two-phase framework in Zhang (2012, Figure 1), and we will use some of its components. Below we will look at the distinction between representation line and measurement line at the two phases (single-source data and integrated source data) before we consider which of the two lines contains the main error for each of the two phases.

One interesting feature of the two-phase framework above is what is the object/unit and at which level. At phase 1, the FTD source has as unit (or object to distinguish from the unit of phase 2) the single transaction. For Italy, this phase 1 is conducted by the Bank of Italy's team, who pre-treated and aggregated the data to send to ISTAT where phase 2 is carried out.

At phase 2, the unit depends on the statistics. As an example, if the aim is to calculate the total sales as in the Norwegian case study, the unit is the local business unit.

The measurement variable usually is the purchase amount or the transaction amount in both phases.

For Italy, Norway and most other countries, the main type of error at phase 1 is not erroneous registration of transaction/purchase amount, but missing units (objects) since e.g. cash purchase is not covered. Thus, the main error is along the representation line.

At phase 2, the main error depends on the statistics. In the same example as above of calculating the total sales, the unit is now a local business unit and all these units might very well be covered, as the situation is in Norway where the debit card data covers all local business units where at least one purchase has been made by debit cards. Thus, there is probably little error along the representation line. On the other hand, the measurement now is total sales for the local unit, and then there is a systematic under-measurement due to the lack of cash purchases. An exception is Portugal.

6. The Italian case study: Italian macroeconomic nowcasts and forecasts

6.1 Purpose

FTD have shown to be quite helpful in macroeconomics analysis due to the wide coverage and the level of detail they provide and, particularly, their timeliness (near real-time basis). Nonetheless, key challenges remain in their use for official statistics macroeconomic indicators concerning their interpretation as well as the traditional aspects of the collection, validation, storage and privacy issues.

As Istat currently produces short-term indicators for economic analysis and disseminates (or develops only for internal use) different models for the study of short-term trends, the Institute is very interested in testing the predictive ability on new series of FTD macroeconomic indicators.

This case study uses the FTD aggregated time series of the Bank of Italy, which has been used in experimental studies for some time, and suitable transformations of new FTD coming from a private operator. The aim is to forecast private consumption, the monthly retail trade index and e-commerce sales. These exercises were carried out before and during the period of the COVID-19 emergency in Italy (February 2020 - May 2020), a period on which these new series contribute to shed some lights on the size of the economic crisis caused by the pandemic.

6.2 The current nowcasting and forecasting activities in official statistics

Besides the flash estimates relating to GDP and other macroeconomic aggregates being traditionally conducted as part of the national accounts estimates, ISTAT has since January 2011 been assigned the responsibility of macroeconomic forecasting activities. This represented an important shift in Istat's functions and has fostered a relevant change in economic research activities.

In national accounts, the early estimate of GDP is officially released at T+30 days (it was at T+45 days until 2017), and it is compiled using the same sources and methods adopted for the full compilation of quarterly national accounts (released at t+60) and also following the same quality criteria. Estimates of GDP are built from numerous sources of information, including business surveys, household and other social surveys, as well as administrative information. Nevertheless, the availability of these sources is limited especially for the third month of each quarter and also for most industries of the services sector. In particular, no indicator related to services is produced sufficiently timely for being a part of the rapid estimates, except for the industrial production index and the matrix of accounts for financial intermediation services. Therefore, the availability of new timely economic series is particularly relevant, and for this reason, there are streams of studies considering new data, like those provided by FTD and the initial exploitation of electronic invoices data.

To implement the macroeconomic forecasting activities, Istat created the Econometric Studies and Economic Forecasting Division that has the responsibilities of carrying out the modelling and research activities, including macroeconomic short-term and medium-term forecasts, as well as micro-simulation models of the effects of fiscal policies on households, businesses and institutions.

The new macroeconometric model MeMo-It of the Italian economy was constructed, and the estimates of the model are disseminated twice per year (in May and November). This is an annual model for the Italian economy that requires two sets of external (exogenous) information over the forecast period. First, consistent assumptions are needed about the developments in the international scenario (such as trade growth, exchange rates, ECB interest rates, and the oil price). The second external information is an annual estimate of key GDP components obtained from short-term models based on monthly and quarterly data available at the time of the forecast.

Moreover, micro-level information is used for microsimulation models for households and firms that could be used to generate additional results to integrate the macroeconomic dataset and enhance model performance.

The economic modelling in NSIs can benefit from the availability of the wide range of raw and validated statistical data as well as from in-house expertise on data management, integration and quality. However, this activity is always seeking for new sources able to improve the performance especially in the context of the short-term models for the study of the GDP components. This type of activity is exactly the one that we will experiment with the use of the new FTD series in Section 6.4.1

6.3 FTD source

A joint Bank of Italy-Istat working group was set up in late 2018 to produce FTD indicators and test their adoption in nowcasting or forecasting macroeconomic aggregates in addition to traditional series. It resulted in the production of monthly and daily time series extracted from two inter-exchange (clearance system) and settlement systems: BI-COMP and Trans-European Automated Real-Time Gross Settlement Express Transfer System (TARGET2) retail. BI-COMP is the Bank of Italy multilateral clearing system created to guarantee the settlement in central bank money of euro-denominated retail interbank payments.

Although these high-frequency series are timely available, such data do not allow having information on the commercial breakdown because they come from predefined interbank procedures at the settlement level. Thus, the Bank of Italy, using FTD microdata coming from both BI-COMP card data and data from a major private card operator, after a long phase of data munging to cleaning it up, derived a more granular set of monthly and daily time series. The new series are based on a grouping into ten Merchant category code groups (clothing, hotels and restaurants, home, work, retail, services, telecom, web and direct marketing, travels and transport, cash advance, not defined) the e-commerce transactions being collected from the acquirer side and containing the expenditure by Italians when purchasing with debit cards from Italian firms.

In the case study, we could apply the following four types of FTD time series:

- monthly time series of BI-COMP and TATGET2 total transaction and by payment instruments (e.g., credit transfer, direct debit, payments cards - both ATM and POS -, cheques, other); series are available since 2000 (both in volume and amount);
- monthly time series of BI-COMP e-commerce transaction in terms of indices (2014=100), available since 2014;
- daily time series of BI-COMP/T2 total transaction and by payment instruments (POS, ATM), available since 2006 (both in volume and amount);
- daily time series of BI-COMP domestic debit card transaction (considering only the proximity channel) by ten MCC groups in terms of indices (2014=100), available since 2014.

6.4 Improving the current official statistics by FTD

We exploit the informative power and the timeliness of new series to improve the accuracy of the nowcasting and forecasting of some Istat models used in the national accounts (NA) framework or for economic research purposes. Models perform a one-month-ahead forecast of the Istat monthly retail trade index and the e-commerce sales index in the first trial. We model the behaviour of the quarterly NA final consumption expenditure with a MIDAS Model (used in Istat for internal forecasts) in the second trial (Ghysels et al. 2007).

6.4.1. Empirical analysis

TRIAL 1 - Monthly Retail trade index and e-commerce sales index

In the context of quarterly national accounts flash estimation**,** retail trade indicators represent a crucial variable for estimating trade margins. The retail trade index is available monthly, but its release has a certain delay. Therefore, it requires to be supplemented by some prediction for the last month. We aim at verifying the size of one-step-ahead forecast errors resulting from modelling monthly retail trade index by two sets of predictors: one provided by traditional business surveys for firms operating in trade, and the other and more innovative predictor is based on FTD series.

Retail trade is tracked by Istat through a monthly 2015=100 index detailed by type of product, employment size and type of sales. The e-commerce index is a sub-component of total retail trade. Total retail trade is available over the period January 2000-March 2020, whereas the e-commerce index is available over a shorter period (January 2015-March 2020).

Business surveys make available several statistics, monthly climate indices, frequencies and balances, adding up to 88 series over the period January 2000-March 2020. However, after a preliminary analysis, our focus in the case study is limited to climate indices, which seems to be the only potentially significant predictor among the 88 series.

In this trial, concerning new FTD series, we used the BI-COMP/TARGET2 set of monthly series on main payment instruments (seven in terms of volume of transaction and other seven in terms of their value in euro), and the BI-COMP/TARGET2 retail components consisting in the monthly POS (BI-COMP) series in euro by ten MCC groups.

Concerning pre-treatment of payments data: For the BI-COMP/TARGET2 data by type of payment instrument, all the outliers (namely, those observations standing clearly out from the rest of the observation due to various recording reasons) from original data had to be removed. Figure 6.1 shows these sets before (red line) and after (blue line) the outlier removal. FTD series by retail components have been backcasted to 2000 to make them homogeneous/comparable to the rest of the predictors. Finally, we computed other two couples of synthetic series resulting from the extraction of the first two factors by principal component analysis applied respectively to the set of BI-COMP/TARGET2 by type of payment instrument and the by retail component[14].

[14] Principal component analysis in forecasting is a method of linear transformation for summarizing many predictors in a small number of factors that account for much of the observed variation of the entire set of staring time series. Therefore, the forecasting with many predictors can be simplified by using a few estimated factors instead of all the available predictors. In this application, the extracted factors are used as predictors in dynamic linear regressions. For an overview see Stock and Watson (2006), and the references there available.

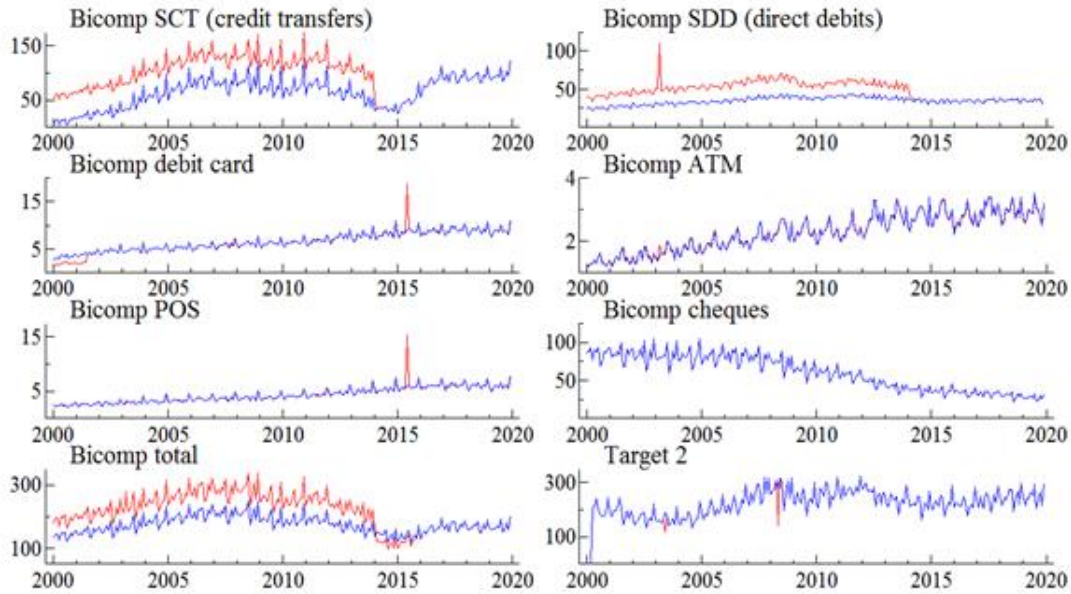

FIGURE 6.1. BI-COMP AND TARGET2 SERIES BEFORE (RED LINE) AND AFTER (BLUE LINE) OUTLIER CORRECTION

The seasonally unadjusted indicators of the two retail trade indicators as levels are shown by red lines in the two left panels of Figure 6.2, the total on the top and the e-commerce component on the bottom. The two right-hand panels show in red the corresponding seasonally unadjusted annual growth rates and in blue the synthetic index of payments by retail component relative to the first-factor component, as annual growth rates.

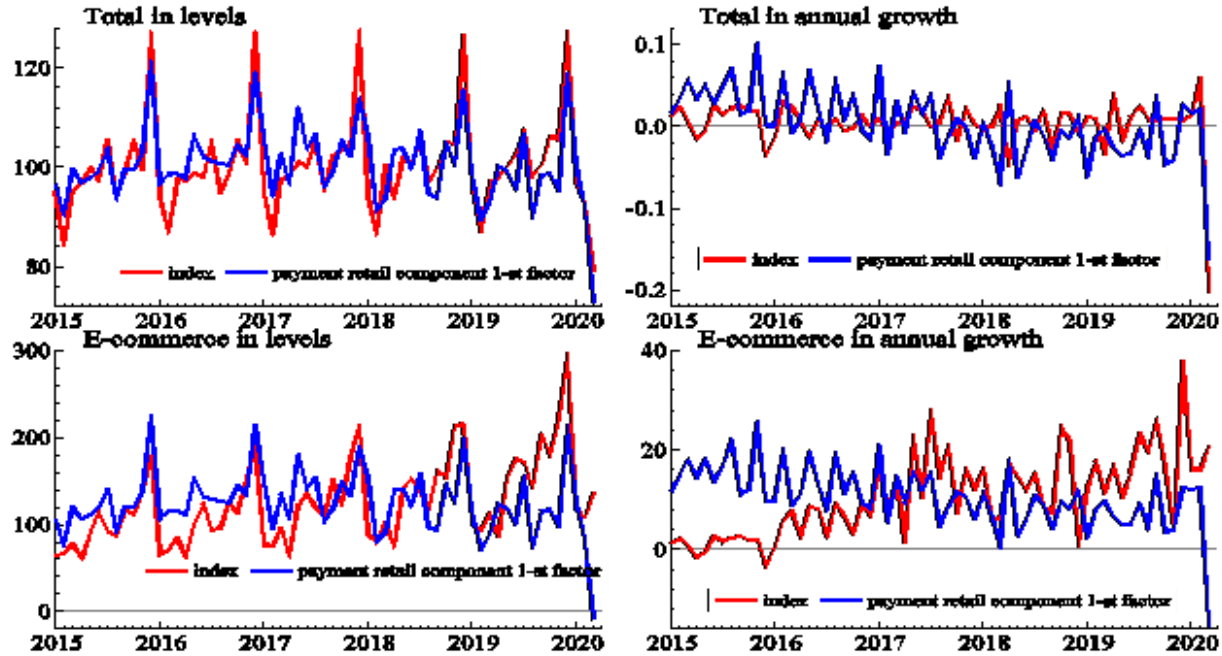

FIGURE 6.2. RETAIL TRADE DATA AND SYNTHETIC INDEX OF PAYMENTS BY RETAIL COMPONENT

The graphic analysis of Figure 6.2 shows that:

1. the first factor of payments by retail component presents higher volatility than retail trade indices, which emerge in particular over data in differences. The same evidence is emphasised for single components of payments data (not shown here for brevity);
2. the first factor of payments by retail component fits generally better to the total retail trade than to the e-commerce component. This could be due to differences in the e-commerce sale boundary definition. For the construction of the Istat e-commerce index, only the pure e-commerce activities are considered and only referring to the sales of goods and not services; moreover, Italian firms can sell not only to Italians but also to foreigners (even on a small scale). In our FTD series, we consider all the Italian firms selling only to the Italians both goods and services through a non- pure e-commerce channel;
3. For both retail trade variables, the fit of the same payment factor worsens when moving from the comparison of data in levels (left-hand-side panels) to the panels on annual growth, especially for the e-commerce index. This could be due to the dominance of seasonal factors in both series under comparison, which is deleted out from the transformation, but it could also be due to a higher order of the trend in the e-commerce component.

Concerning the model strategy, after difficulties encountered within the multivariate *Seemingly Unrelated Time Series Equation* (SUTSE) system[15], we went for the univariate class of autoregressive distributed lag (ADL) models[16]. This strategy has been followed both for variable selection and for the setting of the final equation from which the forecasts are computed. We adopted the general-to-specific (GETS) algorithm, developed into the *Autometrics*, which is an algorithm for automatic model selection within the general-to-specific framework (Doornik & Hendry 2013). The original dataset was not subject to seasonal adjustment in order to mimic as much as possible the strategy applied in the statistical production process for which seasonal adjustment operates in a second step only. ADL models were fitted to data within the period January 2000-December 2017, whereas one-step-ahead forecasts were computed over 27 months, from January 2018 to March 2020.

The strategy followed for the definition of the (general) unrestricted ADL model has been to start from models where the three sets of predictors (e.g. Business survey data; FTD by type of payment instrument and by MCC groups) were all represented. In this way, we can measure the relative contribution of each set of data to the forecasting performance. For FTD series, we conducted several experiments to understand if they favour either the use of single components (alone, in couple, sets of three, altogether) or of the extracted factors.

Other settings of the unrestricted ADL model concerned the choice of the lag order of the predictors (limited to 0 and 1 only), the inclusion of trading days effects (one regressor for the number of working days, Easter and leap year effect), seasonal dummies, constant and linear trend. The presence of the lagged dependent variable among predictors has been also limited to 1. Furthermore, both models of data in levels and annual growth are tested. Notice that in this latter case, data of business surveys were considered in levels, whereas FTD were also differenced. In all the experiments, estimation is carried out by ordinary least squares (OLS).

Table 6.1 presents the results of estimation for the model identified by the GETS Autometrics search for total retail trade, while Figure 6.3(a) the fitted values versus the true ones. The specification is for data on annual growth rates and turns out to be particularly parsimonious. Among regressors, only the first factor referring to the Payments by type of instrument, lagged by one period, resulted statistically significant. The sign of this regressor is negative, as expected for one-period lagged regressors. No regressor of the payment broken by

[15] See Harvey (1989).
[16] See Hendry (1979) and Hendry et al. (1984).

MCC set resulted in being significant, taken neither individually nor through their synthetic indices. Trading day effects are not significant, but we did not remove them from the final specification because they are useful for increasing the forecasting performance.

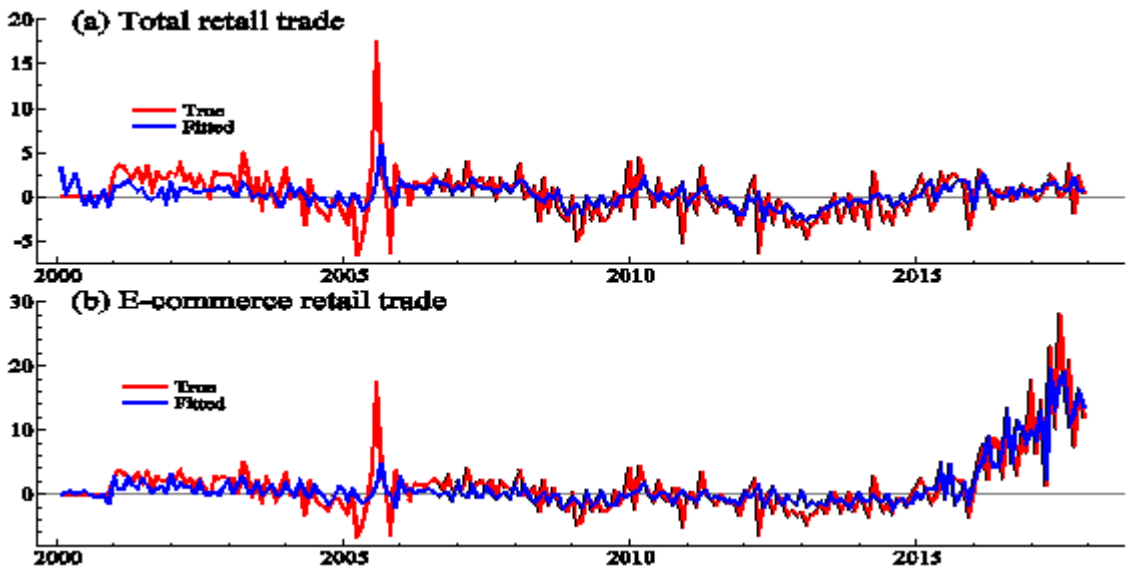

FIGURE 6.3 A AND B. FITTED VERSUS TRUE VALUES FOR SPECIFIED ADL MODELS

Even if the fit of the model to retail trade data appears particularly effective, three negative elements emerge:

- the lack of FTD by MCC groups among the predictors,
- the small impact of predictors measured by relatively low values of the regression coefficient (note that both dependent and independent variables are of similar magnitude),
- the absence among the regressors of FTD payment by type of instrument 1-st factor at lag 0 and the negative sign for the same regressor at lag 1; a possible explanation of this drawback is the high volatility of payments data, which may distort the relationship with the dependent variable.

TABLE 6.1. ADL MODEL FOR TOTAL RETAIL TRADE IN ANNUAL GROWTH RATES. OLS ESTIMATION RESULTS

Predictor	Lag	Coefficient	t-student
Total retail trade	1	0.2618	3.93
BS climate index	0	0.0487	3.92
FTD Payments by type of instrument 1-st factor	1	-0.0460	-3.84
Easter effect	0	1.0157	1.74
Leap year	0	1.4961	1.27
Trading day	0	0.0984	1.78

Note: Estimation period February 2001-December 2017

Forecast errors in mean absolute (MAE) and mean squared (RMSE) terms are in Table 6.2, where they are computed over the annual growth rates of original data. The values of the MAE around one, for months of 2018-19, are similar to values calculated in analogous experiments recently conducted using other monthly

Italian macroeconomic indicators (industrial production, labour force etc.). The gains in terms of reduction of the relative statistics once payments data are removed from the specification of Table 6.1, are shown in parenthesis. The gain is of 12.1% in terms of MAE and 3% in terms of RMSE over the period 2018(1)-2020(1).

TABLE 6.2. ONE–STEP-AHEAD FORECASTING PERFORMANCE OF ADL MODEL FOR TOTAL RETAIL TRADE OVER THE PERIOD JANUARY 2018-MARCH 2020

Test period	Mean absolute error - MAE	Root mean squared error - RMSE
2018(1). 2020(1)	1.1813 (12.1%)	1.6529 (3.0%)
2018(1). 2020(2)	1.3031 (11.7%)	1.8315 (4.5%)
2018(1). 2020(3)	1.9582 (7.9%)	4.0727 (1.2%)

Note: in parenthesis gain for specification without payment data

Not surprisingly, the forecasting performance of ADL model of Table 6.1 worsens when the test period extends to February and March 2020, the beginning of the Covid-19 Crisis period in Italy, as provided in Table 6.2. Notice, in particular, the explosion of RMSE to 4.0727 when the test period spans 27 months until March 2020, the month of the lockdown of the economic activities. However, a positive result of this exercise is that the presence of payment indicators guarantees gains in the forecasting performance of the model.

Results of the ADL model fitted over the *monthly e-commerce sales index* are in Table 6.3, whereas fitted versus true values are shown in Figure 6.3(b). The fit concerns annual growth rates over the period February 2001-December 2017. As on total retail trade, Table 6.3 presents the best performing model obtained by the *GETS Autometrics* procedure.

Main pieces of evidence are as follows:

1. Business survey data are not significant and therefore disappear from the final specification;
2. FTD by MCC groups are here widely represented, they appear by single components instead of their synthetic factors; this is the case of FTD for Hotels and restaurants, and in the same way for Home, Service, Telecommunication, Transports.
3. The sign of payment indicators is counter-intuitive for payment data relative to hotels and restaurant and telecommunication (lag 0). This is clearly because of the strong volatility encountered in this set of data;
4. The presence of significant deterministic effects among selected regressors, including a linear trend, which explains the accelerating behaviour of e-commerce with respect to payments. Notice also the presence of some significant seasonal components, able to explain residual seasonality and trading effects.

TABLE 6.3. ADL MODEL FOR THE E-COMMERCE SALES INDEX IN ANNUAL GROWTH RATES. OLS ESTIMATION RESULTS

Predictor	Lag	Coefficient	t-Student
e-commerce sales index	1	0.2471	4.48
FTD for Hotels and restaurants	*0*	*-0.1600*	*-2.30*
FTD for Hotels and restaurants	*1*	*0.6636*	*8.38*
FDT for Home	1	*-0.3435*	*-5.47*
FTD for Service	*0*	*0.5201*	*5.57*
FTD for Telecommunication	0	*-0.3070*	*-4.41*
Telecommunication	1	*-0.1128*	*-2.41*
FTD for Transports	*1*	*-0.1780*	*-2.98*
Trend	0	-0.0074	-2.69
Seasonal dummy	0	1.6089	2.23
Seasonal dummy	6	1.2383	1.77
Easter	0	1.8724	2.49
Trading day	0	0.1439	2.03

Note: Estimation period February 2001-December 2017

Moving now to the one-step-ahead forecast performance in Table 6.4, we see that for all sample periods considered in the trial, both MAE and RMSE are higher than the case of total retail trade shown in Table 6.2. This occurs because this experiment runs over a specific subcomponent of total retail trade considered in the former exercise, which has an explosive evolution in recent periods. The gain due to the inclusion of FTD by MCC groups is remarkable, larger than 50% in almost all cases.

TABLE 6.4. ONE–STEP-AHEAD FORECASTING PERFORMANCE OF ADL MODEL FOR THE E-COMMERCE SALES INDEX OVER THE PERIOD JANUARY 2018- MARCH 2020

Test period	Mean absolute error - MAE	Root mean squared error - RMSE
2018(1). 2020(1)	6.8772 (66.9%)	8.8650 (57.2%)
2018(1). 2020(2)	6.6367 (73.6%)	8.6937 (59.8%)
2018(1). 2020(3)	7.6178 (54.2%)	10.6501 (31.9%)

Note: In parenthesis gain for specification without payment data

TRIAL 2 - Quarterly final consumption expenditure

In this second trial, the considered target variable is the National Account (NA) aggregate referring to the quarterly final consumption expenditure (chain-linked 2015=100 and working day adjusted). As a first exercise, an *Unrestricted MIxed DAta Sampling* (U-MIDAS) regression has been estimated for the period from 3rd quarter 2014 to 4th quarter 2019, namely before the Covid-19 Crisis (Ghyels et al. 2007).

The FTD series by merchant category code groups (for the period from July 2014 to December 2019) have been considered as an independent variable in the regression to evaluate the forecast performance.

Because the FTD variables are calculated as index number (May 2014 = 100), in the pre-treatment phase, the final consumption expenditure has been calculated as well as an index number with the base 2nd quarter 2014=100. As the second step, FTD and final consumption have been adjusted estimating and removing a linear or quadratic deterministic trend without removing the seasonal component.

The graphical analysis of the combination of the quarterly series of final consumption and the monthly FTD series exhibited a clear correlation between them. As an example, a plot of final consumption and the FTD series referred to clothing sales and hotels (two of the best performers) is in Figure 6.4: the dotted series indicate the quarterly NA final consumption growth rates and the lines are the growth rates for the monthly FTD series for clothing sales.

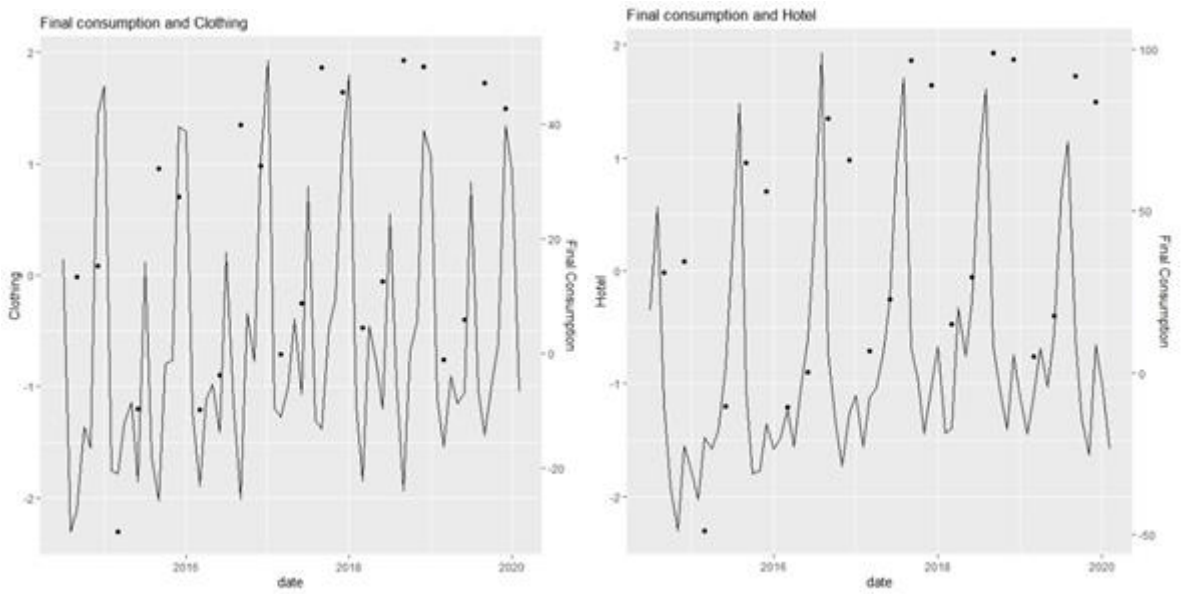

FIGURE 6.4. QUARTERLY SERIES OF FINAL CONSUMPTION AND MONTHLY FTD SERIES

An unrestricted MIDAS model has been estimated for the period 3rd quarter 2014 to 4th 2018 using three lags for consumption and one lag for monthly FTD time series. The forecasts have been evaluated out-of-sample on the period 1st quarter 2019 to 4th quarter 2019 using a recursive and rolling window and the Root Mean Square Error (RMSE) and the Mean Absolute Percentage Error (MAPE) as to accuracy measures.

In Table 6.5 the accuracy measures of our model are compared to the ones obtained estimating an auto-regressive model of order 3 considered as benchmark model (in the columns RMSE and MAPE the ratio between RMSE and MAPE for the model including FTD by MCC and AR(3) model are calculated). For values lower than one in Table 6.5 the predictor variable shows a better forecast performance than an AR (3) model without predictor. Thus, results show that FTD for hotels, clothing and work series gives the best increases in forecasting performance. For example, the gain due to the inclusion of FTD by MCC groups in terms of RMSE for the recursive window experiment is 21% for hotels, 9% for clothing and 18% for work.

TABLE 6.5. ACCURACY MEASURES RESULTING FROM THE MIDAS MODEL USED TO FORECAST THE QUARTERLY FINAL CONSUMPTION EXPENDITURE MODEL IN THE PERIOD 2019 (Q1)–2019 (Q4) WITH RESPECT TO AR(3) MODEL

	Rolling window		**Recursive window**	
	RMSE	**MAPE**	**RMSE**	**MAPE**
FTD for Clothing	0.99	0.87	0.91	0.84
FTD for Hotels and restaurants	0.85	1.02	0.79	1.00
FTD for Food	1.57	1.06	1.77	1.15
FTD for Home	1.22	1.15	1.10	0.94
FTD for Work	0.87	0.78	0.82	0.78
FTD for Retail	1.02	0.91	1.04	0.94
FTD for Services	1.19	1.00	1.29	0.97
FTD for Communication	1.78	1.30	1.72	1.11
FTD for Transports	1.00	1.02	0.84	0.89

The exercise has been repeated including the Covid-19 Crisis phase in the forecasting period; thus, we used 1st quarter 2020 for consumption and monthly data from January to March 2020 for FTD series (Table 6.6). The same approach has been used to evaluate the forecast performance considering a recursive and rolling window and the Root Mean Square Error (RMSE) and the Mean Absolute Percentage Error (MAPE) as to accuracy measures. The hotels and clothing series show a better forecast performance compared to the previous exercise, particularly the improvement of 47% of the hotels series. The reason for this could be partly the autoregressive model of the benchmark, which is less able to predict the strong negative growth rates in the 1st quarter of 2020. Overall, the FTD variables show an improvement thanks to their timeliness and monthly frequency that is more able to capture the strong decrease following the Covid-19 lockdown and the first signals of the consumption crisis due to the unpredictable exogenous factors.

TABLE 6. ACCURACY MEASURES RESULTING FROM THE MIDAS MODEL USED TO FORECAST THE QUARTERLY FINAL CONSUMPTION EXPENDITURE MODEL IN THE PERIOD 2019 (Q1)–2020 (Q1)

	Rolling window		**Recursive window**	
	RMSE	**MAPE**	**RMSE**	**MAPE**
FTD for Clothing	0.85	0.89	0.81	0.86
FTD for Hotels and restaurants	0.53	0.94	0.43	0.91
FTD for Food	1.04	1.05	1.13	1.12
FTD for Home	1.01	1.11	0.95	0.95
FTD for Work	1.01	0.85	1.00	0.85
FTD for Retail	0.95	0.93	0.92	0.95
FTD for Services	1.18	1.03	1.06	0.99
FTD for Communication	1.00	1.22	0.97	1.07
FTD for Transports	0.97	1.01	0.96	0.92

As a final exercise, we calculated a forecast combination of the FTD series using as weights the in-sample root mean square error (RMSE) for the period from 3rd quarter 2014 to 4th quarter 2018. This combination led to a 20% reduction of the RMSE of the forecast for the period 1st quarter 2019 to 1st 2020 compared to the benchmark model (autoregressive model of order 3).

6.5 Discussion and conclusions

Macroeconomic nowcasts and forecasts are relevant tools for policy-makers and their relevance increases with the reduction of the time lag between the period and the publication of official data.

FTD timely availability allows releasing real-time nowcast and forecast. The key advantages of using such sources of information lie both in the coverage and the level of detail they provide and their timeliness. Because of their availability on a near real-time basis[17], they provide a snapshot of current tendencies well before they are recorded in official statistics. Nonetheless, key challenges remain in their use for official indicators. First, we should sign an Istat-Bank of Italy agreement to have data access and correctly interpret the information. Secondly, we should face several concerns about their validation, aggregation and storage.

In our experiments, we could use only some aggregated FTD series to assess how much payments-system data can contribute to the production of more accurate nowcasts/forecasts of macroeconomic aggregates in official statistics.

In our first trial, the automatic General to Specific Selection Procedure for the selection of the best performer series in the ADL model makes new FTD series work in both models and we observed a good fit especially for the period before the Covid-19 Crisis. The gains in terms of reduction of the relative statistics are of 12.1% for the MAE in the case of the retail trade index, and of 67%-73% for the e-commerce. Considering also the Covid-19 period of the lockdown of the economic activities (March 2020), the FTD series still guarantees gains in the forecasting performance of the model larger than 50% in terms of MAE in the case of e-commerce sales.

In our second trial on the forecasting of the quarterly final consumption expenditure, an unrestricted MIDAS model has been estimated for the period 3rd quarter 2014 to 4th quarter 2018 and evaluated both for the period 1st quarter 2019 to 4th quarter 2019 and for the period 1st quarter 2019 to 1st quarter 2020. We compared the accuracy measures of our model to the ones obtained estimating an auto-regressive model of order 3 as the benchmark. In the case of the forecasting out of the sample before the Covid-19 period, results show that hotels and work reduce the forecast RMSE of the AR (3) by 15% and 13%, respectively. When including the Covid-19 Crisis phase in the forecasting period (1st quarter 2020), the hotels and clothing series show a further improvement in the forecasting performance (47% of reduction in terms of RMSE), being the timeliness of FTD series able to capture the determined strong decrease in consumption. The retail and transport series also produced an improvement, although less relevant (less than 5%). Even using a weighted combination of the nine FTD series in the model, the reduction in terms of RMSE is equal to the 20% compared to the benchmark model.

Considering these positive results, we will continue to investigate the relevance of payments-system data and the way to employ this information into nowcasting and forecasting models.

[17] Released 3 days after the end of month T.

7. The Norwegian case study -- Retail sales index statistics of Norway

7.1 Purpose

The purpose of the retail sales index statistics case study below, is to see to what extent and how financial transactions data (FTD) could improve today's current official statistics, either for replacing the current data collection, for improving the estimator, or finally for quality assessment.

7.2 The current official statistics

7.2.1 Data collection

The current Retail sales index statistics are based on a survey that is providing Statistics Norway with the total monthly retail sales for each of the local business units selected to the survey sample. The answers are collected at *legal unit* level.

The survey sample is selected by using two different sampling plans:

- The retail chains are all included in the sample, and for each chain in the sample, all its local business units are selected.
    - Thus, the retail chains are fully enumerated, and the total retail sales are available within the sample at both chain-, legal unit- and local business level.
- For the non-chains, a sample of local businesses are selected.
    - Thus, only some legal units are represented in the sample. Further, if a legal unit is present in the sample, only some of its local business units are usually included in the sample. The total retail sales are available at local business level, but the total retail sales are not available at legal unit level unless all the legal unit's local units happened to be sampled.

In this case study, *the current source* denotes the dataset containing the survey answers on monthly total sales for each of the local business units included in the survey.

7.2.2 Statistical output tables

The statistics output tables are based on the total sales per month by the following tabular groups:

- group 1: Retail sales in non-specialised stores (NACE-code 47.1)
- group 2: Retail sales of food, beverages etc (NACE-code 47.2)
- group 3: Retail sales of automotive fuel (NACE-code 47.3)
- group 4: Retail sales of ICT-equipment (NACE-code 47.4)
- group 5: Retail sales of other types of household equipment (NACE-code 47.5)
- group 6: Retail sales of cultural/recreation goods (NACE-code 47.6)
- group 7: Retail sales of other goods (NACE-code 47.7)
- group 8: Retail sales not made in stores but on web (NACE-code 47.9)

As seen above, each group corresponds to a 3-digit *NACE*-code[18].

7.3 FTD source

The financial transactions data used in this case study are debit card transactions at point-of-sales terminals for one day in September 2016 from the National payment system, c.f. Appendix 5. In the future, data will be available for whole months.

[18] NACE, "Nomenclature statistique des Activités économiques dans la Communauté Européenne", is the statistical classification of economic activities in the European Community.

We will in this case study refer to the FTD debit card source above simply as the *FTD source.*

For our purpose, the main variables of the FTD source are purchase amount, total transaction amount, and bank account number of the receiver. The source is restricted to the transactions made through the debit card scheme above, and this debit card scheme covered averagely roughly 60-70% of the purchased amount in 2016, the remaining purchases being mostly made by credit cards since cash is not used much in Norway. The unit of the FTD source is the single transaction.

7.4 Integration of sources

Consider the two-phase data integration quality framework introduced in Section 5. At the 1st phase, we consider each of the sources by themselves where the unit for the current source and the FTD source is local business and transaction, respectively. The measurement variable at the 1st phase is the transaction amount (in NOK). However, when the two sources are integrated in phase 2, we use instead the legal unit as the unit, for reasons that will become clear below. Still, the transaction amount is the measurement variable.

When we integrate the two sources, we prefer to have a common unit so that we can link the two sources on this unit level and then compare the total sales on micro-level[19]. The natural unit level would be local business unit since this unit is the unit of the current source and also that the FTD source can be aggregated to local business unit (i.e. transaction unit being nested[20] within the local business units).

A linkage problem occurs at the local business level since the FTD source has no local business unit id. However, the FTD source contains the identifier "bank account" which is attached to the the legal unit and not the local business level. The current source has a legal unit id in addition to the local business unit id, and we will below see how the FTD source can be linked to the current source at the legal unit level.

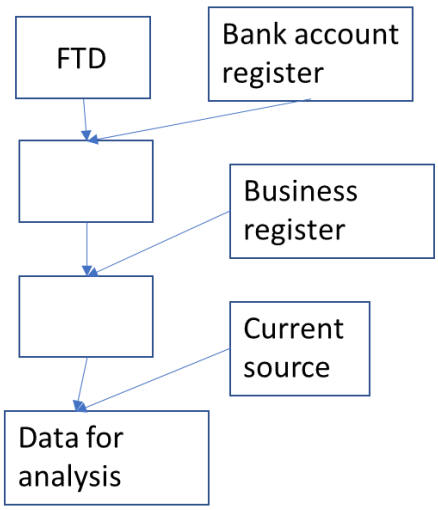

Figure 7.1. The integration of the FTD source with other sources.

[19] Micro-level quality (at the legal unit level) of the FTD source is not necessary for our purpose since the official statistics only needs quality at the aggregated level of the nine groups. Assume e.g. a situation where the publication group level total sales from the two sources were equal but where a $N(0,\sigma^2)$-distributed noise is added to each unit of the FTD source, thus causing a lower micro-level quality of FTD whereas there are no quality issues at the group level. Thus, high micro-level quality is sufficient but not required.

[20] a transaction is related to one and only one business unit.

The integration of sources is illustrated in Figure 7.1. The Tax Authorities' Bank account register contains the legal unit id for each bank account. Thus, the legal unit id is added to the FTD source by linking the FTD source and the Tax Authorities' Bank account register, using bank account as the linkage key. The legal unit id is now used as linkage key for the rest of the integration: We add NACE3 to the FTD source by linking FTD and the Business register, since the Business register contains the NACE3-code of the legal unit.[21] Finally, the FTD source is linked to the current source. This gives us an analysis data set containing, for each legal unit i in the survey:

- from the FTD-based source: the total sales X_i which is restricted to debit cards, and only for one day.
- from the current source: the total monthly sales Y_i restricted to the sales coming from those of the legal unit's local business units that were sampled into the survey.
- The NACE3 code of the legal unit

The current source contains all legal units having at least one of their local businesses included in the survey. Due to the survey sampling plan of the current source (Section 7.2), for each of the legal unit not belonging to a retail chain, Y_i will be systematically too low unless all its local businesses were drawn into the survey sample.[22] Fortunately, the legal units belonging to a retail chain have all their local business units included into the survey, thus avoiding the problem and we therefore restrict our analysis dataset to retail chains.

We cannot compare the local business level total sales, even though the analysis data set now only contains the retail chain legal units and thus includes each single local business total sales from the two sources. As an illustration: Consider e.g. a legal unit with four local business units in each of the two sources. We cannot tell which of these four FTD source-local business units belong to which of the four current source local business units, and thus we use the legal unit level.

In this subsection, we described the construction of analysis data set, and also identified quality issues related to the *integration*: the coverage error if not sub setting to retail chains. In the next section we will present the methodological approach and quality issues related to the aim of our case study (use of FTD for predicting the total retail sales).

7.5 Improving current statistics by FTD.

We will first in Section 7.5.1 look at the challenges that we are facing when using the FTD source for improving the current statistics. Then in Section 7.5.2 we will model the relationship between the FTD source and the current source sales.

7.5.1 Types of potential errors in the FTD source as a source for the retail sales statistics

The ultimate goal of our case study is not to predict correctly for each legal unit, but to aggregate from the legal units to the NACE3-level. Here is a source of error since the legal units unfortunately are not nested within the NACE3-categories, as we can see illustrated in Figure 7.2: a legal unit can consist of several local businesses each belonging to different NACE3-categories, although most of the legal unit's purchase amount will usually belong to the legal unit's NACE3-code which is defined as the main NACE3-code for the local business units of the legal unit. This error is a *domain* error since some local business units are tabulated

[21] The Business register naturally also contains NACE3 for each local business unit, but this is less relevant for us since we have linked on legal unit level.

[22] Since we are in phase 2 of the quality framework, this error is a measurement error, since the measurement variable "amount in NOK" is systematically lower than the true value. Viewed from the FTD-phase 1 perspective, the error would have been regarded as representation error, but this gives no meaning since we are now talking about the current source, not the FTD source…

within an erroneous retail sales (NACE3) publication groups. In our case study, we will assume this error to be small, but this will be an interesting topic for future studies.

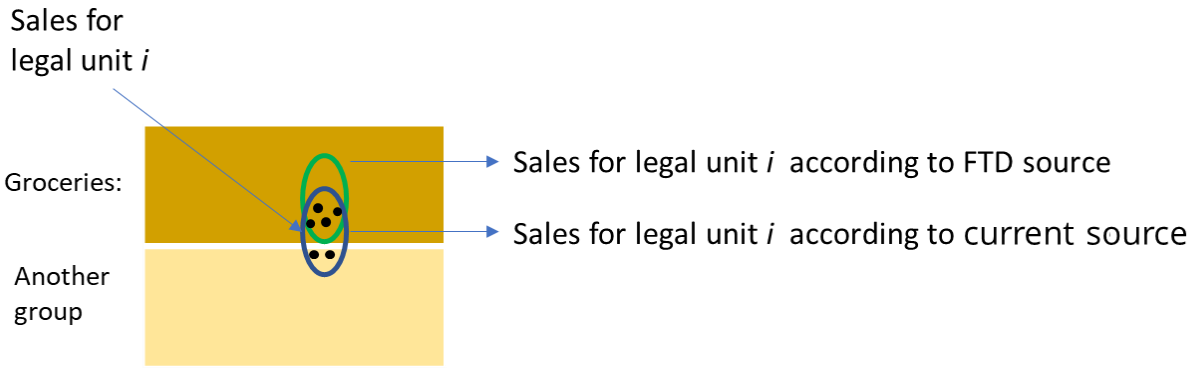

Figure 7.2 Illustration of the domain error caused by using legal units instead of local business units, the former sometimes including local business units from different groups.

A coverage problem of the FTD source is that is is limited to POS-terminals. All the sales for publication group 1-group 8 are covered by POS-terminals, thus no coverage error of this type is related to these groups. However, the FTD source has no observations in publication group 'sale through the web', group 9. Thus we cannot use the FTD for estimating the sales of group 9. We leave this question for future work and concentrate on publication group 1 – group 8 The question that we will address below is then how well the FTD source covers the publication group 1-group 8 sales.

The FTD source is only covering debit cards sales, and neither credit card nor cash. The FTD source thus systematically gives lower total sales than the current source, which from the phase 1-quality perspective is a representation error[23] since some of the transaction units are not covered. However, the phase 1-perspective is the perspective of the data whereas our perspective for the case study is that of the retail sales statistics. Thus, we here adopt the phase-2 perspective where the unit is the local businesses, and then the restriction to debit card sales implies that we have a measurement error in measuring the total retail sales for the legal unit, unless all the legal unit's sale is made by debit cards purchases. Finding the total sale is a question of inference from the observed debit card sale to the total sale.

Until monthly FTD are received at Statistics Norway, we only have FTD for one day in this month. This causes a measurement error in addition to that caused by having only debit cards sales.

7.5.2 Modelling

We let Y_i denote the current source sales total for legal unit i during the month. The FTD source debit card sales total for the single day for the month is given by $X_{D,i}$ and X_i respectively.

We assume Y_i to be correct for our analysis dataset of legal units. This means that the legal units report the correct sales numbers in the survey. We think this assumption is reasonable since the turnover is the perhaps most important information in the local unit's administrative system. The legal unit respondents have no incentive to deliberately report erroneous numbers since the reported numbers cannot be accessed by the Tax authorities or any other governmental institution except Statistics Norway itself.

[23] *Coverage error* if the FTD source is viewed as a population dataset, *sampling error* if the FTD source is regarded is a non-random sample from the population of all transactions. Notice that our FTD source can be viewed as a non-random sample of the population of all the retail sales transactions (at point-of-sale terminals).

The measurement error caused by having only one day could be modelled as a part of the measurement error caused by having only debit card purchases, and consequently our inference could be from debit cards on one day to total sales in the month. However, since monthly data are only temporarily missing, we will divide the inference:

- The 1st-step inference is from day to month
- The 2nd-step inference is from monthly debit cards sales to total sales.

The 1st-step inference will no longer be necessary once monthly data are received, and thus we will in this case study only use a simplistic model to cover this step as a temporary solution until monthly data arrive: we will make the strong assumption that all legal units have 24 sales days, and that for all legal units, the sales during the single day contained in the FTD source, is representative for each day in this month. Thus, we assume that we have

$$X_i = 24X_{D,i}, \text{for all } i .$$

Let R denote the *debit card rate* which we define to be X / Y, i.e. the rate of the total purchase being made by debit cards during the month. In Figure 7.3, we see the debit card rates R_i as a function of X_i for our analysis dataset that we have restricted to groceries, and more precisely to NACE3-code 47.1. In Figure 7.4, we are only considering those legal units where the debit card rate is less than 1.0, this shows the pattern better. We notice that the centre of the debit card rate-distribution is about 65%-70%, but with a large variability, which partly is due to the fact that X_i is estimated from $X_{D,i}$. There are some outliers having very low debit card rates, and which will cause difficulties in finding a suitable model. There are also outliers having debit card rate exceeding 1.0, some of them even being more than 5.0. Such low and high debit card rates will occur when their actual monthly debit card sales are poorly estimated by multiplying the single day by 24.

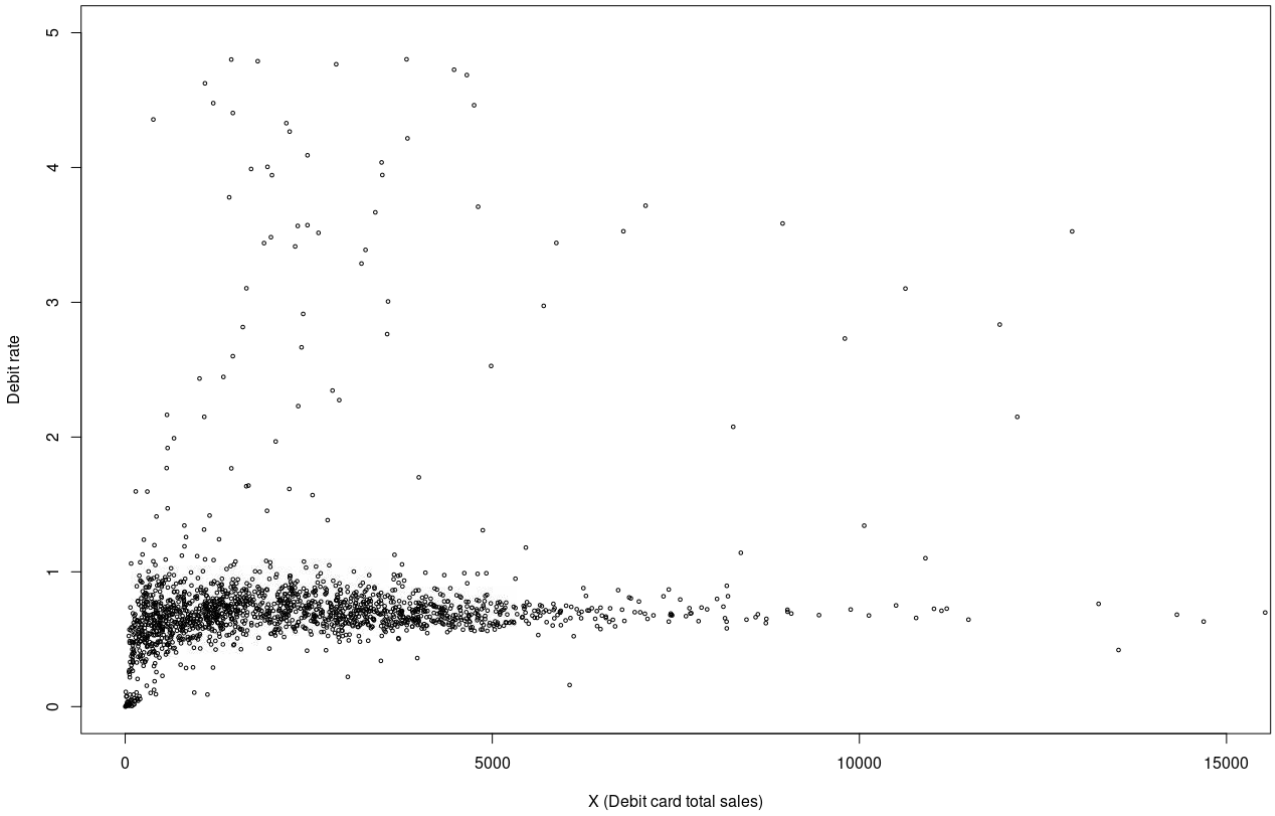

Figure 7.3 The debit card rate for legal units within the analysis data set. Groceries (NACE3=47.1). Omitting the 3.58% legal units having debit card rate exceeding 5.0 and the 68 legal units with debit card total sales exceeding 15 000.

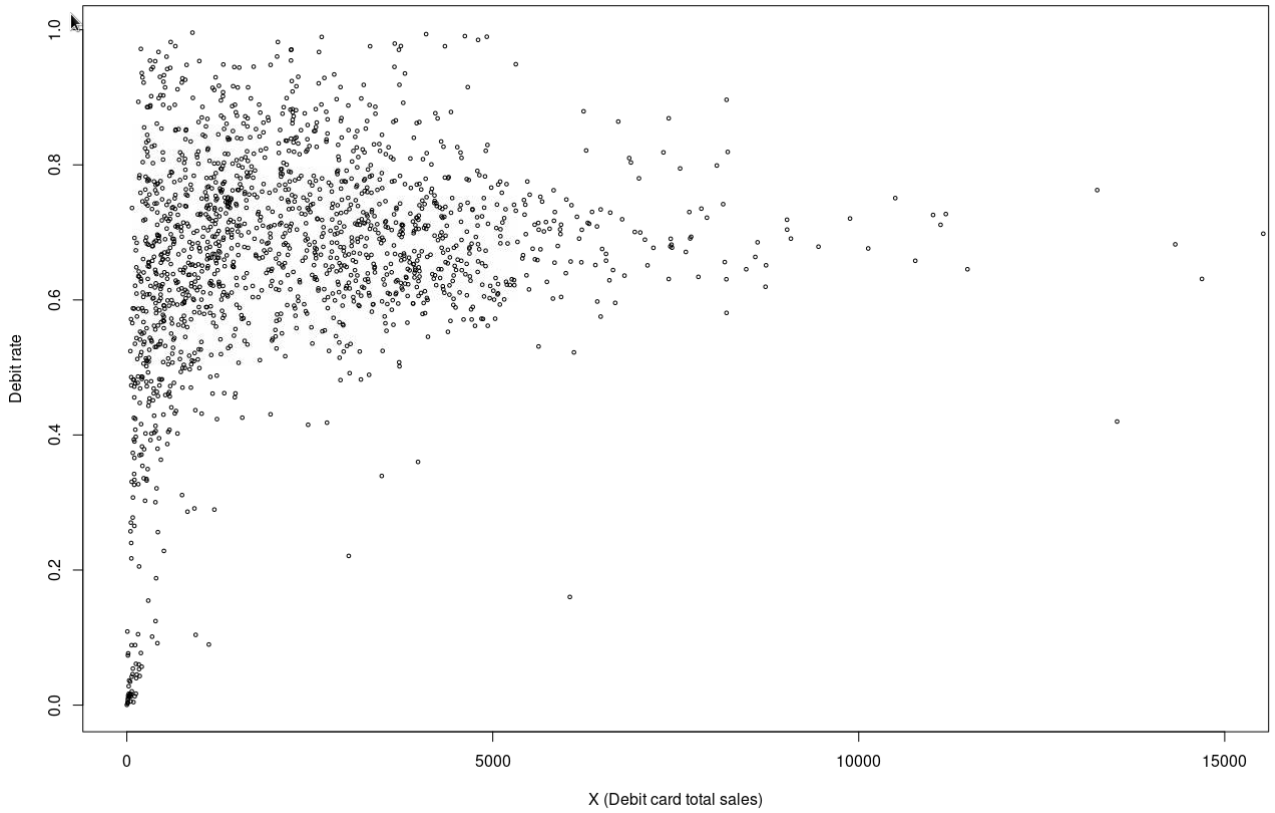

Figure 7.4. The debit card rate for legal units within the analysis data set. Groceries (NACE3=47.1). Omitting the 10% legal units where the debit card rate is exceeding 1.0 and the 68 legal units with debit card total sales exceeding 15 000.

Given the X_i, we can now model the relation between the sales totals Y_i and the monthly debit card sales totals X_i, i.e. we consider the monthly FTD-based totals as non-random. Our aim is to find a model fitting well the relation between Y_i and X_i, so that it can be used to predict the true value Y_i from X_i for the legal units not included in the analysis data set.

As our initial model, we will regress Y_i onto X_i. We remember that we only need to find a model being sufficiently good for prediction at the aggregated NACE3-level, not at the (legal unit) micro level. We will in Section 7.5.3 look at which model to choose, and how to assess the performance of each model.

The predicted value of each legal unit is $\hat{Y}_i = h(X_i;\hat{\theta})$, where $h(\cdot)$ is the regression function and $\hat{\theta}$ is the estimated regression parameter(s).

Once a model is found, the NACE3-level predicted totals are calculated by simple enumeration of all the predicted values for all X_i belonging to each NACE3-category, giving

$$\hat{T}_{g_1} = \sum_{i \in g_1} \hat{Y}_i$$

The performance of the prediction in group 1, g_1, is the relative error of the predicted total $\hat{T}_{g_1}$ compared to the true total $\hat{T}_{g_1}$

$$\eta = \frac{\hat{T}_{g_1} - T_{g_1}}{T_{g_1}}. \tag{7.1}$$

7.5.3 Results

Pre-processing: The presence of FTD for only one day, is as mentioned a problem unless the observed day is representative for the 24 sales days in the month, or more precisely that the monthly debit card sales for this legal unit is 24 times the debit card sales for this day. For some legal units, this is definitely not the case and these legal units are removed from the analysis but will be included in the future after monthly FTD arrives. The following legal units are removed from the analysis dataset:

- those where the debit card rate exceeds 100% (10% of the original analysis dataset)
- those where X is less than 200 (additional 5.6% of the original analysis dataset)
- those were the debit card rate is below 50% (additional 5.3% of the original analysis dataset)

In addition, only chains with more than three legal units are included in the dataset. The final analysis data set contains 1449 legal units.

Requiring observations to have debit card rate more than 50% seems to be a strong requirement, but an attempt of 40% or 45% caused severe violations on the model assumptions for a linear model. Also, this requirement is reasonable since X is not the observed monthly debit card sales, but only estimated from the sales during one weekday by assuming that the sales can be multiplied by 24. This assumption will be erroneous for e.g. tourist region units having their main sales during the weekends, leading to a low debit card rate.

The simplest linear regression model is

$$Y_i = \beta X_i + \varepsilon_i, \tag{7.2}$$

i.e. where the constant term is set to zero, this is natural in our setting where we would anticipate zero total sales when there is zero debit cards sale. For the model without constant term, the model assumptions are better fulfilled than with a constant term, but still the assumptions are far from met, as we can see in Figure 7.5 where

- the upper right panel reveals substantial deviation from the normal distribution assumption.
- the lower left panel shows that the variation of the residuals is a function of the fitted value and several residuals are also too large
- the lower right panel shows that a few of the observations are very influential on the model fit.

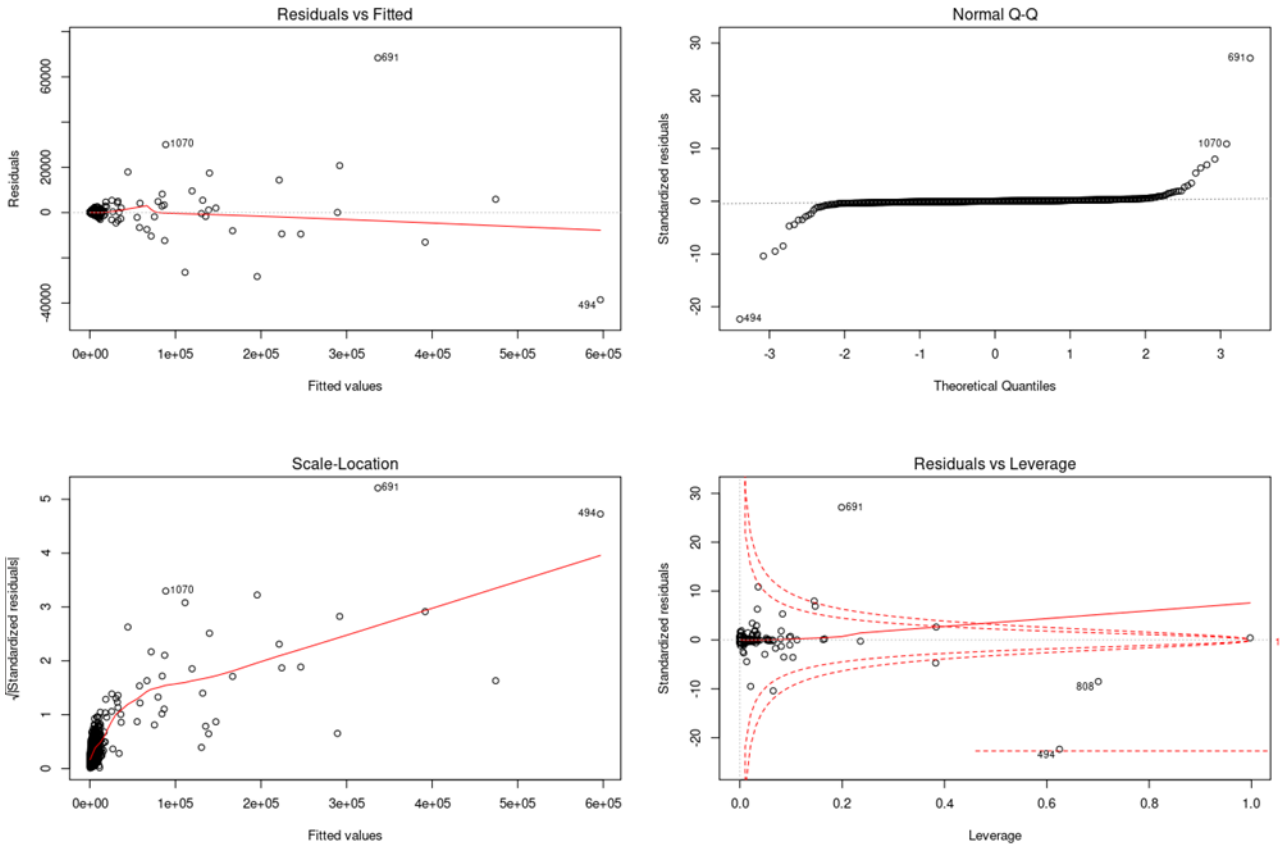

Figure 7.5. Model diagnostics for the linear model (7.2). Outliers removed as described in the pre-processing paragraph.

A log-transformation improves the model, more specifically the model

$$\log(Y_i) = \alpha + \beta \log(X_i) + \varepsilon_i \tag{7.3}$$

better fulfils the model assumptions, c.f. Figure 7.6. A similar model without the constant term shows slightly worse model diagnostics. Figure 7.6 shows that the variation of the residuals is not a function of the fitted value (left panels), the residuals are approximately Normal distributed (upper right panel), and the single observations being most influential on the model, is much less influential than for the model with no transformation.

Figure 7.7 shows the model diagnostics for the model

$$\log(Y_i) = \alpha + \beta_j \log(X_i) + \varepsilon_i \tag{7.4}$$

where j is the chain to which the legal unit belongs. Thus, the model includes the interaction between debit card sales and chain. We notice that the residuals are distributed closer to the Normal distribution compared to the model without the interaction term. However, the residuals are now showing signs of being dependent on the fitted values.

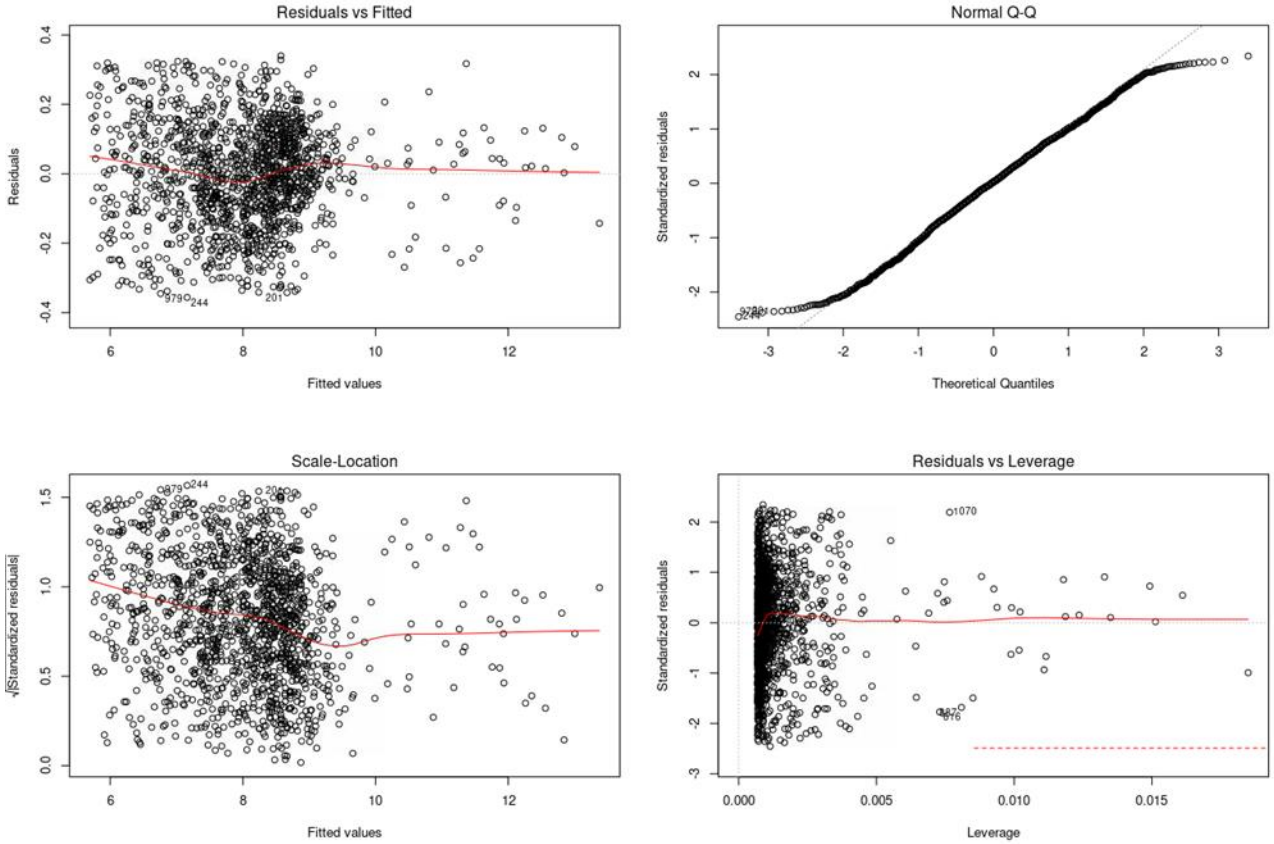

Figure 7.6. Model diagnostics for the log-transformation model (7.3). Outliers removed as described in the pre-processing paragraph.

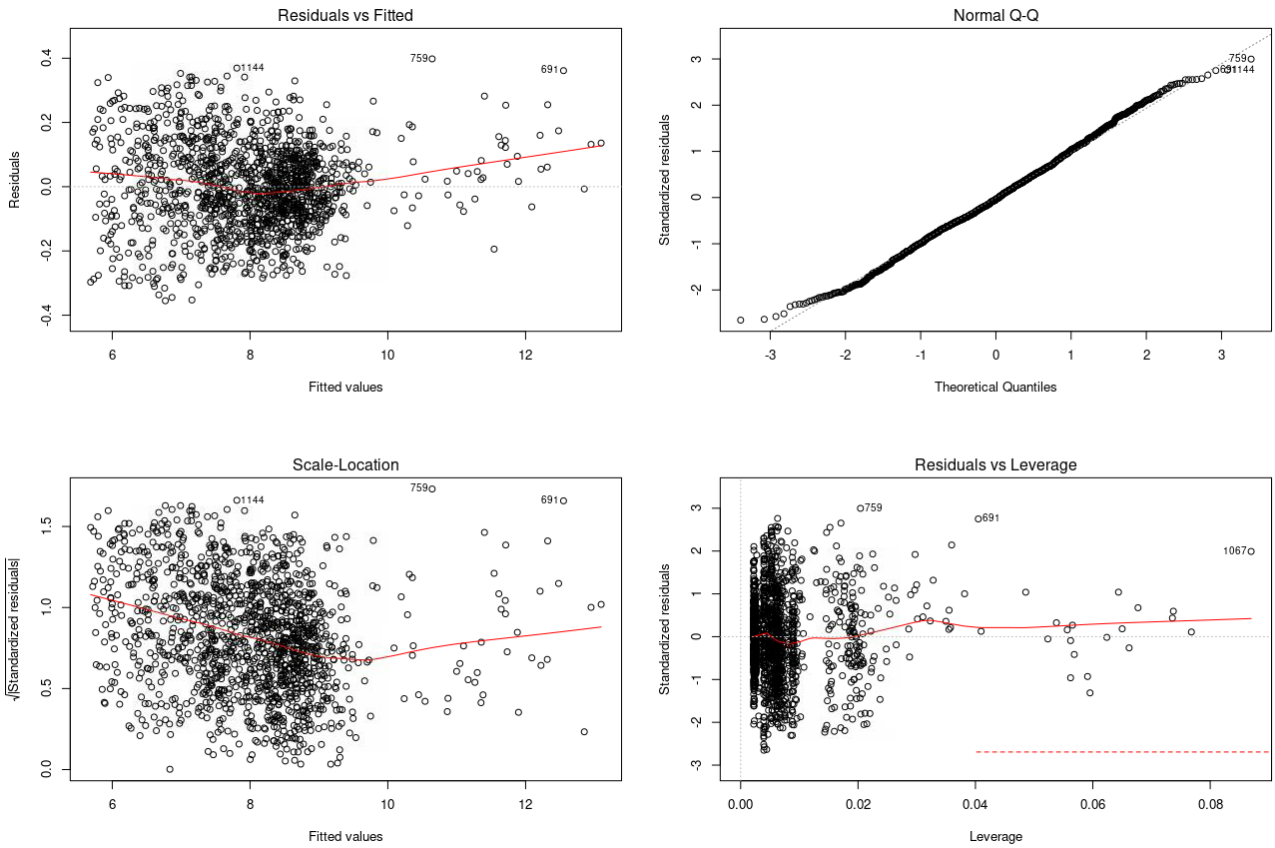

Figure 7.7. Model diagnostics for the log-tranformation model with interaction between debit card sale and chain (7.4). Outliers removed as described in the pre-processing paragraph.

For assessing the performance of the model when it comes to prediction, first the analysis data set is split into two equally sized training data set and test data set. The model is fitted for the training data set, and then the model is used for predicting the total sales Y_i for the test data set, The aggregated total $\hat{T}_{g_1}$ for the whole grocery group is then calculated. The absolute relative error η (7.1) is then calculated. The split into training set and test set, and calculation of η is repeated 1000 times. This leads to a mean absolute relative error of 5.6% for the model with interaction, and the estimated distribution of η is shown in Figure 7.8.

The reason for η being so large, seems to be a consequence of the large variability of the debit cards rate given X, meaning that the model fails to explain the data and also that by the random allocation of observations into training and test datasets, the training set model fails to cover the pattern of the test data set, even in our case when only the aggregated total sales of the whole test data set is of interest.

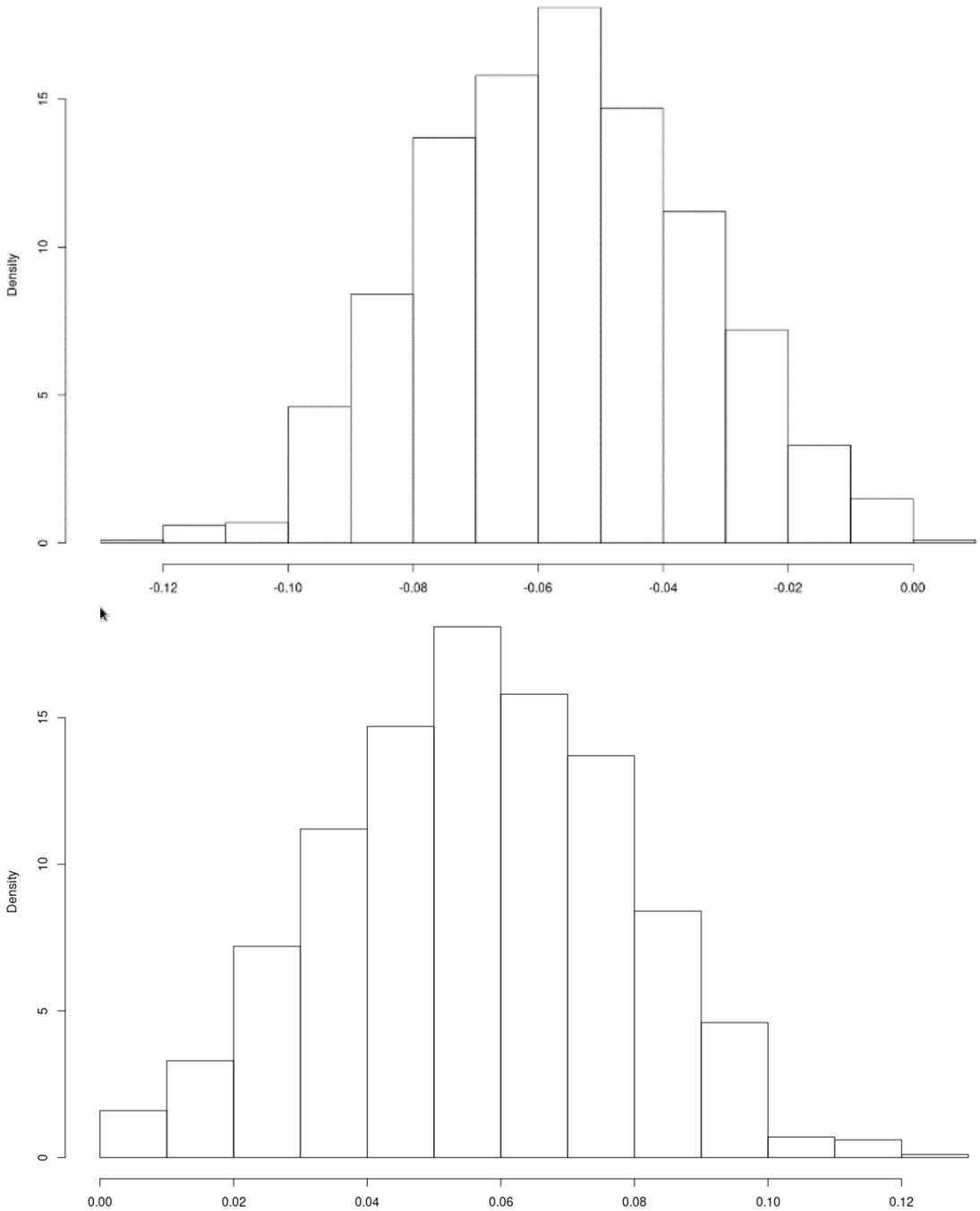

Figure 7.8. The estimated distribution of the relative error η *(upper panel) and the distribution of its absolute value (lower panel)* of the FTD-based prediction of the grocery group total sales. For *the log-transformation model with interaction (7.4). Based on 1000 replications of the division into training data set and test data set.*

7.5.4 A method for future studies

An approach that can be promising once monthly debit card data are received, is to utilise the VAT-register. This register contains the total retail sales, but not monthly. Instead it contains the total retail sales for two months. Its production time is about five months. A preliminary estimator for November that could be estimated very rapidly could be

$$\hat{Y}_{Nov,pre} = \frac{Y_{May+Jun}}{X_{May+Jun}} X_{Nov} \ ,$$

where $Y_{May+Jun}$ is the VAT-register for the period May-June from the same year. We notice that this ratio estimator is based on the debit card source, and the VAT-register is only used for weighting.

A final estimator with about five months production time could then be

$$\hat{Y}_{Nov,fin} = \sum_{\text{legal units } i} \frac{Y_{i,Nov+Dec}}{X_{i,Nov+Dec}} X_{i,Nov} \quad .$$

This is a post-stratification estimator, and the VAT-register is also for this estimator only used for weighting.

7.6 Conclusion

Using debit card FTD for only one day for predicting the total sales for the eight non-internet groups of the retail sales index, often leads to rather large errors in the predicted total sales. Hopefully this problem will to a large degree disappear once the monthly FTD appear. However, the total sales in the group containing internet sales, is not covered by the FTD source, and this is a topic for future studies.

8. The Portuguese case study: FTD in Shared Economy for Lodging and Transport

8.1. Purpose

Financial transactions that correspond to purchases of goods or services are monitored by the Tax Authority, which keeps track of all the FTD that correspond to the issue of an invoice. In this case study we use FTD from the Portuguese Tax Authority provided by the Portuguese Ministry of Finance, in order to assess the demand in two different sectors: Local accommodation and Car Ride-sharing services. The general purpose of this study is to see in what extent it is possible and useful to incorporate FTD from an external source in the production of official statistics.

8.2 The current official statistics and source

Currently, the production of official statistics in the business sector is largely based on data collected from surveys. For example, data for the "Business Turnover, Employment, Wage and Hours worked Index in retail trade" is collected, and the index is computed every month as a Short-term business statistics (STS) based on survey data. Besides, at the European level, all national statistical authorities use statistical questionnaires for collecting STS data. Member States are encouraged to measure economic activity and to transmit seasonally adjusted indices to Eurostat.

In this sense, and in order to reduce the statistical burden on respondents, Statistics Portugal has been spending some efforts in taking advantage of using administrative registers and FTD that are already collected for other purposes. The Portuguese Tax Authority already collects data on purchases and sales of products subject to VAT, and these FTD can hopefully give valuable inputs for the turnover and other variables of Short-term business statistics. On the other hand, information about remuneration[24], number of workers, etc. are available in the Social Security System, and this administrative register source is another source of interest.

The case study will be presented in two different domains:

- **Small accommodation services** –the current survey: IPHH, a Tourism survey that aims at characterising the capacity of hotels, and other type of accommodation services, is not intended to collect data from small lodging. We all know that small accommodation services are having an increase of demand in the tourism industry. We are facing a new era of lodging, since the creation of companies such as Airbnb's with the impact they have on Hotels, Travelers, and Cities. Statistics Portugal aim, in a future survey, to characterise the small lodging / accommodations services, and where the sampling frame is based on the information provided by RNAL - the National Register of Lodging/Local Accommodation which is described in Section 8.3.
    - In the case study, the FTD E-invoice data and the Social Security administrative data will be used for assessing the tourism activity related to small lodgings. These sources are a potential alternative to the future study described above.
- **Car riding services** - the goal here is to characterise small transportation services (ex. Uber). Statistics Portugal does not have access to the information provided by these services. However, information exists on remunerated transport providers through the TDVE system described in Section 8.3.

  There are other data than FTD e-invoice and social security data available for the TVDE companies and platform companies in the Business register database, already available for Statistics Portugal.

[24] Payment for work or services

One of the most important variables is *VVN* (Volume de negócios, Turnover of establishments), which will also be used for analysis in this case study.

8.3 FTD source

In this section we describe the new FTD source and the new administrative source to be used for the analysis: The E-invoice data and the Social Security Data. We also describe other sources that will be used in the case study.

E-invoice

The Portuguese Tax Authority is responsible for collecting the financial transactions related to all purchases taking place in Portugal (based on the invoices issued), regardless of the payment method used. It can be assumed that the transactions from all payment instruments and payment platforms will be included: electronic payments (ATM, bank transfers or Credit Card), as well as non-electronic payments (in cash), but the data do not contain a detailed breakdown by type. After successive reforms of the process of dealing with the invoices, the digitalisation of the administrative systems allowed invoices to be communicated directly by the entity that provides the service or sells the good to the Tax Authority. The "E-Invoice" is the invoice confirmation system issued under a taxpayer number. This system is an integral part of the Finance Portal, under the scope of the Portuguese Ministry of Finance. All the invoices issued by Portuguese sellers of good and services of all kinds are registered in this database. This dataset is provided on a monthly basis, and contains about 80 million transactions every month, corresponding to an overall amount of 50 000 million EUROS of taxable amount per year.

A recent protocol between Statistics Portugal and the Portuguese Tax Authority has been signed last year that allows the access of Statistics Portugal to the E-Invoice data set. The attributes in the database (monthly data from January 2018, updated the most recent month) to which Statistics Portugal has access to, are the following:

Table 8.1 – Attributes of the E-invoice dataset.

Attribute name	Format
NIF (VAT Number or taxpayer number)	String
VAT of the buyer	String
Month of transaction	Date
Country (of the buyer)	String
EFAT (Taxable value aggregated bt VAT Number)	Number

The VAT Number can be useful to derive other attributes in existing datasets of Statistics Portugal, such as the NACE code of the company, turnover, or company status.

The E-invoice dataset contains very important information about retail activity that is an important input for certain surveys in official statistics such as the "Business Turnover, Employment, Wage and Hours worked Index in retail trade". In addition, the dynamics of turnover - a very important variable in measuring economic activity - can be derived from purchases and sales recorded in this database.

Social Security Data

Statistics Portugal already publishes quarterly, experimental statistics on remuneration based on the information of the Monthly Statement of Remunerations transmitted by companies to the Social Security and the Contribution Ratio of Caixa Geral de Aposentações' (General Pension Fund) subscribers[25].

This information concerns around 400 thousand companies and approximately 4.2 million employees. In the future, Statistics Portugal will get the data at employee level and is intended to complement these statistics with others that will make it possible, among other possibilities, to know the distribution of wages and to carry out the socio-demographic characterisation of workers.

Table 10.2 – Attributes of the Social Security dataset

Attribute name	Format
NIF (VAT Number or taxpayer number)	String
DMR (monthly statement of remunerations)	Number
Month of transaction (remuneration?)	Date

The variable *DMR* - monthly statement of remunerations – is of a particular importance in the current case study, since it can provide information on the earnings paid by the companies, which is seen as proxy of the turnover which is a useful additional source to be used in combination with FTD.

Other data

RNAL, National Register of Lodging/Local Accommodation, is a list containing the small hotels (under 10 beds) and it is provided by the Ministry of Tourism, which is the National Tourism Authority in Portugal. RNAL will be used in this case study for the part investigating accommodation.

TVDE – In Portugal, all vehicles with driver that are rented through mobile apps from Bolt, Cabify, Kapten, Uber and It's My Ride are identified in the car's rear windshield with the symbol TVDE[26]. This acronym denotes the remunerated transport of passengers in vehicles that are featured using an electronic platform. The activity of electronic platform operators and TVDE operators can be carried out by legal persons established in the national territory, upon proof that they have the requirements for performing the activity. The TVDE source will be used for the case study part investigating transport.

[25] Available at: https://www.ine.pt/xportal/xmain?xpid=INE&xpgid=ine_statslab
The statistics on "Gross monthly earnings per employee" published with this Press Release are the result of the calculations and analysis performed by Statistics Portugal on the information from the Monthly Statement of Earnings (DMR – Declaração Mensal de Remunerações) from Social Security (SS), following a protocol celebrated between Statistics Portugal and the Instituto de Informática, I.P., and the Contributive Relation (RC – Relação Contributiva) of Caixa Geral de Aposentações (CGA). These statistics are stored in the StatsLab area of the Statistics Portugal website, where new statistical resources that give useful information for economic and social analysis are published. The information received from the DMR includes the sum of earnings reported by enterprises to the SS by "Social Security Contributory Scheme" and "Earnings Components". The information received from the RC includes the sum of earnings of the subscribers of CGA by "Type of earnings". Despite the name differences, both the "Earnings Components" and the "Type of Earnings" refer to the earnings components paid to employees.

[26] A list of TVDE operators is available in https://imt-tvde.webnode.pt/ (Portuguese only)

8.4 Integrating the sources

The integration of the FTD E-invoice source and the social security source into the national statistical system has been done experimentally until now.

There are several issues with the integration of the FTD source into the current statistical production. This dataset contains millions of rows per month, and many outliers. There are also some negative taxable values, corresponding to credit notes and thus not being errors. They correspond to payments that have been cancelled, and therefore the pair (payment, credit note) cancel out the value of the transaction. However, many credit notes have no information on the acquires' VAT number, and therefore finding the pair is a complicated process. Typically, the parts of the pair (payment, credit note) are not recorded in the same month, so the monthly values have many discrepancies when considering each company separately.

It is necessary to pay attention to the existing outliers. Outliers are observations with atypical values that present a marked difference from other observations in the same data set. The dataset contains some very high values that cannot be discarded because they correspond to a large part of the invoicing values of the companies concerned, as well as a large part of the total transaction value of the entire business area. On the other hand, some very low values cannot be discarded as well, because these negative values correspond to credit notes, c.f. discussion above.

After a statistical analysis of the E-invoice dataset we concluded that:

- the invoice value distribution has a positive and a negative part;
- the positive part has a distribution close to the lognormal, typical of monetary values, with relatively low values and, occasionally, high values that correspond to transactions between large LA companies with tourism companies;
- the negative part also has a distribution close to the lognormal when measured in absolute values;

The strategy for outlier detection is being currently defined, and will certainly go in the direction of outlier detection within groups defined by CAE (The Portuguese Classification of Economic Activities); Number of persons employed ; and NUTSII: NUTS II of the issuer; Institutional Sector; Situation of the Company Before the Activity; and Turnover.

After some investigation on the issue of statistical outlier detection, we decided to transform the data by an Inverse Hyperbolic Sine transformation, and compare the use of different scores (such as z-score, modified z-score, boxplot-score and complex-score); finally, we aim at comparing the use of scores with the Generalised extreme Studentized deviate test. (Iglewicz and Hoaglin, 1993; Rosner, 1983) Outlier detection in E-invoice is under development in Statistics Portugal

In the case study, the FTD E-invoice data and the Social Security administrative data will be used for assessing the tourism activity related to small lodgings as well as for assessing the transport activity. Also, the RNAL and TVDE source will be used. The following picture illustrates the way how we explore data for the specific companies related to SEP (from RNAL and list of TVDE players).

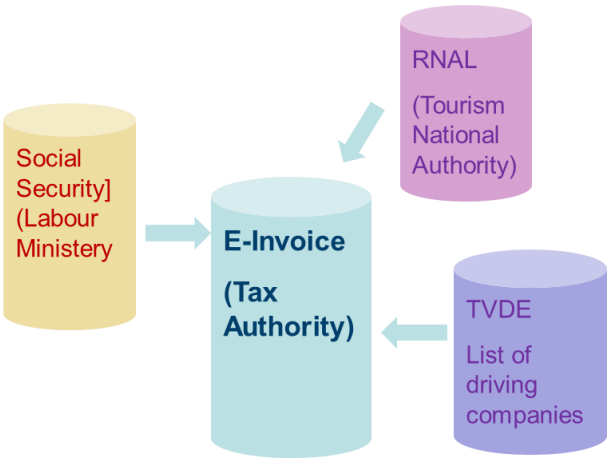

Figure 8.1 –Sources being used in this case study in addition to internal sources within Statistics Portugal.

Just as with the E-invoice data, we are dealing with some problems in the Social Security source. Below we will describe this further.

The data from Social Security (SS) are received by Statistics Portugal every month, containing the remunerations declared by the companies for the previous months. For each month k, four different versions of data are delivered. The existence of different versions results from the fact that companies can correct data previously declared. It turns out that for different months of the quarter, different versions of the history are used in the detection of the companies to be imputed, depending on the quality of the information provided by each company and month. Furthermore, there are missing values in the Social Security related to the remuneration for some of the companies. In order to be able to respond to the needs of different statistical operations of Statistics Portugal, it was agreed to identify and input missing values and to identify in advance the companies that probably will make significant changes to their data in the following versions sent to Statistics Portugal this month. The imputation process aims to fill missing values that are concluded as being non-zero. For this purpose, a remuneration value that (between two versions) change with more than 10.000 (ten thousand) euros between d is considered significant in the sense that we want to to identify these entries in the early version, when they are registered as zero, and impute them instead of waiting for a new version. Other possible values than 10.000 were tested, but this threshold was what enabled a high correction coverage rate to be achieved without compromising the process of detecting companies to be targeted in time, since the use of too low threshold is time-consuming for Statistics Portugal. In addition, this value is associated with a low number of false positives.[27] For a more detailed explanation on how the edit and imputation process was made in the variables associated with the remunerations, number of workers, and contributions, see Santos and Campos (2020)[28].

[27] I.e. low risk of imputing values that in later versions turn out to actually be zero.
[28] Santos and Campos (2020), «Support Vector Machine for Imputation and Editing», Portuguese Statistical Society Bulletin, Spring Issue 2020: (Portuguese Version only) https://www.spestatistica.pt/publicacoes/publicacao/ine-85-anos-de-estatisticas-servir-o-pais

8.5 Improving the current official statistics by FTD - monitor the Shared Economy for Lodging and Transport

In the following, we use the FTD E-invoice data and the Social Security data to assess the demand in two different sectors: Small Lodging/accommodation services and Car Ride-sharing services. The main goal of this study is to see to what extent it is possible to incorporate FTD in the production of official statistics.

We start with Small accommodation services and then proceed to the Car Ride-sharing services. In both cases we used panel regression trees. A regression tree is a tree-structured solution, where data is fitted throughout a regression model. In each node of the tree, a recursive partitioning of predictor space is made. The (generally named) decision tree models are machine learning algorithms that learn to infer the values of the response features and are very attractive models that care about interpretability. Sela and Simonoff (2012), as an extension of the GUIDE method by Loh and Zheng (2013), introduced the *RE-EM*[29] algorithm that expanded regression trees to the panel data which is the the type of data being used in this case study. Basically, the RE-EM algorithm uses a methodology that combines the structure of mixed[30] effects models for longitudinal data with the flexibility of tree-based estimation methods.

Data is organised in such a way that the same companies, *objects*, have been observed in different periods of time, a situation that is commonly used in longitudinal analysis. Longitudinal data refer to the situation where repeated observations are available for each sampled object. A standard methodology in the statistics literature for this type of data is the mixed effects model, where these differences between objects are represented by random effects.

We tried to investigate whether attributes of the transaction values from the E-invoice data (EFAT , the taxable value from the FTD source), are correlated with the business attributes of DMR (remuneration; from the social security data) and VVN (turnover; from the business register), and further to understand to what extent the variable EFAT can be used as a proxy for the turnover in official statistics.

8.5.1 - Small Accommodation services

The goal of the case study part on small accommodation services is to know to what extent can E-Invoice be used to measure the activity of small accomodation services, by comparing the FTD taxable value (EFAT) with the social security system remuneration value (DMR). If these variables are strongly related when correcting for other variables, this is an indication that EFAT can be used for this purpose.

As described in Section 8.5.1, the current survey (IPHH) does not cover accommodation services with less than 10 beds. To study such accommodation services, 46652 companies with values in E-Invoice from January 2018 to December 2019 (a 24 observation monthly series) were identified in RNAL for our case study.

Data Cleansing

Some 6,912 records contain missing data and therefore, a total of 39,740 companies remained to be analysed.

In terms of outliers, after analysing the corresponding monthly E-Invoice data, thresholds have been chosen to examine the values of EFAT (taxable amount):

- the lower threshold was -25000 €, the upper threshold was 250 000 €;

- there are five or six companies that pass most of the credit notes over € 25000

[29] 'RE' being an acronym for random effects, 'EM' referring to the EM-algorithm.
[30] Combination of fixed effects and random effects.

- a singular NIF exceeded an invoice value of 500 000 €

In order to solve these problems, we proceed with different implementation strategies:

- analyse case by case the credit note values for those E-invoices exceeding a fixed limit (€25000) or exceeding a dynamic parameter (20% of monthly invoicing). However, this analysis requires pairing of the invoice and the corresponding credit note, which was not always possible due to the lack of VAT number for many buyers;
- consider only the aggregated monthly values being the sum over all companies and expecting the random effect of the population to correct for the issue of credit notes not being attributed to the correct original transaction.

Figure 8.2. depicts the evolution of small accommodation in terms of the three sources under analysis, showing some correlation between them

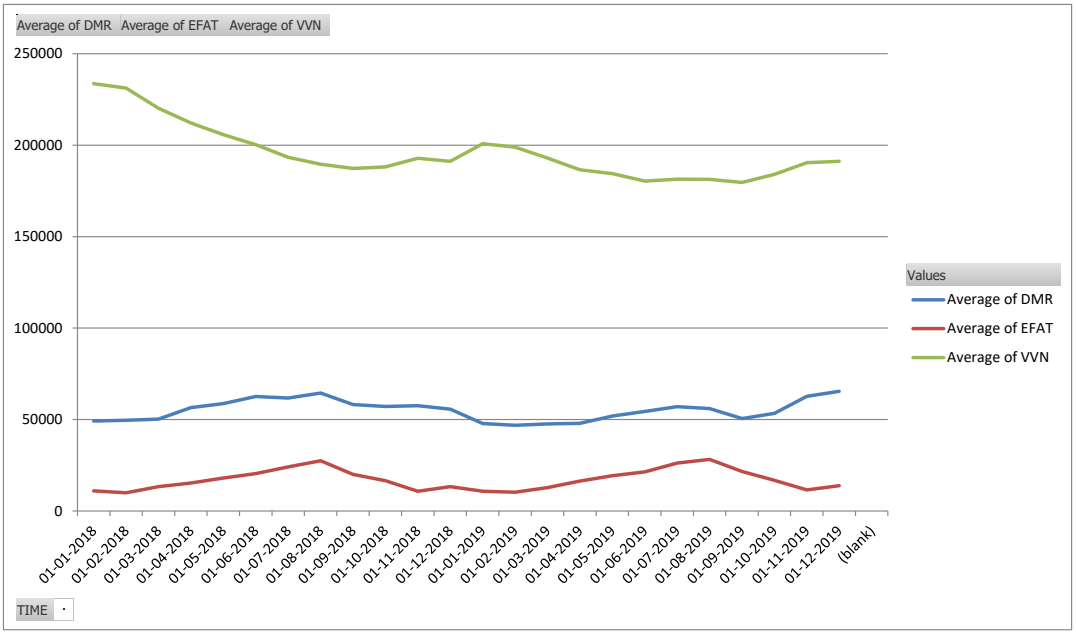

Figure 8.2 – Small accommodation: evolution of DMR, EFAT and VVN throughout the period of analysis

Correlation

The correlations between EFAT and other variables of interest in our existing datasets,VVN - from the business register and DMR – from the social security dataset, show the degree of closeness between the different sources. We may state a stronger association between EFAT and VVN, when compared to other correlations.

Table 8.3 – Correlations between EFAT (FTD), DMR (Social Security) and VVN (Business register).

	EFAT	DMR	VVN
EFAT	**1**	0.58	0.73
DMR	0.58	**1**	0.65
VVN	0.73	0.65	**1**

Modelling

Linear mixed models (LMM) have been used in addition to the Panel Regression Trees described in Section 8.5.

In LMM, we combined fixed effects with random effects (effects at the subject level)[31]. This model assumes linearity, so it is necessary to check in advance whether data should be transformed before using the LMM model. Furthermore, there are specific R packages for analysing LMMs, such as *lme* and *lme4,* and LMM can be used in datasets with a size in the order of thousands of subjects.

A first overview of the residuals, taking into account the model: DMR ~ EFAT is presented in Figure 8.3. The first five figures of the panel indicate that the assumptions of normality are reasonable. The last figure indicates that the model is well adjusted, that is, the actual DMR values are close to the fitted values of the model.

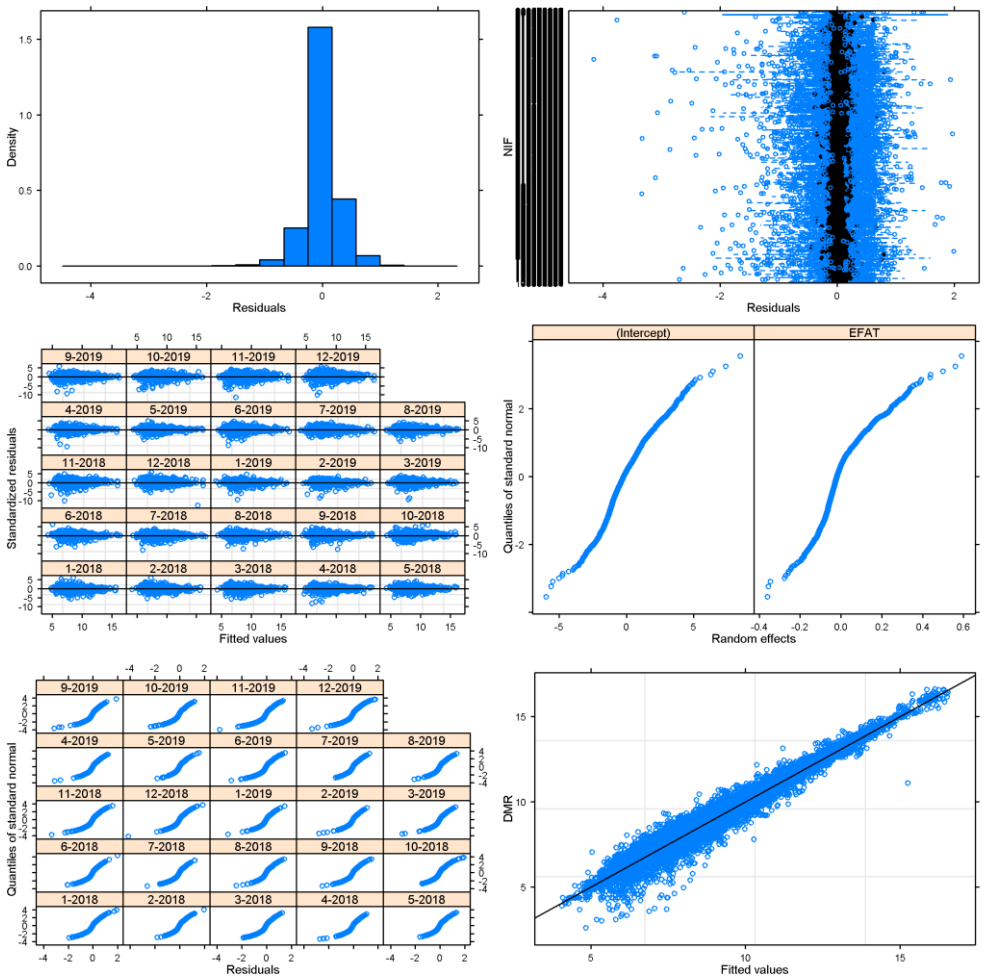

Figure 8.3. Residuals of the model DMR~EFAT. These graphs indicate that the assumptions of the model are met

[31] The fixed effects are the usual independent variables in a regression analysis.

The model fit of the model shown in Table 8.4 is carried out with the following mixed effects: TIME as a fixed effect : NIF (the VAT payer number) as a random effect.

Table 8.4. The model fit DMR~EFAT+TIME+EFAT:TIME

```
Model fit:
            numDF denDF  F-value p-value
(Intercept)     1 39196 68919.71  <.0001
EFAT            1 39196   407.66  <.0001
TIME           23 39196   165.20  <.0001
EFAT:TIME      23 39196     4.93  <.0001
```

An apparently good fit is provided by the LMM. Therefore, the Taxable value EFAT (aggregated by VAT Number) is strongly correlated with the DMR, since the remunerations paid seems to very well explained by the other variables, mainly by EFAT.

For the Panel Regression Tree, a new variable is considered: tipo, based on the type of business: 1=individual (one-person business(?)); 2= collective (.....); 3= other.

Results and corresponding plots have been obtained for the following model, where we capture the effects of EFAT on DMR: `DMR~EFAT+TIME+VVN+Tipo+NIF`

Random effects are TIME and NIF (the VAT payer number).

For each node of the tree (numbered 1 to 31), we can see the predictors VVN (the turnover from the register database), EFAT (the taxable value from E-invoice), and TIME and the results of the predicted variable (DMR, the Social Security variable of the monthly statement of remunerations) in red, corresponding to the value represented on the right-hand side of each row.

In the following, the regression tree is described and depicted. A regression tree is interpreted in this way: The bottom lines of Figure 8.4 represents data subset categories (defined by combinations of predictor values), each of which represents a certain impact onto DMR. Also, by starting from the top of Figure 8.4, following the line/branch (data subset) downwards, each division into two branches represents the most important impact by a predictor onto DMR for the subset that is split. At the top, the first division is when VVN < 5.52e+6. The vertical line on the left side represents this situation and for this subset, the next split into branches is when VVN < 1.129e+6, and when this is true, we see that this line (branch) ends, i.e. that none of the predictors are relevant for dividing this subset further for predicting DMR. In red colour we have the corresponding average of the DMR in each combination of variables.

```
1) root 39740 7.739838e+15    55241.50
   2) VVN< 5530103 39620 1.301068e+15    35931.39
     4) VVN< 1129186 38640 1.201915e+12    17336.82 *
     5) VVN>=1129186 980 7.597367e+14   769088.50
      10) EFAT< 242795.1 829 1.430607e+14   644402.00
        20) VVN>=1279605 717 4.661895e+12   568988.10 *
        21) VVN< 1279605 112 1.082162e+14  1127185.00
          42) VVN< 1254658 91 1.392064e+11   655343.30 *
          43) VVN>=1254658 21 2.485298e+10  3171834.00 *
      11) EFAT>=242795.1 151 5.330305e+14  1453626.00
        22) VVN>=3113497 59 4.128383e+10   333512.90 *
        23) VVN< 3113497 92 4.114925e+14  2171959.00
          46) EFAT< 446526.4 84 1.592379e+14  1662294.00
            92) TIME=3,6,7,8,9,12,16,17,18,19,20,21,22 77 1.113536e+12 1249895.00 *
            93) TIME=4,5,10,15,24 7 9.769426e+11  6198683.00 *
          47) EFAT>=446526.4 8 1.328394e+12  7523437.00 *
   3) VVN>=5530103 120 1.546270e+15  6430798.00
     6) VVN>=8136730 24 3.474157e+10  1490389.00 *
     7) VVN< 8136730 96 8.140063e+14  7665900.00
      14) VVN< 7274720 72 4.904875e+14  6786305.00
        28) VVN>=6335943 24 1.293220e+11  3513749.00 *
```

```
      29) VVN< 6335943 48 1.048117e+14  8422583.00
        58) TIME=1,2,3,4,12,13,14,15,16,24 20 4.650248e+11  6685743.00 *
        59) TIME=5,6,7,8,9,10,11,17,18,19,20,21,22,23 28 9.198545e+11  9663183.00 *
    15) VVN>=7274720 24 1.006969e+14 10304690.00
      30) TIME=1,2,3,4,5,6,9,11,13,14,15,16,17,18,21,22,23 17 3.723618e+10  8992593.00 *
      31) TIME=7,8,10,12,19,20,24 7 3.158384e+11 13491190.00 *

[1] "Estimated covariance matrix of random effects:"
            (Intercept)
(Intercept)  1894134490
[1] "Estimated variance of errors: 240676276.658473"
[1] "Log likelihood:  -483113.66809398"
```

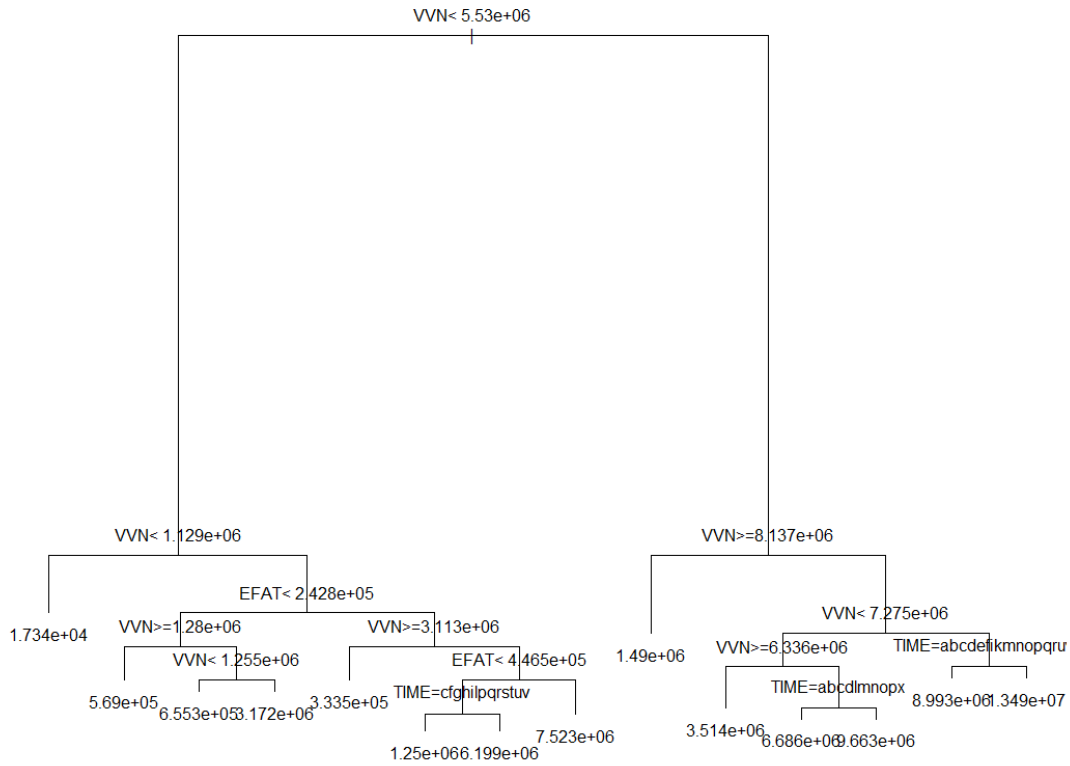

Figure 8.4. Panel Regression Tree - text and picture versions (months during the period 1 to 24 are represented by letters A to X).

Based on the results above, we may see the impact of EFAT on DMR, and also how other variables are either 1) contributing to this impact (as interaction terms) or 2) having a direct impact on DMR. When EFAT increases, DMR increases in the same sense, which we can see from the tree itself. The two main branches in the tree (uppermost branches) correspond to lower and higher values of VVN. For lower values, we can see that the months with higher corresponding DMR values within the 24 month period of analysis are: 3, 6, 7, 8, 9, 12 ,16, 17 ,18, 19, 20, 21, 22. It means that in December (month 12) and in the whole summer (months 6,7,8, 9 and 18, 19, 20, 21) the values of DMR increase. For higher values of VVN, the months with higher values of DMR are 7,8,10,12,19,20,24, which confirms the trend.

An evaluation of the panel regression tree has been made, based on cross validation (holdout method), where the data set is separated into two subsets, called the training set (70% of the original data) and the testing set (containing the remaining 30%) . The model fits a function using the training set and then we predict the output values for the data in the testing set. The Mean Absolute Error (MAE) has been used to provide a measure to evaluate the model. In this case, MAE is approximately 75000. The corresponding Root

Mean Square Error (RMSE) is 12000. For a comparison and utility of MAE and RMSE see Chai and Draxler (2014).

8.5.2 Car Ride-sharing services – TVDE

A list containing 6600 TVDE vehicles were extracted from the Business Register and followed during the period under analysis (January 2018 until June 2020, i.e. a 30-month period) was considered. We found 691 missing values in the variable DMR. So, the remaining 5909 companies have been used in the analysis. We started by observing the evolution of the variables during the period of analysis. We can clearly see the effect of the crisis caused by the COVID-19 pandemics during months 26-30 (February 2020 to June 2020)

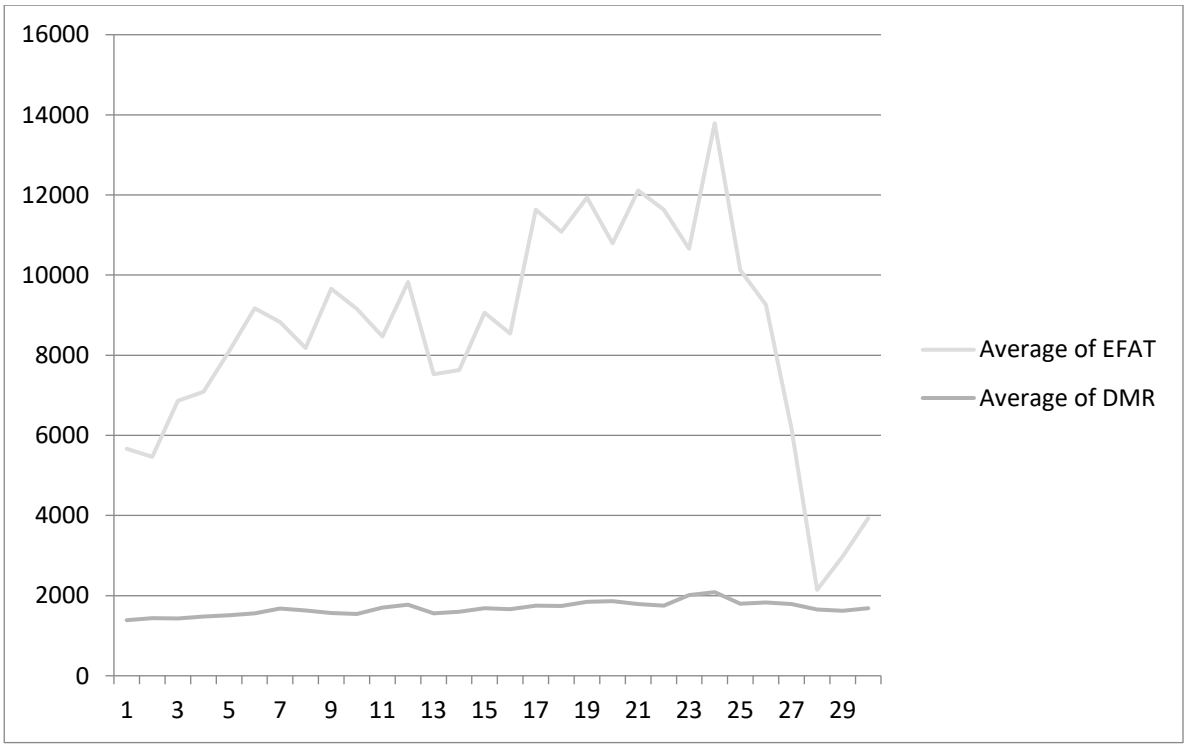

Figure 8.5 – TVDE: evolution of DMR and EFAT throughout the period of analysis.

It seems that at the end of this period of analysis, the fall of the retail (measured by EFAT) was not followed by a corresponding fall in DMR, since many companies continued to pay their workers' wages.

Correlation

Following the previous section we proceeded by finding the correlation between the businesses'attributes of EFAT (transaction value) from the FTD, with of DMR (remuneration total) from the social security system. We see that the correlation between DMR and EFAT is strong.

Table 8.5 – Correlations between EFAT (FTD) and DMR (Social Security).

	EFAT	DMR
EFAT	**1**	0.61
DMR	0.61	**1**

Modelling

Again, two different types of models, Linear mixed models (LMM) and Panel Regression Trees have been used for this analysis

A first model using LMM was produced, DMR~EFAT+TIME, and with NIF (VAT number) as random effect, wit h the following results:

```
Fixed effects:
             Estimate Std. Error t value
(Intercept) 1.130e+03  1.715e+02   6.591
EFAT        2.412e-02  1.522e-03  15.847
TIME        1.562e+01  1.270e+00  12.301
```

We see that the effect of EFAT on DMR is positive, although weak. TIME exerts a stronger effect on DMR.

Next, we produce a regression tree to analyse de predictive capacity of EFAT onto the DMR. We applied RE-EM trees to the data using the model DMR~EFAT+TIME, and obtained the following results:

```
1) root 5909 25322870000  1683.791
  2) EFAT< 67852.12 5857  3144705000  1501.505
    4) EFAT< 18972.75 5185    99173340  1248.980 *
    5) EFAT>=18972.75 672   163737300  3449.930 *
  3) EFAT>=67852.12 52    63038370 22215.430 *
[1] "Estimated covariance matrix of random effects:"
            (Intercept)
(Intercept)     3826952
[1] "Estimated variance of errors: 488256.482374562"
[1] "Log likelihood:  -53508.0091937246"
```

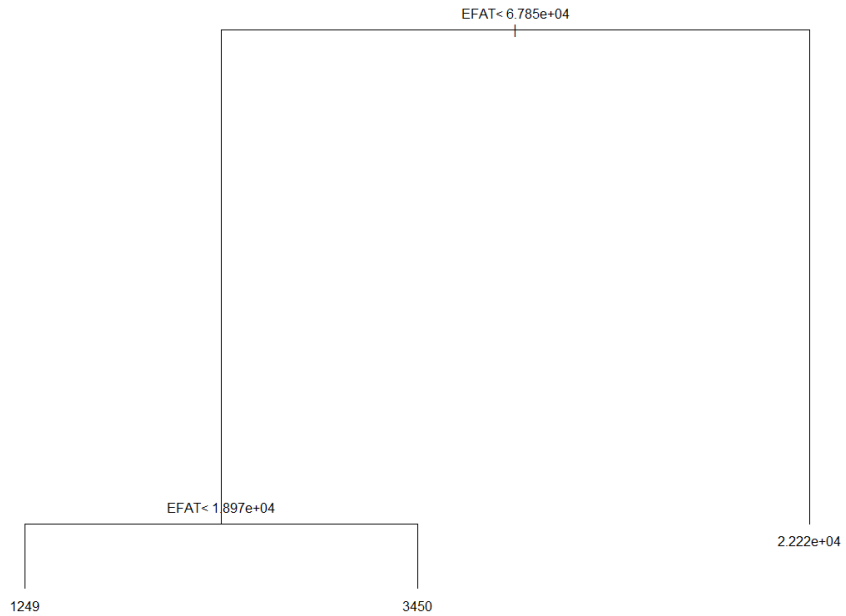

Figure 8.6 –Panel Regression Tree (months during the period 1 to 24 are represented by letters A to X).

It is possible to observe a similar trend, when comparing these results with the section before: when DMR increases, EFAT increases as well and shows a good predictive capacity of this variable. The regression tree shows that EFAT is the best predictor. Furthermore, we can also state that the prediction abilities of the model are higher than in the previous case.

The measure MAE, based on cross validation, has been applied here and a value of 1100 was obtained, much more lower than the one obtained in the model of the small Accommodation services. The Root Mean Squared Error (RMSE) is 90.

8.6 Discussion and conclusions

This case study sought to analyse the relationship between some variables used in current statistical production and a new source: the tax authority's E-invoice. Based on companies from two specific sectors linked to the shared economy (Local Accommodation Services, and Car-ride sharing), we sought to understand whether the sales of companies registered in the E-invoice have any relationship with other company variables, such as the remuneration paid to workers (DMR). Two different time periods have been observed, one for each sector: in the former case a period of 24 months from January 2018 to December 2019 was analysed. In the latter, the period allowed us to consider the COVID-19 crisis because it covered 30 months from January 2018 to June 2020.

It was noticed that the variables are closely related. The relationship between EFAT and DMR makes it possible to consider that we may use these variables as predictors in the future. Although the linear mixed models are generally good, the panel regression trees models applied in the two different sectors (Local accommodation and Car Ride-sharing services) differ. The former model shows a lower quality than the latter.

Table 8.6 – A summary of the panel regression tree evaluation measures.

	MAE	RMSE
Local accommodation	75000	12000
Car Ride-sharing services	1100	90

Furthermore, it seems to us that at certain times, namely in times of COVID-19 crisis, wages cannot be measured by EFAT, as despite the fact that the volume of sales has decreased, many companies continued to pay their workers' wages.

We understand that incorporating FTD from the external sources mentioned above in the production of official statistics needs to be better explored. We think that the need for further studies, namely in detecting EFAT outliers, is necessary. In addition, it is also needed to learn how this information can be incorporated into the forecasts. Perhaps, more than predicting future values, it makes more sense to use external sources to predict the trend of future values, so as to be able to anticipate information and view on the economic situation and short-term indicators.

Part II.2 – Sharing economy platforms (SEP)

9. Definition of Sharing economy and Sharing economy platforms (SEP)

Sharing economy is a phenomenon as old as human beings are. People always shared possessions, efforts, etc., whenever these assets were available from their regular use. However, before the internet boom, people tended not to share with strangers outside their local community. With the internet expansion, sharing with strangers turned possible and the number of participants in the sharing of a single asset, as well as the variety of goods and services subject to sharing, has increased substantially. Development of technology extends the assets subject to sharing. For example, the penetration of distributed energy systems opened doors to share electricity produced on sunny roofs with houses in a position less suitable for solar energy production. Technology development made it possible for households to produce and share electricity. All the changes due to internet and other technology improvements, lead to the current sharing economy that we are now interested to measure in official statistics,

The changing technology not only extends the assets but also has led to new business models that to some extent displaces traditional business models. The sharing economy reduces substantially transaction costs and brings about profound changes in labour market and income distribution. These changes have brought to life new business models like economy on demand, gig economy, etc. but digital technologies have been profitable for some and disruptive to others. On the one hand, Uber, Lift, Airbnb and likes have constantly increased their market shares in taxi services and tourism. On the other hand, last year Thomas Cook declared insolvency after 178 years in business. Online technologies underlain Thomas Cook demise. Almost everyone could design her holiday online and does no longer need Thomas Cook's service to book tickets and accommodation.

The changes in the business market due to sharing economy and related business models poses challenges to regulators. The regulators are keen to regulate the business market to eliminate free riders from the market but the sharing economy allows for digital platforms to become market incumbents bypassing industry regulations. For instance, many tourist accommodation establishments are joining Airbnb and other platforms for short-term renting in order to have access to the advantages of an online marketplace. As a result, both platforms and renters save a lot of money that traditional hotels have to pay to comply with industry regulations. Similar developments can be observed in other industries, like the taxi industry.

An additional issue is that of the social – not just economic – consequences of the diffusion of the new "sharing" business models. Many empirical studies found out short term renting via Airbnb, Homeaway, VRBO, etc. underlies higher prices for long-term renting. A rush in short term renting reduces the supply of houses for long term renting and prices go up. Uber, Lift and likes made possible many independent drivers to enter taxi services market. The result is increasing the share of self-employed in this industry and reducing that of employees. However, the process of reducing employees/self-employed ratio is far from being smooth. Uber spent a lot of time in the courts around the World pretending to be just a technology company and rejecting the view of having a direct control on the operations of its cars and drivers. It was argued that the company just provides drivers and passengers with an app to connect each other. On the other hand, drivers pretend that they have little control on the transactions: Uber controls the price, Uber controls clients, Uber penalises drivers when someone complains, and drivers are not allowed to take customer who hails them on the street. Shortly, they are subject to too much control to be self-employed. In a sense, Uber drivers are somewhere between employees and self-employed but the European Court of Justice finally ruled in 2017 that Uber is a transport company, not just a technology platform.

According to recent UNCTAD data[32], in 2018 more than 1.4 billion people at global level did buy goods or services online for a total amount of around 25.6 trillion US dollars. Eighty percent of the total sales could have been managed by digital platforms even though most enterprises do sell via their own websites too. In the EU, 60% of citizens – according to Eurostat - have purchased goods or services online during the year 2019 (with great differences across EU countries: e.g. 22% in Bulgaria and 79% in Germany) and 20% of enterprises have sold their output online (42% among large firms). An increasing number of customers and firms are getting in touch within the largest online marketplaces: those digital platforms that are profiting the most from the diffusion of the sharing economy. Last, by not least, the COVID crisis is expected to have increased – largely on a permanent basis – the penetration of digital platforms in a number of key industries (food, consumer goods, entertainment, education, etc.) albeit negatively affecting some well-established digital businesses like intermediation in tourist accommodation or flight transportation.

Overall, the demand for statistics on the above-mentioned market and social consequences of the sharing economy is on the rise and NSIs have to compete with a broad range of data companies and institutions in monitoring such processes. Main users of sharing economy data are, of course, policy makers and regulators who are dealing with a constantly evolving phenomenon by using normative and regulatory frameworks designed in a different economic period and by relying only a partial and often outdated data. There is a clear information asymmetry between public authorities and digital platforms which makes the provision of timely data a major issue in the on-going negotiations about the limits should be applied to the activities by digital platforms[33]. Thus, the demand for data by public institutions about the sharing economy is essentially about:

- The identification (mostly, profiling data) of the national residents – both individuals and businesses – selling or buying goods or services via digital platforms.
- The income generated the the digital platforms' activities in a territory and the information about who should be charged of the related income taxes.
- The intensity of such activities at industry or territorial level in order to prevent negative effects at economic (market – including labour market - distortion or disruption) or social (e.g. cities' touristification).

Other potential users of sharing economy data include business people and researchers/analysts. Both groups have the need of defining the relationships between the actors of the sharing economy and the existing infrastructures, either economic infrastructures (suppliers of goods and services, logistic and trade networks, financial institutions, etc.) or cultural infrastructures (economic and social theories, conceptualization and classification systems, policy objectives, etc.). Thus, the demand of indicators by these groups of users will mainly focus on[34]:

- Indicators on the economic impact of digital platforms (sales, employment, trade, value chains, etc.).
- Indicators on the social impact of digital platforms (changes in behaviour by customers/users, digitalisation of procedures and relationships among individuals, production and diffusion of knowledge and culture, etc.).

[32] https://unctad.org/press-material/global-e-commerce-hits-256-trillion-latest-unctad-estimates

[33] As an example, in 2018 Danish tax officials and the international home rental site Airbnb have announced a World-first agreement that will ensure that short-term rental income in Denmark will be automatically reported by the company to the authorities.

[34] The description given in this paragraph is based on the findings of the literature review undertaken as a source of data the WPG's Deliverable 1 had to rely upon. Most of scientific papers taken into consideration were focusing on a single platform's (or a few platforms) activity by highlighting specific issues like: market conditions, customers' relations, employment policies, business models' features or the impact of digital platforms on local communities/urban planning.

- Indicators on the impact on territories/environment (seasonal, including touristic, or permanent displacement of people, Internet-related globalisation and environmental impact, sustainability of the sharing economy paradigm, etc.).

What are the relevant data sources for producing indicators related to sharing economy? Surveys are burdensome for the respondents, so alternatives are welcome. Some of the above statistics could be produced using administrative registers as the source. For example, administrative registers keep records of social security payments of both employees and self-employed. As another example, many cities across the EU and the USA put in place registration of short-term rentals, and this is a source for new or existing administrative registers. An alternative source to administrative registers on sharing economy are the mediating platforms. For example, the number of houses available for renting could be assessed from Airbnb, or Homeaway websites by web scraping. Another example is to address the platforms directly asking for data on their business activity. A third type of source for sharing economy indicators is the FTD that we discussed in Part 1. We will mention some issues related to the use of these sources in Section 6.

Literature on sharing economy show that there is no a single definition of sharing economy used by economists and business people (Codagnone & Martens 2016; p.8). This might seem strange in light of the far origin of the sharing economy concept. Also, there are other business models that are overlapping with sharing economy, such as collaborative economy and economy on demand, the latter often referred to as 'gig economy'.

We will adopt a definition of sharing economy that is suitable for the use of SEP data or other sharing economy FTD in the production of official statistics, either for improving, quality assessment or replacing existing sources for official statistics. Our definition of sharing economy should be focused on official statistics and accommodate emerging models of sharing.

Our basis for the definition of sharing economy is the literature on sharing economy and on SEP. However, there is a vast amount of literature on this topic, and our ambition is not to cover all relevant literature, but instead to study enough literature to be able to make a suitable definition.

A *theoretical definition* describes what sharing economy is in terms of economic theory. However, the *operational definition* focuses on the measurement of sharing economy for official statistics production and is thus our target. The relation between theoretical definitions and operational definitions are studied in Appendix 8. Below, we will first discuss a theoretical definition of sharing economy. Then we will discuss different operationalisations.

9.1 Theoretical definition

Frenken & Schor (2017) defined sharing economy as persons *granting each other temporary access to under-utilised physical assets ("idle capacity"), possibly for money.* Important fundamentals of sharing economy are temporary access, idle capacity and that the ownership of the asset is not changed.

When renting a spare seat of a car ride or renting a car/apartment during a week when the owner doesn't use it, the rented asset would otherwise not be used. The notion of underutilised capacity is central when drawing a border line between sharing economy and economy on demand. When the house owner rents out while being away on vacation or business trip, or he/she has one or more spare room, the asset is underutilised and he/she rents it out as a sharing transaction. When the owner buys a second house and rents it around the year this is an accommodation business and no sharing is involved. The same reasoning could be applied to ride-sharing.

The definition by Frenken & Schor (2017) points to certain patterns that may or not be defined within the scope of sharing economy and related economies:

- whether temporary access that is paid by money is a part of sharing economy: required (R), allowed (A), or not allowed (N);
- whether businesses should be allowed (A) to be one of the two peers in the transaction or not (N). In the latter case, both parties must be persons;
- whether non-physical goods (services) are allowed (A) or not (N) to the above definition;
- whether the existence of a platform mediating between the two peers is required (R) or allowed (A).

Above, *required (R)* means that without the presence of this property, the transaction is outside the definition.

Frenken & Schor (2017) argues against allowing services, which they argue are *on demand economy* or *gig economy,* and which lack the property of using an asset that would otherwise not been used. Then, if we represent the definition's relation to these four patterns in a vector, we can represent the definition suggested by Frenken & Schor (2017) as (A,N,N,.), i.e. not considering the fourth pattern. An extension of Frenken & Schor's definition is to add "A" to the fourth pattern, i.e. (A,N,N,A).

EU Commission (2016) considers that 'sharing economy' is often interchangeable with 'collaborative economy' and its definition of the latter considers "…activities are facilitated by collaborative platforms…" for temporary usage of goods but also services, that "… generally do not involve a change of ownership and can be carried out for profit or not-for-profit". Its definition can be represented by the vector (A,A,A,R), where the 'R' means that any "activity" that is not mediated by a platform, falls outside of the definition. Thus, when a person A hires out her apartment to person B for a week directly, this is not a part of the collaborative economy.

We notice that both definitions above focus on making available under-utilised assets on a temporary base with no change in ownership. However, contrary to the definition of sharing economy, the definition of collaborative economy

- Allows businesses to be one of the parties in the transaction.
- Sometimes allows change of ownership.
- Allows services in addition to physical goods.
- Insists on the presence on a mediating platform.

We will adopt the exclusion of service from the theoretical definition of sharing economy and adopt the (A,N,N,A) as our chosen theoretical definition: when a person is not using a service, the service provider can use his/her time differently. Rising capital through platform is a relatively new way to rise funds from many supporters and by-pass traditional sources of capital like banks, venture capitalists, and likes. The motives of participants in the fund rising platform (capital seeker and supporters) are a mixture of fund raising, hefty return, expanding awareness, being part of a community, support a cause, etc. (Gerber & Hui 2013). None of these has to do with sharing underutilised assets.

9.2 Operational definition suitable for official statistics production

We need an operational definition that, in theory, lets us measure the sharing economy output using different sources of information: FTD, SEP data or other sources, for example administrative data. The part of the economy that is not related to monetary transfer, are usually kept out of the economy since it is difficult to measure. A famous joke was made by the 1970 Nobel prize winner Paul Samuelson: "When a man marries his maid GDP falls" refers to the difference in production boundary in economic theory and in national accounts.

We start the operationalisation by adopting an operational definition where

- the transaction for the temporal access should involve payment in money. The sharing is then done *for profit* which is here defined as covering at least the costs, e.g. fuel consumption.

An argument for allowing businesses to be providers is the difficulty of separating legal persons from natural persons. Remember that self-employed and one-person companies share characteristics of both natural and legal persons. By allowing businesses to be providers of services, our operationalisation extends the theoretical definition of sharing economy that we chose above.

We now have two options for the operational definition: One is to include businesses and services inside the scope of SE, which would lead to defining sharing economy as the bluish circle of Figure 9.1. This definition is closely related to definition of collaborative economy as described in Section 9.1. The other is a narrower definition in line with the theoretical definition, except for assuming that all transactions have to include money transfer which is necessary in order to be operational. Thus, our second alternative is to restrict sharing economy to the clear blue part of Figure 9.1.

The first alternative described above (bluish part of Figure 9.1), means including transactions that constitute economy on demand. This is a much more pragmatic approach that has the advantage of not having to differentiate between whether the receiver is a private person or a business, and whether the object of transaction is a good or a service. Neither of these differentiations can be easily implemented, it is e.g. difficult to assess whether the dwelling rented out was bought for a business purpose (business activity) or it is used by the owner and just rented out when being empty (non-business activity).

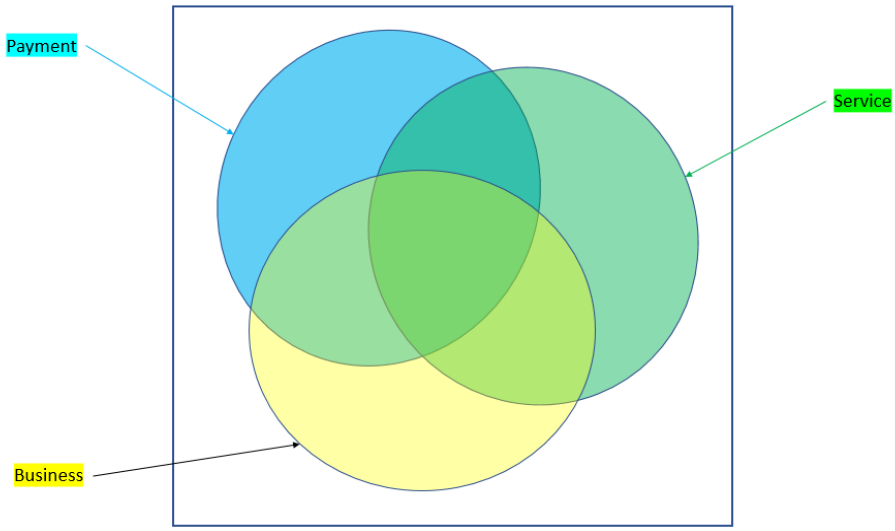

Figure 9.1. Illustration of the contents of the sharing economy concept depending its inclusion/ exclusion of payment (blue), businesses (yellow) and service (green).

The difference between the narrow and wide operational definitions are summarised in Table 9.1.

Table 1. Theoretical definition and operational definition of sharing economy.

Narrow operational definition	Wide operational definition
People grant each other temporary access to underutilised assets possibly for money	People share assets for money.
Participants: sharing took place between natural persons	Participants: sharing took place between natural and legal persons
Shared transactions are negotiated directly between natural persons or an online platform mediates transactions.	Transactions are always mediated by an online platform. Only transactions mediated by an internet platform are subject to measurement.
Assets subject to sharing are temporarily underutilised. Example: a household rents spare rooms during vacation season.	Assets are built to be rented out. Example: a household build a house and rent it out all the time if possible.

9.3 More on definition of the platforms

Considering the scope of the term 'SEP' (sharing economy platforms), we will include any internet site that is acting as mediator of sharing economy transactions. Notice, however, that from the perspective of the narrow definition in Section 9.1, many SEPs are also mediating in transactions not being part of the sharing economy, and some SEPs are even *mainly* intermediating in non-sharing economy transactions. Difference between Blablacar and Uber are instructive. Both companies provide shared car-ride services. However, business models are quite different. Blablacar employs cost sharing business model. Drivers make no profit but share costs with strangers for gas and car wear and tear. Uber drivers are independent contractors who work for profit. Blablacar transactions fall into sharing economy domain, Uber transactions are frequently considered to fall into on-demand economy domain.

9.4 Conclusions

As a theoretical definition, we chose that of Frenken & Schor (2017) since it restricts the definition to idle/excess capacity goods, excluding services and sharing. However, for the operational definition, we need to choose an alternative that could work in practice. We have a narrow alternative and a wide alternative, where the latter is closely related to the definition of collaborative economy. Which alternative to choose depends on the usefulness of each alternative and will be studied in Section 10.1.

10. Sharing economy platform (SEP) data: data accessibility and SEP-related indicators based on FTD

10. 1 Introduction

By considering the difficulties in defining and delimiting the SEP domain for statistical purposes, the definition of a methodological framework aimed at identifying the potential sources for new SEP-related indicators[35] (and designing an effective production process for them, from data access to data processing and quality checking) has to be seen as highly challenging.

Some questions might help to frame the issues currently at stake when deciding on an operational definition:

- Which platforms exist and if they can be classified into operational categories.
- Under which conditions SEP-related data can be accessed/used.
- Whether a set of SEP-related metadata can be produced from the accessed data.
- Which uses in official statistics are more promising for SEP-related data.
- Whether empirical analysis of one or more SEP-related FTD (Financial transactions data) datasets could be performed.

10.1.1 The options developed by the WPG members

Notice that in Section 9, the "theoretical definition" as well as a "narrow operational definition" of sharing economy on one hand, and the "wide operational definition" loosely corresponding to that of "collaborative economy" on the other hand, represent two different points of view on what the SEP field essentially is about:

A. Activities whose main purpose is that of allowing the sharing of under-utilised assets among individuals and households through the neutral (free) intermediation of technology platforms; or
B. Activities performed, for profit, by technology platforms implementing a business model based on the intermediation, or sharing, between peers (a provider and a user, i.e. operating in a double-side market).

These two definitions imply addressing mainly two subpopulations within the whole population of actors in the SEP arena: subpopulation A consists of web platforms dealing with only C2C sharing activities; subpopulation B consists of large multinational (less commonly, national-only) platforms featuring a combination of C2C and B2C sharing activities. Therefore, different strategies must be taken into consideration in order to set up a consistent measurement framework:

1. Adopting a narrow definition and then limiting the analysis to 'pure' sharing platforms (possibly, at country level), i.e. subpopulation A, being the only subpopulation to allow for the straightforward identification of platforms' activities, being within the narrow definition.
2. Adopting a wide definition and then allowing the analysis to cover all digital platforms operating in a country, irrespective of where their headquarters are located, i.e. the whole population.

A choice between these two alternatives has great consequences in terms of methods and data sources to be used, as well as on the comparability of the results that could be achieved in EU countries (at both pilot and full-scale level). In principle, alternative 1 covers only a part of the population and will lead to bias since it cannot be assumed that the C2C part of subpopulation B would be fully represented by subpopulation A. Alternative B will, on the other hand, cover the whole population of A and B actors, thus supports choosing the wider definition instead of the narrow definition.

[35] or Statistics.

Data collected directly from platforms ('direct approach') are the most relevant data for SEP analysis and thus preferable as far as it is possible to access the data. Assuming that data were to be collected directly from platforms as a WPG task, only a small sample of platforms could be realistically addressed. And alternative 1 is the only feasible alternative that could allow for a good access to platform data relying on both legal obligations and moral suasion (and then only to national platforms' data, since national legislation is not working on multinational companies with the main office in another country). However, alternative 1 could reveal only national patterns of SEP activity and thus there will be the risk of deriving a model of quantitative measurement of the SEP phenomenon that would not be relevant for other countries in the ESS. Further, even in a national perspective, it is important to select platforms that could be representative for the population of national B2C platforms, as well.

Instead of adopting alternative 1, we could adopt alternative 2 but then implementing indirect methods for data collection as e.g. survey among users since the NSI cannot demand data from the multinational platforms using the Statistical Act. We could focus on a small number of big players operating, for instance, in all (or almost all) EU countries, and thus allow for the access to a common evidence base for the whole ESS (which is the current Eurostat approach). However, the advantages of using this approach (few large platforms) could turn out to be partially or totally balanced out if the selected platforms fail to be representative of the population: heterogeneity is often observed in the use of the services delivered by international digital platforms[36], and then the sample can easily lead to biased conclusions.

Table 10.1 is aimed at describing how the feasibility (rightmost column) of data collection types (the leftmost column "DATA TYPES") of a digital platform is affected by two factors: (a) the typology of platforms under observation ("Target population" in the table); (b) the availability of direct vs. indirect indicators. For instance, by comparing the two data collection methods "survey on platforms' users" and "data from platforms", several differences can be observed. Data collected from a given population of platforms allow for a full enumeration of the target population, although for practical purpose only a sample of platforms would be usually selected. A survey among users will be affected by the sample variance of the amount of use of different platforms. Moreover, the survey approach leads to indirect indicators since only the provision of data from the platforms themselves can generate direct indicators. Finally, it has to be pointed out that surveys are always constrained, in scale and scope, by the amount of resources available. For the "get data from the platforms" option, on the other hand the costs of dealing with platforms should not be trivialised: to find an agreement with one or more digital platforms in order to share some data, more or less aggregated, about their business activity implies a number of legal and managerial issues not easy to deal with.

Data potential and access to sources are key factors to be taken into consideration when defining a data collection strategy. Still, with reference to Table 10.1, a preference should be given to sources that could make available relevant information while being easily accessible by users.

Overall, it can be argued that large platforms are already diffusing a big amount of information on their activities (to be potentially addressed through a range of sources), while small local platforms need to be directly addressed to collect some relevant information. From a broader perspective, a trade-off has to be noticed between the potential of a direct access to platform data and the reluctance of platforms to allow such an access to users. This is an essential point to be considered when defining a data collection strategy: the more a direct access will become difficult, the more the alternative sources will become convenient and attractive in order to produce statistical indicators.

[36] With a higher comparability when data are centrally collected and checked by a single institution as currently tested by Eurostat with reference to the activities in EU countries of big multinational platforms offering accommodation and travel services.

A trade-off between accessibility and informative contents sets the context where data sources can be compared. It is essential to find the right balance between the resources needed to extract and process the data and the informative contents of the expected output.

Three options have been compared by WPG members in order to define a suitable strategy to produce SEP-relevant indicators:

a. Investigating if - in the literature – knowledge and issues about SEP transactions are addressed with reference to: accessibility of data, data infrastructure and potential indicators.
b. Collecting SEP data from various data sources according to national interests and specificities.
c. Developing some FTD-based SEP indicators as a basis for a quantification of platforms' economic impact.

Table 10.1. Target populations of digital platforms and potential data sources. "Relevant", the data source has been already used in one or more studies and has proven to be effective in measuring the phenomenon; "Partially useful": examples can be found in the literature about the use of the source but only to produce proxies of the desired indicators; "Irrelevant": the data source is either unavailable or uncorrelated to the phenomenon to be measured

	TARGET POPULATIONS			INDICATORS	FEASIBILITY
DATA SOURCES	Small local platforms	Large multinational platforms	Both populations		
Surveys of individuals (users)	Partially useful	Partially useful	Partially useful	Indirect indicators	Affected by resources and time constraints
Surveys of enterprises (providers)	Irrelevant	**Relevant**	Partially useful		
Public administrative data (e.g. fiscal data)	Partially useful	**Relevant**	Partially useful		Confidential issues to be considered
Financial trans. data (FTD)	Irrelevant	Partially useful	Partially useful		
Web data collected from platforms	**Relevant**	**Relevant**	**Relevant**	Direct indicators	Agreements hard to set
Web data collected by web scraping/mining	Partially useful	Partially useful	Partially useful	Indirect indicators	Data protection issue to be considered

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10.1.2 Option a. The literature review

Option A is the one initially envisaged when the WPG was approved. Later, some questions were raised about the actual availability of a substantial amount of research activity whose results could be effectively used to meet the WPG requirements. All contributing countries have been involved in this activity and encouraged, in parallel, to identify the mixed population of large and small (local) platforms in their country.

A further issue is that about how scientific papers on one hand and non-scientific informative literature on the other hand could be used for getting information on platforms' measurement. In principle, only scientific literature could be helpful for deriving (non-systemic) indicators, as 'grey' literature or media sources do not

match the basic requirements to be used as reliable sources for development of indicators in the official statistics framework[37].

Thus, by considering both small and large platforms, WPG members have realised that a scientific literature review could provide useful - although very partial - material to support future NSI activities about:

- Digital platforms' data accessibility.
- Metadata detailing the use of data for digital platforms' measurement.
- Development of indicators which could feed official statistics activities.

The participants of WPG searched for scientific papers, and also other literature, on data accessibility and platform metadata. The conclusion was that the existence of papers is too limited to pursue further the original plan of a literature review of accessibility and metadata.

A preliminary screening of scientific articles made available in *ScienceDirect*[38] over the last five years (2015-2019), carried out by Istat, gives a preliminary (and probably partial) picture of the most common options for producing quantitative evidence on the digital platforms' activities as a result of scientific research activities.

By focusing only on scientific articles describing issues related to the measurement of any phenomenon related to digital platforms (articles about jobs contracts, and jobs-related issues in general, have been excluded by considering them a field of its own), 48 articles have been identified. Among them, 35 articles are about AirBNB, 3 about Uber, 1 about the Mechanical Turk and 1 about Lyft. References to national-only or local platforms have not been found.

Not surprisingly, most of the articles are reporting about data accessibility, metadata and new indicators in a way that makes it difficult to identify a specific focus on one of the above-mentioned needs on data production and analysis. By considering AirBNB-related studies, almost half of them originate from concerns about the economic and social impact of short-term house renting in large cities. The source of data mostly used is the legal scraping of the AirBNB website. It is remarkable that other sources are used, as well: from surveys of AirBNB users, to social media where opinions on AirBNB are given and shared among users.

Although based on a limited evidence, the preliminary review allows for arguing that a literature review on digital platforms could hardly provide a knowledge base that would be useful for developing reliable experimental statistics in the ESS framework. Some points can be highlighted.

- A systematic review of potential sources of data on digital platforms can be hardly based on a literature analysis, because of the constrains researchers have in accessing relevant databases. By definition, a whole range of data sources – chiefly, administrative data – are only accessible for NSIs.
- Most of the research activity focuses on the 'local impact' of 'multi-national platforms'. The poor attention paid to local platforms is a minor issue compared to the limitation of such analyses to highly localised phenomena. This prevents a comprehensive approach to the measurement of platform-related phenomena which are, of course, locally differentiated (at least, as far as their impact is concerned). Nevertheless, they have common features which should be addressed when defining a broad (European) statistical framework.
- For a methodologically-sound exercise, the scientific literature in the field of economics (or, broadly, of social sciences) is of poor use as issues about how to process the available data are much more relevant in the literature than those about the extraction/collection of data from the original sources

[37] By taking AirBNB as an example, an assessment should be done whether commercial sources of AirBNB-related data, like AirDNA.co, or freely accessible repositories of (unauthorised) scraped data sources (like insideairbnb.com) could provide a useful basis for developing new official indicators by NSIs.

[38] ScienceDirect is a website which provides subscription-based access to a large database of scientific and medical research. It hosts over 12 million pieces of content from 3,500 academic journals and 34,000 e-books (https://www.sciencedirect.com/).

and their quality monitoring. Of course, data quality is an essential requirement when exploring new and unstructured sources of data for statistical purposes.

10.1.3 Option b. A non-systematic collection of platforms' indicators at national level

In addition to a set of studies produced by the European Commission[39], some European NSIs – while waiting for a harmonised initiative by Eurostat in this area – have started to address the issue of measuring the platform economy at national level.

In addition to a number of articles and press releases from EU's NSIs, we believe that the state of the art of such efforts with reference to the UK and Dutch cases, are best described in Office for National Statistics (2017) and Heerschap et al. (2018). Just considering these two papers, we can realise that most of the methodological information that should have been extracted from a literature review is already made available to users there in a comprehensive and usable form. The logical process undertaken by the authors has been, in fact, that of reviewing all available sources - either under direct control by or made accessible to the NSI – of information on digital platforms (or any other activity related to the sharing economy phenomena).

The evidence presented in these papers is not only available in The Netherlands and in UK but similar indicators are regularly produced by most EU NSIs as part of their current statistical activities. It implies that the data collection exercise done by British and Dutch colleagues could be easily replicated in other EU countries. As a consequence, a specific effort of methodological harmonisation at EU scale could substantially improve the prospects for launching a EU-wide data collection on SEP-related issues, mainly in specific areas of statistical production like tourism statistics.

In this perspective, the current activities by Eurostat of starting a process of co-ordination – restricted, for the time being, to tourism statistics – raise the question whether NSIs should invest in systematically collecting data on the activities of big platforms in their countries. In fact, these activities could be soon displaced as a result of a centralised data collection of big platforms' activity data under the Eurostat responsibility (already negotiated and partially agreed with some digital platforms with tourism-related activities all over Europe[40]).

By excluding the methodological assessment of current NSI's activities, on the one hand, and the development of new methods of harmonised data collection with reference to big platforms, on the other hand, the purpose of the WPG activity could have been that to refocus on assessing the availability of data on small, national or local, digital platforms. The argument for pursuing this approach is that this area (which should include the largest part of the narrowly defined sharing economy, c.f. Section 9.2) has been largely neglected so far although its relevant social, if not economic, impact.

10.1.4 Option c. Developing FTD-based SEP indicators

A third option taken into consideration to fulfil the requirements of WPG was that of assessing to what extent the two parallel streams of activity included in WPG could be bridged in some way. Two factors can be pointed out to support the idea of investing in this activity. First, the potential for exploiting synergies within the WPG group. As FTD experts and SEP experts are working shoulder to shoulder, the exchange of information and, even more, a common development of methods and practices is easier than with reference

[39] Several DG JRC papers can be mentioned (e.g.: Codagnone, C., Biagi, F. and Abadie, F. (2016). *The Passions and the Interests: Unpacking the 'Sharing Economy*. JRC Science for Policy Report, Institute for Prospective Technological Studies, EUR 27914 EN, doi:10.2791/474555; Codagnone, C. and Martens, B. (2016). *Scoping the Sharing Economy: Origins, Definitions, Impact and Regulatory Issues*, Institute for Prospective Technological Studies, Digital Economy Working Paper, 2016/01. JRC100369) together with the 2018 EC Report on *Study to monitor the economic development of the collaborative economy at sector level in the 28 EU Member States* (doi 10.2873/83555).

[40] https://ec.europa.eu/commission/presscorner/detail/en/ip_20_194

to any alternative source. Second, FTD have a potential as a source of indicators on SEP as financial transactions' management – in a context of large heterogeneity among platforms, peers and goods/services – is the only comparable process to be consistently undertaken by all platforms.

It has to be anticipated that the strict confidentiality rules to be enforced when processing FTD will make almost impossible to extract a set of transactions (micro-data) based on the ID/bank account codes of single platforms. This implies that the targeting of a specific sub-population of firms managing platforms is barely possible. On the other hand, large platforms can be hardly targeted by NSIs not just because strict confidentiality rules but rather because of their multinational structure, almost impossible to monitor from a purely national perspective.

Some positive alternatives will include the targeting of 'platform users', 'platform providers' and transactions. The FTD-related statistical activities are based on the implementation of ad-hoc selection criteria which could allow for extracting the key information available from the FTD without breaching confidentiality. On the platform side, the proposal could be that of developing a process of aggregation for FTD which could make the aggregated information relevant to the platform economy measurement (i.e. its volume of activities and related GDP impact).

In order to carry out a methodological work aimed at linking FTD and SEP in line with the WPG work programme, the themes of accessibility, metadata and new indicators had to be properly addressed.

As far as the issue of the accessibility of SEP-related FTD data is concerned, some activities have been undertaken, even though only at experimental level:

- focusing on internet transactions (including e-commerce) and collecting available (aggregated) data with SEP potential, as a first step;
- highlighting specific patterns of online FTD, included sectorialisation, average value and international flows, as a second step;
- exploring to what extent online FTD could be accessible, at micro-level, compared other FTD, as a third step;
- testing the collection of related information on the use electronic payments online (web analytics, e-commerce statistics, etc.), as a final step.

The focus was, of course, on aggregated data and on the availability of a detailed breakdown of FTD when coming to features which are needed in order to identified SEP-related transactions. Similar accessibility tests should have been carried out in all countries involved in the ESSNet (even beyond WPG contributors) providing they had achieved and even limited access to FTD and could be able to produce a list of variables available (with an associated level of breakdown) to describe FTD in the national territory. This comparative assessment was based on a common methodology which relied on a classification tree to make the selection easier and consistent across countries (Figure 10.1). The aim was that of identifying, as a by-product of the FTD work, those financial transactions which could involve – with a given probability – a large digital platform.

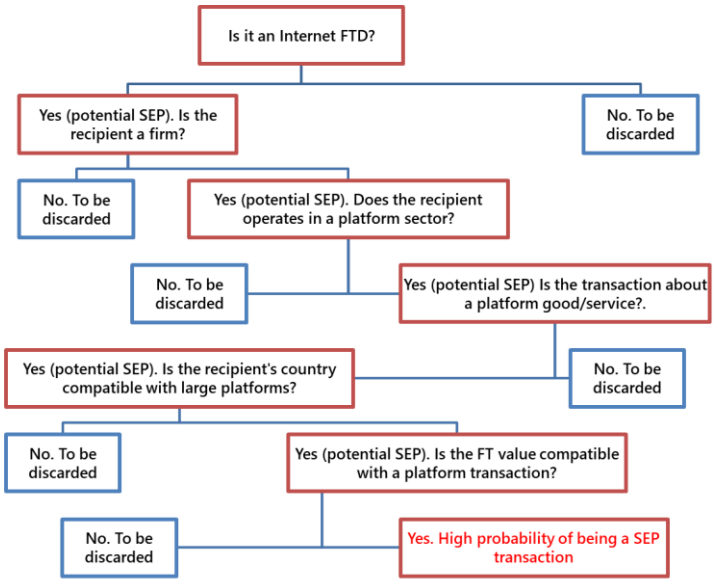

Figure 10.1 Classification tree proposed for the classification of aggregated FTD in order to select those FTD with a SEP potential.

As a second stage of the process, some specific methodologies were developed in order to deal with the SEP-related FTD data previously identified.

Classifications provide the basis for data analysis. As an essential part of the methodology to be developed in order to extract from the FTD population those observations with a high probability to be SEP-related, a number of concepts and related classifications have been produced, but only partially tested and standardised.

New classifications were aimed to address nature, sector and features of platforms, peers and transactions themselves (as implemented in the proposed classification tree).

Table 10.2. Main classification criteria of FTD to be codified in order to be adopted in the SEP FTD classification tree

Electronic / non-electronic
Nature of peers (firms / individuals)
Sector of activity of peers
Good / services transferred
Nationality of peers
Value of transaction

A high level of co-operation was requested to countries contributing to the ESSNet in order to assure that the classification framework would be realistic with reference both to the actual availability of FTD data in all concerned countries and with the same activities of digital platforms as they emerge from available indicators.

10.2 The operationalisation of FTD-based indicators on SEP.

After having found an agreement among the WPG members about the potential feasibility of the above-described FTD-based indicators on SEP ("Option c"), a range of activities have been undertaken in order to develop a methodological framework to test the option c basic concepts.

This option was about measuring the SEP activities in terms of (monetary) transactions. It implies that only platform-mediated transactions with an economic value and a related transfer of money could be taken into consideration. In addition to this requirement, by adopting a definition of "sharing platform" which focuses on "digital platforms" (operating over the Internet), the connected definition of transactions will have to be restricted to "digital financial transactions", i.e. those taking place between Internet operators (including a digital platform as a mediator between a user and a provider, both operating on the Internet).

The role of platforms in the e-commerce framework

This definitional framework raises the issue of the relationship between e-commerce[41] – as a broad phenomenon – and the more specific SEP activities. By assuming that e-commerce encompasses all economic transactions taking place electronically, a subset of them should include internet transactions and, among them, a further subset of digital transactions mediated by platforms. This is relevant to the extent internet financial transactions (IFTs) should be used to measure the platforms' activities. That is because IFTs are actually measuring, at a broad level, the rate of digitalisation of a (country's) payments system and, more narrowly, the diffusion of e-commerce in a given economy (as far as the financial transfers would be an evidence of commercial transactions).

By assuming that it could be possible to identify those IFTs taking place as payments for the provision of goods or services, the actors involved in the transaction processes are those included in Table 10.3. In the e-commerce framework described in Table 10.3, platforms can be identified both as "online marketplaces" and "e-retailers".

The former are big commercial platforms, mostly with a multinational scope, which are acting as intermediaries between multiple sellers and a very large population of potential customers. A key feature of them is that of controlling the flow of payments for the goods/services purchased through the platform. Thus, they can be associated to a transactions' flow from a crowd to a single player (and, at a further stage, from the single player to many providers). In the category of e-retailers most of the other digital platforms acting as intermediaries in the commercialisation of goods and services can be found. The financial transactions associated to them are very similar to those already described for the big platforms, but they differ from them as to the number of potential customers: no longer an undifferentiated crowd but specific groups of customers. The transactions associated with this group of actors are linking, in terms of transactions, "many to a few".

[41] **E-commerce** can be defined generally as the sale or purchase of goods or services, whether between businesses, households, individuals or private organisations, through electronic transactions conducted via the internet or other computer-mediated (online communication) networks. The term covers the ordering of goods and services which are sent over computer networks, but the payment and the ultimate delivery of the goods or service may be conducted either on- or off-line.

Table 10.3. A simplified outline of the «Strategic assessment of the three pillars of e-commerce» in FOSTEC & Company GmbH (2021, Figure 4)

	Types of Actors		
	Online marketplaces	Third party e-retailers	Direct sales including affiliates
Examples of actors	Amazon ebay Alibaba	wayfair vente-privee home24.de	A brand-name eShop with different affiliates.
Key components	International ecosystem infrastructure Large customer base	Companies acting online only, with websites having a focused product relevance. Highly relevant target group	Online brand offering seamless purchasing options over different media channels. brand-strengthening affiliates

Finally, some basic e-commerce activities can be identified by considering a third group of actors: those involved in direct sales through their own e-commerce websites. Under this case, platforms are excluded by definition and the transactions can be defined as relationships from “many to many”. As a consequence, when deriving some platform indicators from FTD data, these transactions should be excluded as far as possible.

The above arguing is summarised in Table 10.4 where the potential for developing e-commerce and platform indicators by using FTD is compared between the different groups. Under very demanding constraints, it can be assumed that FTD have indeed a potential to provide some broad information about the commercial transactions generating online bank transfers, usually by card. Of course, the main condition is that statisticians will have access to a significant amount of FTD (either at micro or aggregated level) and, more specifically, to data about online bank transfers by card. Additional information about selected bank accounts’ IDs, MCCs and cross-border transfer could help to finalise the develop of a set of statistical indicators. The potential availability of such data has been tested with reference to the six EU countries contributing to the WPG.

Table 10.4 The three patterns of e-commerce transactions' relationships and related indicators.

E-commerce actors	Transactions' relationship	Potential indicators	Conditions/constraints
E-commerce total	Any	Volume of e-commerce sales	Ability to target Internet (cards) payments. Identification of the nature of transfer by using MCCs.
Online marketplaces	Huge number of customers transferring money to a few large online retailers (crowd to a few)	Volume of e-commerce sales intermediated by marketplaces	Identification and selection of bank accounts ID used by marketplaces to receive payments (by number of transactions).
		by destination country	If breakdown by country available, "crowd to a few" transfers to selected low-taxation countries could be targeted.
Third Party E-retailers	Selected groups of customers transferring money to a few specialised online retailers (many to a few)	Volume of e-commerce sales intermediated by specialised e-retailers	Combination of intensity indicators (number of transfers to a single bank account) and MCCs.
Direct sales	Groups of customers transferring money to a large number of online sellers (many to many)	Volume of e-commerce sales by producers	Identification of C-to-B bank transfers. Best suited to target intra-national transfers. Low reliability.

10.2.1 The WPG mini-survey on the availability of FTD to develop SEP indicators

In the period September-October 2019, the national teams contributing to the WPG of the ESSNet Big Data had been asked to report in a comparable and detailed format about the actual (and expected) availability of FTD which could be used to produce some indicators as proxies to measure the activities of digital/sharing platforms in EU countries.

The mini-questionnaire included three sections, respectively about data availability and sources (1), data breakdowns (2) and complementary information (3). The blank questionnaire is attached as Appendix 9.

The survey was answered by Bulgaria, Germany, Italy and Norway. Portugal reported that the NSI does not have currently access to credit card data (thus, no information on online payments by card were available). While for other WPG countries the main sources of data are public or private financial institutions, Portugal is currently collecting, directly from the Portuguese Tax Authority - a wide-ranging evidence including all the financial transactions related to all purchases taking place in Portugal regardless of the payment method used. It can be assumed that all types of transaction will be included: electronic payments (ATM, bank transfers or Credit Card), as well as non-electronic payments (in cash) but without a detailed breakdown by type. Further investigation will be performed about the possibility to get in the future from the Portuguese Tax Authority a detailed information about the payment method of each transaction. For a similar lack of detailed information about online payments, as well as transactions by credit card, also Slovenia declined the request to join the mini-survey.

As a result, four countries were able to provide, at least partially, the requested information: Bulgaria, Germany, Italy and Norway (the questionnaires sent back by these countries are attached as Appendix 10, the results are compared in Table 10.5). It has to be pointed out that information on online transactions by credit card is available only in three countries (Bulgaria, Italy and Norway[42]): Germany does not have

[42] In Norway, the information is on all cards, and credit cards cannot be separated from debit cards.

payments cards transactions - including detailed information as for example the Merchant Category Code (MCC) – currently available which limits the scope for empirical analyses on estimating shared economy transactions.
As mentioned, the first section of the questionnaire asked the WPG countries about data availability. Questions 1.1 to 1.3 were indeed about data sources. All countries reported about the practical constraints limiting the ability of WPG national NSI teams to deal with multiple data providers and, moreover, to assure the consistency (formats, meaning, coverage, etc.) of collected data. In this respect, a preference has been often given to deal with a single data source – usually, that assuring the largest coverage of the phenomenon - even when the coverage of 100% of the requested information from that single data source could not be assured.

FTD are in most cases available at national level (i.e. for the whole country), at least with an annual frequency. Only Bulgaria, among the respondents, will collect just data on the aggregated 'value' of financial transactions; the other countries will access both data on value and number of transactions.

As to the availability of FTD detailed data on online credit card transactions (a key condition to derive e-commerce/platform indicators from FTD), only Bulgaria, Italy and Norway positively replied, thus answering also the specific questions in Section 2 about which information will available about credit card data.

On the 'payer' side of the credit card online transactions (question 2.1.1), it is very difficult to identify a breakdown achievable in all three concerned countries. Nature (individual/business) and nationality of the payers are inconsistently available between the respondents. Quite encouraging is the possibility to identify the country where the credit card of the payer has been issued in most cases.

Much broader is the coverage of the 'receiver' side (question 2.1.2). First, information allowing for a clear distinction between residents and foreigners will be given by providers in all three countries. Under some constraints, also the nationality of the receiver could be guessed. About discriminating individuals from enterprises, this seems to be very difficult for receivers, too. Nevertheless, as far as MCCs will be made available (probably, not in Bulgaria) some additional information could be extracted from them.

By focusing on the 'transactions' (question 2.1.3), the availability of the information on their total value can be given for granted. About the nature of the transactions, the MCCs should be the main source to exploit (hard to gauge, at this point, how much information could be extracted from them).

Moving to the question 2.2 of the questionnaire, the possible combination of different breakdowns is taken into consideration. In Norway there is a huge potential for matching payers' and receivers' data with reference to the transactions' value. This is also possible in Italy but only considering the 'nationality' of peers, not their nature (individuals or businesses). Unlike Italy, data matching in Bulgaria is only possible with reference to the peers' nature. A matching based on the peers' nationality is either impossible or uncertain.

A similar matching between peers but with reference to the 'domain' (commercial sector) of the transaction seems even more problematic. Actually, it is highly feasible in Norway but totally impossible in Bulgaria and severely limited in Italy by the limited availability of information on the nature of peers (much better it is the availability of data split by residents/foreigners).

When asking the respondents to make proposals for developing new indicators, only Norway was able to develop its own vision about what could be achievable, by proposing four indicators:

- Payments with Norwegian cards over the Internet in Norway and abroad, and payments by foreigners' cards over the Internet in Norway (coverage 100%)
- Country where the card has been issued, and country where the card has been used (coverage 100%)
- Merchant Category Codes (coverage 100%)

- Use of cards issued abroad on user locations in Norway broken down geographically by country, and for Norway also by postal code (coverage 100%)

Even though Italy was not in the position to develop any specific proposal for new indicators, it can be argued that – on the basis of the answers to previous questions – most of the indicators proposed by Norway could be also produced for Italy, at least on an experimental basis.

Finally, Bulgaria forwarded a single proposal that has to be mostly seen as a potential denominator for further indicators:

- Aggregated online payments across residents and across residents and non-residents (coverage 100%)

The last section of the questionnaire (Section 3) was dealing with a number of issues related to the expected delivery of new SEP-related FTD indicators. Overall, Norway is quite open to experimenting new approaches (also because of their actual experience in processing FTD), while Italy and Bulgaria (and Germany) which are still struggling in the process of defining the contents of the datasets they will be able to access, are understandably more cautious.

For instance, all countries but Norway are pessimistic about using indirect information in order to identify online transactions (for instance, by restricting the analysis to online operators. Norway is the only country considering the identification of payments from mobile devices possible. Similarly, only Norway is open to an experimental identification of the 'top receivers', for instance by looking at the rate of concentration by MCC (even with a geographical breakdown).

Two countries, Italy and Norway, are indeed optimistic about the future use of MCCs to deduct the nature of receivers (much less that of payers). The remaining countries are either pessimistic or have no access to the MCCs. Similar arguing affects the options to produce a data breakdown by the 'receiver's economic activity' and the 'transaction's domain' both strongly dependent on the availability and reliability of MCCs.

Also, the identification of the peers' nationality will be largely based on data availability (still under discussion in all countries). Whether extending the basic breakdown between residents and foreigners (the only option for Germany) will depend on both the amount of information which will be made available to NSIs and even on the number of observations available for each country to be taken into consideration as a small number of observations will make the information useless because of the confidentiality constraints.

Final topics considered in the questionnaire's Section 3 refer to:

- Expected frequency of data release: arguably yearly in order to get comparable results.
- Potential use of FTD microdata: excluded by all countries but Norway.
- Timing (trends, seasonality, etc.) of FTD transactions: to be taken into consideration by Norway and Italy mostly relying on a potential daily delivery of transactions data.

Last but not least, the quality issue – quite sensible when dealing with big data – has been apparently already addressed by all countries compiling new indicators based on FTD

10.3 Feasibility assessment

Addressing the ultimate aim of the SEP component of WPG, i.e. testing the feasibility of developing a set of new indicators as a by-product of the mainstream uses of FTD for statistics, the results of the mini-survey mostly suggest using FTD as just a complementary source for the development of SEP indicators in addition to other statistical (see Section 10.1.3), administrative or internet sources.

As discussed below, some ideas can be developed around the potential for integrating a subset of FTD – those about online credit card transactions – with internet-based indicators to tell us how the e-commerce – and,

partially, the narrower platform economy – is evolving in EU countries. A basic distinction between these two sources comes from their very different degree of accessibility by users.

A general remark has to be made about the constraints of the ESSNet exercise. The NSIs contributing to the WPG are experiencing many difficulties to get access to the FTD with a level of detail that could allow for the standard production of SEP-related indicators, which we will describe in Example 1 and Example 2 below in this paragraph. It has clearly been revealed from the mini-survey that fiscal administrative data or aggregated data on national or international payments are useless when considering the sharing economy phenomenon, since they do not provide information on the sharing economy as such, but rather on the broader ecommerce domain (see e.g. Table 10.5). Instead, a direct access by statisticians and analysts to the data stored by banks and other financial institutions issuing credit cards would be essential to identify individual payers/receivers (i.e. buyers/sellers) in order to discriminate platforms from ecommerce operators and to qualify the underlying commercial flows.

As far as NSIs won't be allowed to access FTD individual data, new SEP-related indicators, either FTD-based or as a combination of FTD and other data, could be expected to feature shortcomings like:

- to be highly aggregated, as all countries report difficulties to further breaking down Internet FTD indicators by destination country or MCC;
- featuring a high error coefficient about the relative share of ecommerce transactions in a country being platforms-related transactions, as only speculations are possible at this stage.
- being only partially comparable across sectors and countries because of the heterogeneity of classification practices used by the national data providers.

Nevertheless, they could have the potential to be very timely since the data exists practically immediately after the transactions, and still after allowing for some processing time at the central bank and the NSI, it is a short production time. Also, the transactions are highly comparable over time although the data received by the NSI may be susceptible to changes of structure because of the rapid technological changes in the infrastructure for transactions: data structure may change due to restructuring (change of legal status, mergers and acquisitions, etc.) of the data provider, leading to a need for a new data delivery agreement for the NSI, and also adaption of the NSI's data-handling to align with the new metadata that might have been introduced by the new data provider. In the presence of new payment platforms appearing and increasingly used, the adaption of the NSI's data-handling might be even more challenging.

Concerning the comparability across countries, the mini-survey results suggest negative prospects for it, at least without a comprehensive framework which could allow a minimum number of countries to access the same, or highly comparable, data sources. From Table 10.5 the message can be drawn that, so far, hardly any indicators can be compared between more than two countries, being in contrast to our target of comparing the indicators between at least three countries. Notice that this is the a priori (potential) situation based on the survey, we will later in this section present the results when assessing the indicators' potential of being comparable in practice.

Table 10.5. Proposed SEP-related FTD indicators for international comparison (only indicators available in at least two countries are included)

Indicators	Comments	Norway	Bulgaria	Italy
Aggregated total **value** of FTD credit card data transactions on an annual basis	Available in all countries			
Aggregated total **number** of FTD credit card data transactions on an annual basis	Available also in Germany	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis	Proxy of the total volume of e-commerce in a country	Yes*	Yes	Yes

Aggregated total **number** of FTD credit card data transactions **online** on an annual basis	To be used as a denominator of more complex indicators	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by resident vs. foreigner **payers**	Possibly relevant for the purchase of tourist services in the country by foreigners		Yes	Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by nationality of **payers**	As above (with a breakdown probably limited to a few large countries)	Yes*		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by nationality of **payers**	With a breakdown probably limited to a few large countries	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by resident vs. foreigner **receivers**	Key indicators to assess the relevance of foreign service providers and digital platforms	Yes*		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by resident vs. foreigner **receivers**	Same as above	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by nationality of **receivers**	Essential to target platforms based in selected countries (with a breakdown probably limited to a few large countries)	Yes*		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by nationality of **receivers**	Same as above (with a breakdown probably limited to a few large countries)	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **economic activity** of **receivers**	Economic activity breakdown only based on MCCs. Potential matching with economic statistics	Yes*		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **economic activity** of **receivers**	Economic activity breakdown only based on MCCs. Potential matching with economic statistics	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **type of purchase**	Type of purchase breakdown (e.g. goods/services) only based on MCCs. Potential matching with economic statistics	Yes*		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **type of purchase**	Type of purchase breakdown (e.g. goods/services) only based on MCCs. Potential matching with economic statistics	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' nature** and **receivers' residence**	Key indicator for e-commerce (B2B vs. C2B). Partly related to platform activity	Yes*	Yes	
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nature**	Key indicator for e-commerce (B2B vs. C2B). Partly related to platform activity	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nationality**	Mostly on cross-border purchases online	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' economic activity**	Key indicator for e-commerce	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nature** <u>and by **type of purchase**</u>	Key indicator for e-commerce	Yes*		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nature** <u>and by **type of purchase**</u>	Key indicator for e-commerce	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nationality** <u>and by **type of purchase**</u>	Key indicator for e-commerce and platform activity	Yes*		Yes

Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nationality** and by **type of purchase**	Key indicator for e-commerce and platform activity	Yes*		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' economic activity** and by **type of purchase**	Key indicator for e-commerce	Yes*		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' economic activity** and by **type of purchase**	Key indicator for e-commerce	Yes*		Yes

* For FTD payment cards, i.e. including debit card data. Also, information is not available about the nationality of the card holder, only the country where the card is issued as well as the country where the cards are used.

The indicators listed in Table 10.5 have been used for improving and implementing the classification tree designed in order to allow for a quantification of the SEP-related activity in a country (Figure 10.1). As described in Figure 10.1, by going down through the classification tree, i.e. using increasingly detailed information on the concerned transactions to classify them as "potentially SEP-related" or not at each stage, SEP-related FTD are more and more precisely identified as we move from stage to stage in the classification tree. At the same time, availability of time series of FTD could further improve the available information by help tracing the evolution of the SEP phenomenon. The process of deriving the classification variables used in each step of the classification tree of Table 10.5 is definitely an iterative one as every step, in combining different pieces of information (for generating the classification variable at this stage of the classification tree), shed some light on specific features of the SEP phenomena and makes available additional knowledge to improve the understanding of classification variables at the other stages of the tree, both the previous and preceding ones

It has to be stressed that the classification exercise above is not to be considered valuable per se but as a component of, and a step forward, the construction of a broader framework for the statistical measurement of internet-related phenomena. As a matter of fact, the economic context (in terms of behaviour of both businesses and consumers) is equally important in this perspective than the technological context: how many businesses and potential customers have access to the internet, how many of them are using internet in an effective way, to what extent technology is influencing their behaviour etc. The whole context will have to be taken into consideration when comparing the structure, as well as the performances, of SEPs in different sectors or countries.

In terms of *actual* feasibility, after comparing the potential feasibility as described in Table 10.5 to the actual feasibility tested during the first half of 2020 – rather bleak results have to be reported. Among WPG member countries, only Italy had both the opportunity and succeeded in getting in touch with FTD data providers on this topic, to ask questions about the actual options to produce some "aggregated" SEP-related indicators that did not break any confidentiality rule. As a result of these activities, three main hampering factors have been identified before proceeding with the pilot production of SEP-related FTD-based indicators. They are the following:

Problem 1: The empirically available indicators (either actually available or that they can be produced later), are measuring ecommerce and not sharing economy (even less SEP-related phenomena), since the available relevant data (web transactions) cannot separate sharing economy transactions from the rest of ecommerce. The indicators are related to amount of web purchase.

- Example 1: Statistics Norway has published cross-border transactions by MCC over the internet and has data for publishing total cross border transactions by MCC, which e.g. gives total expenditure on

accommodation abroad paid by cards issued in Norway. However, the MCC cannot separate sharing economy from the rest of ecommerce, which can be illustrated by accommodation: despite the existence of separate mcc-codes for many of the hotel chains (aggregated to the MCC-group "Named hotels" in the published table for confidentiality reasons), the most widely used MCC for accommodation is "*7011 Lodging – Hotels, Motels, Resorts, Central Reservation Services (not elsewhere classified)*" which cover both hotels and most of the sharing economy alternatives since the other MCC-accommodation codes only cover *Timeshares, Sporting and Recreational Camps*, and *Trailer Parks and Camp Grounds*. Thus, it is not possible to identify the sharing economy part of ecommerce, and then what can be measured is only the broader ecommerce concept.

- Example 2: ISTAT is testing, in co-operation with the Bank of Italy, the production of statistical indices aimed at measuring the evolution of electronic payments made by cards issued in Italy. Since such indices, if they will be made available to ISTAT, will provide only a poor detail about the merchant category they are related to and it is not expected that a breakdown by destination country could be available soon, the use of them to measure more specific phenomena than just the trend of the overall amount of web transactions is highly improbable.

Problem 2: It seems not possible to find comparable empirical indicators in Italy and Norway within the scope of WPG, and neither between other countries within WPG. Main issues include: a different institutional and normative setting in the two countries; national specificities in the use of electronic means of payment; different kinds of agreement between the FTD data owner(s) and the NSI (in Italy, the Bank of Italy is the mediator between these two actors); different aggregation methods that lead to incomparable aggregated data. Some of these obstacles could have been solved if time and resources were available to produce indicators based on a different aggregation method. Notice also that our findings of international comparability could have been different if we were able to compare different countries.

Problem 3: It is conceptually difficult to separate platform economy from the rest of ecommerce: a firm typically performs some of its sales on one platform, some on another platform, and some directly on web. It is highly probable that all of these sales refer to the same typology of goods/services.

As a conclusion, a pilot harmonised data collection – at least just for Italy and Norway - of the indicators listed in Table 10.5 is not possible. A few indicators were produced, but a number of slight differences about how they were calculated, their actual contents, and the classification detail used, made them useless for a systematic comparison.

This raises the issue about the potential use of the information collected in the WPG's SEP exercise and the lessons to be learnt for future activities in this field.

10.4 Are SEP-related transactions measurable?

As we noticed already – early in this chapter - our approach to the measurement of sharing economy ("option c") is to extract SEP-related data from the FTD with the purpose of calculating some sector-specific indicators. The proposal discussed in Section 10.1.4 (decision tree) was based on the hypothesis that at least three classification criteria of FTD (over the Internet FTD, country of destination and merchant code) could be jointly used to identify, as a proxy, SEP-related transactions.[43] This approach was not successful, as we have seen in the previous sections. One of the causes for this could perhaps be removed by an improved institutional framework enabling a useful pre-processing of FTD data. However, even in this case, the question is still whether our approach could have been successful in the presence of the increasing

[43] Of course, the access to FTD micro-data that could have allowed for a direct identification of digital platforms was excluded a priori since most countries will not be able to access such data.

uncertainty in the identification of digital platforms that are implementing any business model either based on asset-"sharing" or just on the intermediation between peers.

The information contained in FTD is about e-commerce, not about the sharing economy part of e-commerce. Thus, what FTD can be asked to provide in this domain is just an approximate quantification of the e-commerce activity in a country. E-commerce is a subset of *E-business*[44] which we in our context will define as a trading mode, thus excluding the interpretation of it as an industry of its own. An option for future studies will be that of focusing on FTD-based e-business (or, at a lesser scale, e-commerce) indicators rather than more focused SEP-related indicators. A great advantage of this shift in the focus of the analysis from sharing economy indicators to e-commerce indicators, it is that it could allow for an integration of several alternative data sources with FTD data[45], all being sources of information on e-commerce features and trends: official payment statistics (totals), Web analytics data and administrative registers if available. This approach – relying on data integration – is strongly consistent with the Eurostat/ESS Strategy and Roadmap for implementing a Trusted Smart Statistics (TSS) approach[46]. TSS is going to influence in a substantial way the production of official statistics in the EU over the next decades and its principles can be seen as a reference for the development of every new statistical domain where official statistics is challenged by the increasing digitalisation of the economy and the society. Among the TSS key principles (a. Multi-purpose data sources; b. Multi-source statistics; c. Layered organisation of data processing workflow; d. Modular methodological frameworks; e. Pushing computation out; f. Using data without sharing data; g. Trust in Smart Statistics; h. Benefits of enhanced collaboration) the objective to jointly exploit multi-purpose data sources in order to produce multi-source statistics is paramount.

What has been envisaged so far in the WPG context is just the potential for integrating FTD data (multi-purpose) with Web analytics[47] (as platforms operates through websites) and administrative registers (as far as fiscal data on VAT would be available) to test the development of a set of SEP-related multi-source indicators. This approach has not been operationalised yet in the WPG but has emerged, mostly during the last WPG meeting in June 2020 as a possible alternative to get rid of the unpleasant condition where the problems in accessing FTD data caused the project to get stuck. The way out can be identified in extending the data sources mainly integrating the poor FTD evidence about SEP with Big Data, mostly Web analytics, about the internet traffic generated by the websites of digital platforms.

The new approach to measurement of e-business/e-commerce based on the exploitation of Big Data as outlined above is not without risks of being unsuccessful.

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[44] The following definitions are given by Eurostat:
E-business, a contraction of 'electronic business', refers to the use of information and communication technology to support a business strategy. E-business is more than just E-commerce, and involves business processes spanning the entire value chain from purchasing and logistics management to customer service and corporate partnerships - all of them making use of electronic communications.
E-commerce can be defined generally as the sale or purchase of goods or services, whether between businesses, households, individuals or private organisations, through electronic transactions conducted via the internet or other computer-mediated (online communication) networks. The term covers the ordering of goods and services which are sent over computer networks, but the payment and the ultimate delivery of the goods or service may be conducted either on- or off-line.

[45] Possibly, as indices in order to deal with confidentiality concerns.

[46] See: *Implementation of the Bucharest Memorandum on "Official Statistics in a datafied society (Trusted Smart Statistics)" Trusted Smart Statistics Strategy and Roadmap*, Document presented at the 40th Meeting of the European Statistical System Committee, Luxembourg, 16 May 2019. The principles of Trusted Smart Statistics are described in: Fabio Ricciato, Albrecht Wirthmann, Konstantinos Giannakouris, Fernando Reis and Michail Skaliotis, "Trusted smart statistics: Motivations and principles", *Statistical Journal of the IAOS* 35 (2019) 589–603.

[47] Web analytics is the measurement, collection, analysis and reporting of web data for purposes of understanding and optimizing web usage. It can be used to assess and improve the effectiveness of websites (from Wikipedia).

First, a range of different data sources must be integrated to capture the heterogeneity of these phenomena, and this is a challenging task *per se*.

Second, another challenge is the harmonisation of the classifications used in various sources/domains, these might be inconsistent. An example is the poor comparability of industry classifications used for statistical, administrative and market monitoring purposes. The missing conversion of economic activity (or industrial) codes, like NACE or ISIC or NAICS, with MCC is a case in point. No agreement has been negotiated yet between the institution responsible for official industry classifications (like Eurostat or the UN Statistics Division) and VISA or Mastercard, the most intensive users of MCC codes. Only private companies offer non-official conversion tables that have also been taken as subject of a US Patent (US6819270B1): "Method and system for universal conversion of MCC, SIC or other codes".

Third, traditional inferential methods should be reviewed and adapted to data based on combination of many sources, e.g. during the constant revision of time-series data the ability of new indicators to follow clear identifiable trends should be evaluated, mostly when using as source sets of unstructured Big Data.

Fourth, digital processes are intrinsically very dynamic. Therefore, in a time-series modelling a high level of flexibility is needed to deal with sudden changes of the short-term indicators, as well as the long-term trend towards a saturation of the market for specific technologies.

Fifth, another challenge is the rapidly changing structures in the financial market and system which makes the different sources susceptible to after a while disappearing in its usual form and turning into a new form with new population coverage, variables other metadata as well as new data providers.

Based on the e-commerce approach above, a two-stages measurement process for deriving SEP-related indicators could be envisaged[48]. The first stage is the derivation of e-business/e-commerce indicators from the combination of FTD and other sources of data according to their availability. The second stage is the calculation of specific SEP indicators and could be implemented either by adopting a unit enumeration approach (where, of course, individual data should be drawn from administrative registers or extracted from the Web rather than taken from privately-held or owned databases but the objective is that of targeting a specific group of platforms) or by estimating SEP-related flows to the extent available data could allow for isolating financial transactions to specific countries or related to trade in defined commodities. The latter process could include: using payment statistics to quantify card payments over the Internet; collecting product data for a large number of payments and calculating product indices; identifying the outflow from a country to potential platforms as a component of financial transfers to selected foreign countries. This could be an interesting test of combining FTD indicators, like card payments by MCC, with trade statistics based on the standard product classification (CPA/CPC + HS + SITC).

10.5 Perspectives for future developments

Before addressing any consideration about the future prospects for SEP measurement (by using FTD or not), a few remarks about the overall economic and social context have to be done. A key issue is that the platform economy has been highly affected by the COVID-19 crisis and it is still undergoing a thorough adaptation to a new environment[49]. Some trends can be highlighted.

COVID-19 has been the catalyst for fostering a digital transformation strongly affecting the balance between physical and online retail. Consumers are increasingly buying online (also from e-stores of *brick&mortar* retailers) and this behaviour is expected to become structural even after COVID crisis. This implies that e-

[48] The usefulness of such a two-stage process should be tested as a future follow-up to the WPG project.
[49] "The sharing economy will have to change", The Economist, June 4th, 2020.

business will erode the current share of physical (retail) trade and will be adopted as a standard by most retailers.

Large digital platforms are not necessarily winners in this process as the effects of the crisis have been very heterogeneous across the whole industry. Three cases can be taken as examples. First, large non-specialised e-retailers and their digital market-places have been profiting most of the COVID crisis by increasing both their sales and their stock-market value. On the other hand, the paradigm that the "winner takes all" – proposed by several economists – has not been working as, since large retailers have not been able to keep up with the huge demand of people restricted at home, local/sectoral (online) retailers have found their role in the e-market as well. Mostly food e-retailers and local platforms have attracted a number of customers who did not have their demand met by large platforms. Second, some sectors have been devasted by the global lockdown, mostly those selling travel and tourism services. Nevertheless, a distinction has to be done between more "traditional" activities (like airlines, travel agencies or hotels) - that could expect to recover as soon as people will be allowed to come back to travel and to move for leisure - and those business models based on a "sharing" paradigm (Uber, Lyft, AirBNB, etc.) which have won a large share of the market over the last years but whose businesses are at risk because of a drop in the customers' propensity to share vehicles or houses with strangers. Third, a growing e-commerce demand is benefitting both platforms and e-retailers selling their own goods and services. This is a strong driver of change in the e-business where the use of multiple marketing channels (physical, own e-shop, one or more platforms, etc.) is becoming a need for many firms both in manufacturing and service industries. In turn, this is becoming a new and very profitable market for large digital platforms that – rather than focusing on providing an intermediation service between producers and customers – are restructuring their activities offering consultancy and technical support (including cloud services) to e-retailers: a huge potential new market.

These relevant changes in the economic context for e-business – in parallel, to some extent, to a real change of paradigm – will have to affect the measurement of the SEP phenomenon as well. It can be argued, for instance, that the SEP domain has to be re-defined. Its "sharing" component is at risk of becoming irrelevant in the near future (a clue is the tremendous effort of diversification undertaken by big actors like Uber and AirBNB). This will raise the issue about the relevance of the concept of "collaborative economy" itself. At the same time, as e-business becomes more pervasive and mixed business models spread around, platforms are losing their specific features and can be increasingly assimilated to other digital intermediaries or, as far as Amazon and other OTTs are concerned, to large technological platforms like Microsoft, Apple or Alphabet.

Refocusing measurement to e-business in general, it has to be acknowledged that available data mostly refer to e-commerce. Thus, a bottom-up approach could be recommended starting from a substantial improvement of e-commerce indicators, thus a better coverage of e-business activities and, as a specific focus, a close monitoring of the role of digital "sharing" platforms. As argued in the previous paragraph, coverage and timeliness of data can be assured only by the integration of data sources.

It is also worth considering that new taxonomies are urgently needed in order to meet new measurement challenges. A *univocal identification of business models* adopted in the digital economy is not yet available even though it could allow for preventing any confusion about e-business/e-commerce as an economic activity and as a sales strategy (still common in official statistics). Additionally, the fragmentation of the service market taking place as an effect of digitalisation (the emerging of multiple Web-related services) raises the issue of *revising the official classifications of services* with obvious implications in the measurement of international trade in services. Finally, on the workforce side, the development of new businesses has brought to the emerging of new roles for enterprises' employees and collaborators. *Skills, tasks and contracts of these new workers* (mostly in the area of the gig economy) are worth to be properly classified and regulated.

In order to recall the main issues that have affected the SEP component of the WPG project, it can be said that:

a. Limited and extremely constrained access to data.
b. Inability to define the procedure for processing the data, and
c. A generic and elusive object of analysis, as SEPs undoubtedly are

have negatively influenced the final outcome. A substantial amount of methodological work has been carried out and is described in detail in this Report. This could be the basis for launching a new project that could address the shortcoming of the activities undertaken so far. A few conditions have, nevertheless, to be met in order to make it possible:

a. The definition of an effective institutional and methodological framework to allow statisticians and researchers to access FTD as privately-held data in a safe and collaborative environment, such as that of a European Data Space.
b. A methodological work aimed at redefining the system of official statistics on the digital economy with more detailed description of what is included in the "collaborative economy" and how the different actors involved in it interact each other. Some new taxonomies could be extremely helpful tools to allow statisticians to improve the current measurement approach.

The principle of data integration has to be fully endorsed in areas of statistical production where multi-faceted phenomena are under scrutiny. Again, the matching of data – including Big Data - from different data sources is another organisational and methodological challenge for official statistics but a challenge to be definitely taken up.

Part III – Conclusions and recommendations

11. Conclusions

An inventory of the FTD infrastructure has been performed for the different countries of WPG: Bulgaria, Germany, Italy, Norway, Portugal and Slovenia. Also, the possibility of extracting information on sharing economy from the FTD, has been investigated.

We have seen that the infrastructure of FTD is different in different countries although there are some similarities. For Portugal and Slovenia, the only source that can provide access to FTD is the tax authorities. For Italy and Norway, both banks, card companies and service providers are potential sources, but for Italy, this access must go through the National Bank, which by legislation has the right to access microdata and can produce aggregated data for Istat.

Only in Norway and Portugal is micro-FTD going to be available for the NSI. The reason for this is the differences in the legislation. It is worth noticing that micro-FTD from the complete population are large data and the NSI must have enough server capacity, and also a good system for pseudonymisation of any identificatory variables. Further, FTD are widely regarded as sensitive, and thus a GDPR requirement is that the NSI makes risk assessments before getting such data. One element in such assessments is to consider the need for official statistics against the perceived burden of the persons whose transactions are included in the FTD.

The restriction in most countries to only accessing aggregated data, means that an important process lies in getting the provider to aggregate the data in a way being useful for the NSI. This is a part of the cooperation that is needed with data providers to agree on data delivery to the NSI and comes in addition to having the necessary legislation for getting data. The data providers need to use resources on preparing data for the NSI, and the NSI in most of the countries cannot compensate for these resources. Thus, a good cooperation between the NSI and the data providers is essential for a successful outcome. Even in the presence of a good cooperation, the access to FTD takes some time. Some important suggestions came also from the work of the EU NSI's Directors Expert Group on privately held data on how to overcome the critical issues related to the accessibility of FTD, concluding that NSIs should aim at getting in touch with data providers without intermediaries, which could substantially improve the level of co-operation and the quality of data transmitted.

In Part II we presented the applications that we made both in national Case studies and in the use of FTD in measuring sharing economy platforms. The case studies show some of the potential of financial transactions data for official statistics. The Italian case studies show the gain of using time series of total daily card transactions by aggregated mcc-categories for the purpose of forecasting and nowcasting of economic indicators, which is both an improvement of quality and production time. The Portuguese case study looks at how to improve the tourism statistics by addressing lodgings being too small to be covered by the current tourism survey. It also looks at how to measure transport by the recent alternatives to taxi. One challenge is how the data are organised at the Tax Authorities as well as being pre-processed before the NSI receives the data, making it not possible to identify the invoices that eventually should be removed since the credit notes cannot identify the corresponding invoice. The Norwegian case study looks at to what extent debit card transactions at legal unit level can be used to estimate the monthly total retail sales by the ten industry classification groups used for the Retail sales index statistics. This case study had to use FTD for only one day since it was not possible to receive monthly data in time for this project, and this illuminates that receiving financial transactions data often is a complicated process that takes long time.

Results of case studies show that FTD indicators or variables can be used as predictors in modelling the near future, but incorporating FTD in the production of official statistics needs to be better explored. Larger databases being more complete in terms of enclosed activities (firms, groups etc.) are needed to test models.

The timeliness of FTD during the Covid-19 crisis was relevant for improving the forecast of some selected macroeconomic indicators (e-commerce, final consumption, retail trade index), especially when we could use FTD broken down by Merchant category code. This is because different payment flows have registered different trends in the economic terms during the crisis.

Sharing economy is an increasing part of the economy. We have laid a framework for extracting information on sharing economy from the FTD using indicators on sharing economy activity, defining sharing economy broadly as collaborative economy since the narrower definition is less relevant for official statistics and also more difficult to measure. Further, we have investigated what relevant data exist in the countries for being input to such indicators. Given the available data, it was not possible to derive indicators that could be compared between countries. The work on sharing economy also revealed that sharing often cannot be measured separately from e-commerce, suggesting that future measurement should focus on the whole e-commerce instead of only collaborative economy.

12. Recommendations

The provision of payment services – mostly as an intermediation between the banks of the payer and that of the payee – is granted by private institutions or firms in most countries. Beyond efficiency, reliability and accountability, data protection is one of the pillars of such business. Financial intermediaries, which take care of up to millions of transactions on a daily basis, are usually technology leaders and set the standards for effective data storage, data management and data protection (last but not least because they are preferred targets of attacks and frauds). Confidentiality is such a key requirement in this business that easily permeates the whole industry, and even the information to be regularly provided to Public Authorities in charge of preventing inappropriate or fraudulent behaviours is managed to be kept at the minimum level of detail allowed. Below some recommendations to favour the data access and the leverage of the use of FTD for NSI.

HOW TO MOTIVATE PRIVATE FTD OPERATORS TO COLLABORATE

- In light of the need for data protection, for an NSI to get financial transactions data from a potential data provider, a necessary part is to either have a decision based on the Statistical act, or if the Statistical act is not applicable, there must be a formal agreement between the NSI and a potential data provider. In the latter case, an institutional agreement with the National Bank typically is appropriate, since the National Bank often has access to more data than the NSI**.** Under either of the solutions above, a prerequisite is that the data provider should in some sense be motivated for providing data to the NSI.

- Good cooperation through a good relationship is essential for obtaining financial transactions data, whether the Statistical Act grants access to data or not. It is always more tempting for a potential data provider to use resources benefitting someone with whom you have a good relationship. The importance of a positive relation is increased due to the data provider's need to use resources in preparing data for the NSI, as well as sharing and explaining metadata. The NSI usually cannot pay for these expenses and should acknowledge the burden put on the data provider. The data provider should in addition get something in return, and the NSI should ensure telling about these benefits to the data provider. In addition to contributing to the development of official statistics in the society, the benefits could be to get reduced response burden from shorter or fewer official statistics questionnaires. In addition, the NSI can help improving metadata description, developing data

infrastructure, or performing relevant analyses of the data. One approach to achieving the latter benefits, is by letting NSI staff perform tasks within the data provider's office, or just to participate in a common working group that is e.g. developing the routines for processing of data.

- Knowledge transfer to the data provider is another potential outcome. A well-founded description of the society's need of the financial transactions data, is necessary for motivating the potential data provider to use resources on providing data, especially if the resources should be used within a rather limited calendar time interval.

- The data provider is concerned about the secure handling of the data that they possess, especially since financial transactions data are widely considered as sensitive. Reassuring the data provider of the secure handling of data within the NSI is an essential part of the contract with the data provider. A related topic is to inform the data provider of the role of the NSI. Contrary to most other governmental organisations, the NSI is not interested in the contents of the transactions for a single person or a single business, but to use data for statistical purposes only. The purpose of the NSI is to deliver aggregated numbers.

- Central steps in the process of getting data are the study of metadata, making a delivery agreement, and in the situation where no microdata can be obtained by the NSI, specifying how data should be aggregated before sending them to the NSI. The thorough study of metadata in close dialogue with the data provider is an important step that could reveal that presumably interesting data in the end turn out to be less interesting for the NSI. Not only the available variables and their descriptions should be studied, but also what is the unit of the microdata, which part of the population is covered, and known quality issues. The process of getting data usually takes a lot of calendar time as well as personnel resources within the NSI. A thorough study of metadata can be time-saving in cases where this study leads to termination of a data acquiring process that would only end in data with little usability within the NSI.

- A key issue is that of the classifications to be potentially used to match FTD and statistical data. Beyond the undisputed availability of information on the localisation of both payers and payees (at least, at country level) and their economic nature (whether individuals or businesses), a key issue is that of the actual use of "merchant category codes" (MCC), especially when using FTD for the purpose of SEP and ecommerce. Merchant codes – not fully matching the official economic activity (industry) codes used by NSIs – can be in principle very detailed and have been identified as a key information in order to measure e-commerce and SEP-related activities by using FTD. Unfortunately, according to the WPG members' experience, it cannot be assumed that all institutions dealing with financial transactions were applying MCCs in a coordinated and harmonised way. On the one hand, some codes are seldom used in some countries and more systematically used in others, which make comparison difficult. On the other hand, data providers are reluctant to release data - even though aggregated by other dimensions, like time or space – with a detailed information based on MCCs. As a consequence, several different methods of aggregating MCCs are used across the financial industry which, again, prevents any comparability of the results of FTD aggregations. Finally, for the use of FTD for SEP, the usefulness of MCCs are limited, as we discovered in Section 10.3.

ROLE OF ESS IN ENHANCING THE USE OF FTD

- The definition of a new regulatory and methodological context for a public use of privately-held data is a *conditio sine qua non*[50] for an effective use of FTD data for statistical purposes and even more for producing some FTD-based SEP-related indicators. ESS should make an extra effort to arrange for a framework agreement between the NSIs of ESS and the most important companies managing financial transactions in Europe. A starting point could be to target the network of financial institutions currently negotiating an agreement to market an EU-wide debit card[51]. Without a strong political support, any bilateral agreement mostly on a national-only basis, will be affected by limitations and data owners' idiosyncrasies.

WPG members are advocating the inclusion of FTD among the priority areas identified by Eurostat for exploring potential options of public-private co-operation[52]. A role for FTD should be found also in the definition of the European Data Spaces designed by the European Commission in order to enhance the access to privately-held data, via industrial data platforms[53]. With reference to the financial sector, companies could find there a secure and trusted environment to share their data based upon voluntary agreements. A procedure that has been already proposed to assure a high level of cooperation between data providers and data users (i.e. NSIs) in such context (and even tested, in a few cases) is that of asking data providers to process (or pre-process) the data extracted, before delivery, by running software routines designed by the data users.

[50] absolutely necessary condition.
[51] https://www.ecb.europa.eu/press/pr/date/2020/html/ecb.pr200702~214c52c76b.en.html
[52] So far four priority areas have been identified for action: Mobile network operators, Digital platforms (mostly related to tourism statistics), Trusted Smart surveys and Smart traffic.
[53] Planned activities include data spaces in the domains of industrial (manufacturing), Green Deal, mobility, health, financial, energy, agriculture, public administrations, skills and open science.

References

Chai, T. and Draxler, R. R. (2014), 'Root mean square error (RMSE) or mean absolute error (MAE)? – Arguments against avoiding RMSE in the literature', Geosci. Model Dev., Vol. 7, https://doi.org/10.5194/gmd-7-1247-2014, pp. 1247–1250.

Codagnone, C. and Martens, B. (2016): 'Scoping the Sharing Economy: Origins, Definitions, Impact and Regulatory Issues', Joint Research Center Institute for Prospective Technological Studies, Digital Economy Working Paper 2016/01.

Commission (2016), 'COMMUNICATION FROM THE COMMISSION TO THE EUROPEAN PARLIAMENT, THE COUNCIL, THE EUROPEAN ECONOMIC AND SOCIAL COMMITTEE AND THE COMMITTEE OF THE REGIONS - A European agenda for the collaborative economy'.

Doornik J. A., (2009), 'Autometrics', *in* J. Castle and N. Shephard, eds, *The Methodology and Practice of Econometrics*, *A Festschrift in Honour of David F. Hendry*, Oxford University Press, pp. 88-121.

Doornik, J.A, Hendry D.F. (2013). Modelling Dynamic Systems, PcGiveTM 14, vol. II, Timberlake Consultants Ltd.

European Commission (2018), 'Study to monitor the economic development of the collaborative economy at sector level in the 28 EU Member States', Eurostat.

Eurostat Unit G-4: Innovation and information society (2018), 'Note on Measuring the Digital Collaborative Economy', Note updated for the Eurostat Workshop 29-30 May 2018.

Fosen, J. & Zhang, L.-C. (2011), 'Quality evaluation of employment status in register-based census', Proceedings of the 58th World Statistical Congress, 2011, Dublin (Session STS024), pp. 2587 ff.

FOSTEC & Company GmbH (2021), 'B2C E-Commerce Distribution Strategy', accessed 8 April 2021, https://www.fostec.com/en/competences/e-commerce-strategy/e-commerce-distribution-strategy/b2c-e-commerce-distribution-strategy-2/

Frenken, K. and Schor, J. (2017), 'Putting the sharing economy into perspective', Environmental Innovation and Societal Transitions, Vol. 23.

Gerber, E.M. and Hui, J. (2013), 'Crowdfunding: Motivations and deterrents for participation', ACM Transactions on Computer-Human Interaction, Vol. 20, no. 6, 34. https://doi.org/10.1145/2530540.

Ghysels, E., Sinko, A. and Valkanov, R. (2007), 'MIDAS Regressions: Further Results and New Directions', Econometric Reviews, Vol. 26, no. 1, pp. 53-90.

Group on privately held data (2020), 'Group on privately held data – Draft Document for the ESSC meeting on 13-14 May 2020', Draft document for item 5 on the agenda of the DIME/ITDG steering group meeting 12 of February 2020, https://ec.europa.eu/eurostat/cros/system/files/05_-_group_on_use_of_privately_held_data.docx.

Harvey A.C. (1989), *Forecasting, structural time series models and the Kalman filter*, Cambridge University Press.

Heerschap, N., Pouw, N. & Atmé, C. (2018), 'Measuring online platforms', CBS in co-operation with UvA, Amsterdam.

Hendry, D.F (1979), Predictive Failure and Econometric Modelling in MacroEconomics: The Transactions Demand for Money, *in* P. Ormerod, ed, 'Economic Modelling', Heinemann, London, pp. 217–242.

Hendry, D.F., Pagan, A., Sargan, J.D. (1984), Dynamic Specification, i*n* Z. Griliches and M.D. Intriligator, eds, 'Handbook of Econometrics II', North-Holland, Amsterdam, pp. 1023–1100.

Iglewicz, B. & Hoaglin D. (1993), 'How to Detect and Handle Outliers. The ASQC Basic References in Quality Control: Statistical Techniques', *in* E.F. Mykytka, ed, 'How to Detect and Handle Outliers', ASQC Quality Press, Milwaukee, Vol. 16.

Loh, W.-Y. & Zheng, W. (2013), 'Regression trees for longitudinal and multiresponse data', Ann. Appl. Stat. 7, no. 1, 495-522, https://projecteuclid.org/euclid.aoas/1365527208

Office for National Statistics (2017), *The feasibility of measuring the sharing economy: November 2017 progress update*, London.

Narasimhan, C., Papatla, P., Jiang, B., Kopalle, P.K., Messinger, P.R., Moorthy, S., Proserpio, D., Subramanian, U., Wu, C. and Zhu, T. (2018), 'Sharing Economy: Review of Current research and Future Directions', Customer Needs and Solutions, vol 1, no. 2.

Rosner, B. (1983), 'Percentage Points for a Generalized ESD Many-Outlier Procedure', Technometrics 25, no. 2, 165-172, https://doi.org /10.1080/00401706.1983.10487848.

Santos, F. & Campos, P. (2020), «Support Vector Machine for Imputation and Editing», Portuguese Statistical Society Bulletin, Spring Issue 2020, *in Portuguese,* https://www.spestatistica.pt/publicacoes/publicacao/ine-85-anos-de-estatisticas-servir-o-pais

Schor, J. (2014), 'Debating the Sharing Economy', Great transition initiative – Toward a transformative vision and praxis, October 2014, https://greattransition.org/publication/debating-the-sharing-economy.
Sela, R. J. & Simonoff, J.S. (2012), 'RE-EM Trees: A Data Mining Approach for Longitudinal and Clustered Data', Machine Learning, vol. 86, *169-207*, https://doi.org/10.1007/s10994-011-5258-3.

Stanoevska-Slabeva, K., Lenz-Kesekamp, V. & Suter, V. (2017), 'Participation, Privacy, and Power in the Sharing Economy Platforms and the Sharing Economy: An Analysis', a report from the EU H2020 Research Project Ps2Share.

Stock J.H. and Watson M.W. (2006), Forecasting with many predictors, *in* 'Handbook of Economic Forecasting', Volume 1, 2006, chapter X, pp.515-554.

Sung, E., Hongbum, K. & Lee, D. (2018), 'Why Do People Consume and Provide Sharing Economy Accommodation? - A Sustainability Perspective', Sustainability, vol. 10, no. 6.

Vera, D (2015), 'Competition in the Sharing Economy', Institut der deutschen Wirtschaft Köln IW Policy paper 19/2015.

World Economic Forum (2017), 'Collaboration in Cities: From Sharing to 'Sharing Economy'', World Economic Forum white paper.

Zhang, L.-C. (2012), 'Topics of statistical theory for register-based statistics and data integration', Statistica Neerlandica, Vol. 66, issue 1, pp. 41-63, https://doi.org/10.1111/j.1467-9574.2011.00508.x.

Appendices

Appendix 1. Terminology related to financial transactions

- *Acquirer* A bank or financial institution to which the acceptor (usually a merchant) transmits the information necessary in order to process card payments. When cash withdrawals take place over the counter or from ATM's, it is the entity (bank or financial institution) which makes cash available to the cardholder (either directly, or via the use of third-party providers).
- *Automated clearing house (ACH)* an electronic clearing system in which payment orders are exchanged among participants (primarily via electronic media) and handled by a data-processing centre. The ACH is pan-European (PE-ACH) in case of a platform and rules for the processing/clearing of euro payment instruments.
- *Automated teller machines (ATMs)* are terminals that allow authorised users, typically by using a card, to access a range of services such as cash withdrawals, balance enquiries, transfers of funds and/or acceptance of deposits. Note that not all ATMs need to have cash withdrawal functionality.
- *Cards* are payment instruments based on a unique number that can be used to initiate a payment, cash withdrawal or cash deposit that is processed using/over a card scheme or – for withdrawals and deposits at the ATM – within the network operated by the issuer of the card. The number can be stored on a plastic card (one of the traditional meanings of the word "card"), on another physical device (e.g. key tag, sticker, smartphone) or can be held virtually without a physical device.
- *Card schemes* are a single set of rules, practices, standards and/or implementation guidelines for the execution of card payments.
- *Cheques* are negotiable payment instruments based on written orders from one part (the drawer) to another (the drawee, normally a bank) requiring the drawee to pay a specific amount from a specified transactional account held in the drawer's name with that institution to be drawee, or a third part specified by the drawer. Cheques may be used for payments as well as settling debts and withdrawing money from banks. Cheques include traveller's cheques and banker's drafts.
- *Chips* are microprocessors embedded on a physical device, which are loaded with the information necessary to transmit payment information.
- *Contactless technology* allows for the transmission of payment information from a physical device to point pf sale (POS) terminals or ATMs without the need for physical contact between the physical device and the terminal. Information used for contactless payments can be stored on a chip, plastic card or other physical device (e.g. smartphone, key tag, sticker, watch). For the purpose of this data collection, the concept of contactless extends beyond that of near field communication (NFC) and also includes other forms of technology applied to perform contactless payments, e.g. Bluetooth low-energy and QR codes.
- *Credit transfers* are payment instruments based on payment orders or possibly sequences of payment orders made for the purpose of placing funds at the disposal of the payee. Both the payment orders and the funds move from the payer's institution to the payee's institution, possibly via several other institutions as intermediaries and/or one or more payment systems.
- *Direct debits* are payment instruments based on preauthorised debits, possibly recurrent, of the payer's account by the payee.
- *End users* are users of payment services as opposed to institutions offering payment services, for example consumers, corporates and merchants. The institutions providing payment services are also end users whenever they use payment services offered by others for their own retail payments (e.g. utility bills and salaries).

- *E-commerce* defined generally as the sale or purchase of goods or services, whether between businesses, households, individuals or private organisations, through electronic transactions conducted via the internet or other computer-mediated (online communication) networks. The term covers the ordering of goods and services that are sent over computer networks, but the payment and the ultimate delivery of the goods or service may be conducted either on- or off-line.
- *Issuers of payment instruments* are entities that (i) make payment instruments directly available to end users, (ii) authorise payments (on physical or online points of sale, over the counter or from ATMs) and (iii) in the case of card payments, guarantee payments to the acquirers.
- *Merchant category code (MCC)* is a four-digit number listed in ISO 18245 for retail financial services. MCC is used to classify the business by the type of goods or services it provides[54].
- *Non-banks offering payment services/instruments* are entities not licensed as banks according to their jurisdiction's legal framework which offer payment services, such as payment institutions, e-money institutions etc. This may also include the government if it offers payment services.
- *On-us payments* are payments in which the payer's and payee's accounts are held by the same institution. In the case of card payments, this means that the issuer of payment cards and the acquirer are the same institution.
- *Payment scheme* a single set of rules, practices, standards and/or implementation guidelines agreed between payment service providers for the execution of payment transactions across the Union and within Member States, and which is separated from any infrastructure or payment system that supports its operation.
- *Payment instruments* are instruments that give access to money, and are used for fund transfers, i.e. credit transfers, direct debits, card payments and cheques by use of bank account money, credits as well as e-money.
- *Payment services* refer to any service that is required for payments or for facilitating them.
- *Payment service provider: a bank, financial corporation, payment institution or e-money institution licensed for* offering payment service.
- *Payment system* this term has two meanings: 1) in some cases, it refers to the set of instruments, banking procedures and interbank funds transfer systems which facilitate the circulation of money in a country or currency area; 2) in most cases, it is used as a synonym for "funds transfer system".
- *Point of sale (POS) terminals* are devices typically used at a retail location to capture payment information electronically and – in some cases – on paper vouchers.
- *Prepaid cards* are cards on which a monetary value can be loaded in advance and stored either on the card itself or on a dedicated account on a computer. Those funds can then be used by the holder to make purchases.
- *Processing* the performance of all the actions required in accordance with the rules of a system for the handling of a transfer order from the point of acceptance by the system to the point of discharge from the system. Processing may include clearing, sorting, netting, matching and/or settlement.
- *Retail payments* are payments where at least one of the parties is an end user. This includes payments by institutions offering payment services when they use payment services offered by others to pay their own utility bills, salaries etc.
- *SEPA (Single Euro Payment Area)* process initiated by European banks and supported, inter alia, by the Eurosystem and the European Commission with a view to integrating retail payment systems and transforming the euro area into a true domestic market for the payment industry.

[54] https://www.iso.org/standard/33365.html

- *Settlement* is the completion of a transaction or of processing with the aim of discharging participants' obligations through the transfer of funds and/or securities. A settlement may be final or provisional.

Appendix 2. Bulgaria: financial transactions data (FTD)

The more monetised an economy is the more helpful FTD are in enriching existing statistics on domestic and foreign demand. Monetisation refers to the size of financial sector in the GDP. Economies with huge financial sector has a lot of financial instruments: cash, loans, tradable and non-tradable securities, etc. A deep financial sector provides a huge variety of financial instruments to business and households and less cash is used in transactions. The most obvious example is paying with plastics even small purchases like bus ticket.

Bulgarian is a rather poorly monetised economy and cash dominates in payments. According to Bulgarian National Bank POS payments and online payments for goods and services amounted to BGN 10.1 billion in 2017. (EUR 1 = BGN 1.956). The private consumption amounted to BGN 61 billion. In 2017. Assuming that all POS and online payments associate with consumption these payments still account for only 16 percent of overall consumption expenditures. Most likely several companies pay through POS terminals and/or online for goods and services supply (intermediate consumption): water supply, electricity supply, internet and mobile services, etc. It means that BGN 10.1 billion refers to both intermediate and final consumption. Hence, more than 84 percent of private consumption is paid in cash. A huge chunk of cash payments makes FTD not very informative about private consumption. Why? Because cash payments, unlike online and plastics payment, do not leave digital footprint in settlement system. Cash payments leave digital footprint in the national tax authority servers. In Bulgaria, each cashier's machine is connected with National Revenue Agency servers. When cash payments dominate the economy tax authority is a much better source of information than FTD.

A2.1 The different FTD

Available financial data

It is reasonable to consider the available financial data before moving to more detailed FTD. Bulgarian national bank publishes regularly data about monetary aggregates, credit to government and non-government sector (non-financial enterprises, households, and financial enterprises). The following monetary data are readily available:

- Monetary aggregates (M1, M2, M3)
  M1 consists of cash in circulation and time deposits. This is the most liquid part of money stock in an economy.
  M2 includes M1 plus demand deposits
  M3 includes M2 plus up to two years tradable debt securities.
- Loans to non-financial enterprises by amount and economic activity
- Loans to households and Not-for-profit-institutions-serving-households (NPISH) by amount
- Deposits of non-financial enterprises by amount and economic activity
- Deposits of households and NPISH by amount
- Card payment through POS terminal and online payments

Loans to and deposits of households and NPISH includes two institutional sectors: households and NPISH. Household sector consist of individuals/group of individuals as consumers. Also household sector includes households that produce goods/services for market or for their own final use.

Publishing data about households and NPISH is a shortage of banking statistics. It would be better to have data about bank savings of households only. Household sector is the net creditor in an economy and it is good to have data about savings and financial assets and liabilities of the sector excluding NPISH.

Deposits include overnight deposits, time deposits and deposits redeemable at a notice. Loans include overdraft; loans other than overdraft and repurchase agreement.

A2.2. Getting FTD

Until now, Bulgarian NSI has gained no access to FTD outside public data published regularly by National Bulgarian Bank. We get in touch with the Bulgarian National Bank and discussed possibilities to gain access to more granular data about C2B payments. No result until now.

Looking at the legislative framework it seems Bulgarian NSI should include the FDP based indicators in the Annual statistical program and passed it through the Parliament.

Art. 20 of the Bulgarian Statistical Act (cf. Section 2.2) addresses two important aspects of data access: first, databases we are interested in should be stipulated by a law and second, the survey that relies on that data bases should be included in the National Statistical Program.

The very first step toward access to FTD is to define indicators based on FTD, to put them in the National statistical program and to pass the program through the government. Having FTD included in the National Statistical Program increases chances to gain access to granular FTD.

A2.3 Metadata

The data about claims and liabilities of financial sector are reported by all banks in Bulgaria, including foreign banks branches. Reporting period is one month for monetary survey and quarter for deposits and loans by amount and economic activity.

Bank statistics is in line with the ECB requirements on monetary and financial statistics and with ESA 95. The economic activity breakdown follows Statistical Classification of Economic Activities in the European Community (NACE Rev.2).

Bulgarian National Bank collects and submits to the ECB statistics on POS payments and online payments on the annual base in line with the Regulation (EU) 1409/2013.

Data source of card payment statistics are licensed payment service providers on the territory of the country; the settlement finality payment system operators licensed by the BNB; the Real-time Interbank Gross Settlement System (RINGS) and the national system component of the Trans-European Automated Real-time Gross Settlement Express Transfer system (TARGET2-BNB).

A2.4 Official statistics potentially benefitting from FTD

The list of monetary data in Section A2.1 could prove helpful in improving quality of official statistics. For example, credit dynamic provides useful additional information to national accounts: GDP growth, investments, consumption, etc. In Bulgaria private credit to GDP ratio tends to decrease over last couple of years. This is an indication for poor financial intermediation and non-sustainable growth.

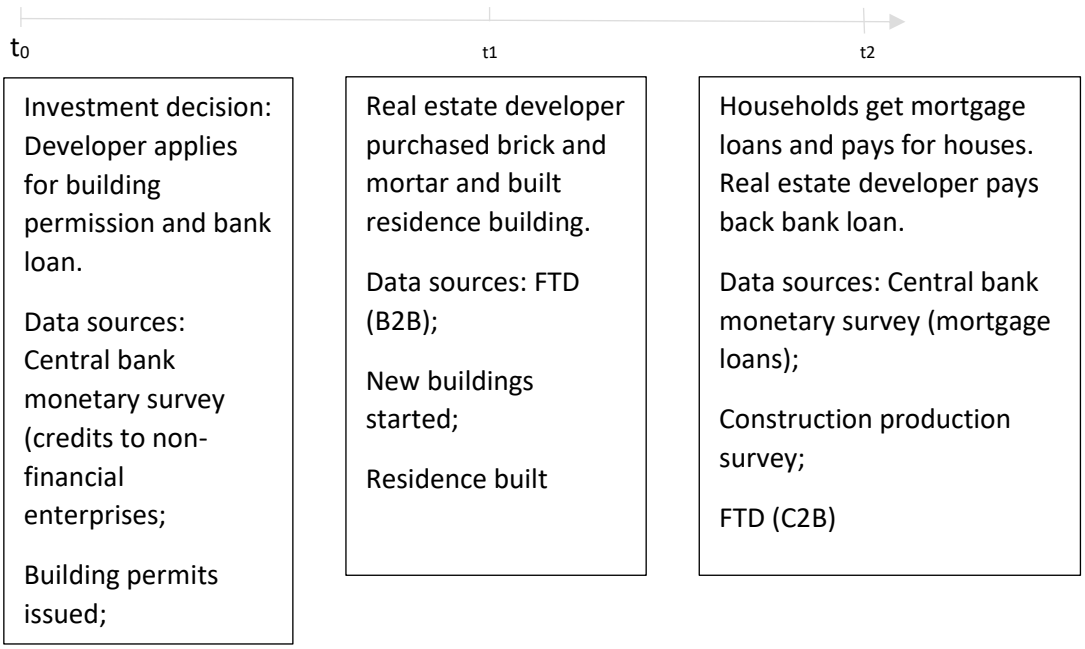
Deposits of households and NPISH provide useful information about saving rates and inequalities of households in terms of bank savings. In Bulgaria deposits and cash are the main instrument for saving because of the shallow financial market and small number of financial instruments. From deposit data available we see that 98 percent of deposits are up to BGN 50 000 (EUR 25 510). Those deposits accounts for 56 percent of the overall money in households & NPISH deposits. Only 0.01 percent of deposits are bigger than BGN 1 000 000. This small percentage accounts for 4.5 percent of overall deposits value. Quite unequal

distribution of deposits! Having in mind that many rich people save abroad the inequality is even bigger. One could easily park her/his savings abroad when domestic market is underdeveloped and do not provide a variety of instrument to save in.

The concept of inequality is multifaceted. People are unequal in terms of income, education, qualification, health, financial and non-financial assets possession, clean environment, etc. Deposits distribution is only a fraction of inequality phenomena. Understanding inequality implies a set of indicators rather than one indicator. FTD are informative only in the context of a broader set of indicators. Let us consider the following example.

A real estate developer decided to invest in a residence building. Before purchasing bricks and mortar, she/he applies for credit and after business plan approval the bank extends a loan to entrepreneur. After having the loan entrepreneur hires a construction company and pays for bricks, mortars, etc. Later on households get mortgage loans to pay for houses and developer pays back bank loan. Households repay mortgage loans within decades.

We can present the whole residence building process alongside time axis.



There are at least three players in the residence investment: real estate developers, construction company, and households. A data set could capture overall process from investment decision up to the buying houses by households in a more comprehensive way than FTD alone. Data about real estate developer activity address the supply side of the transaction or resources from the national accounts perspective. Data about households' mortgages and payments for residence refer to demand side of the real estate market. FTD are a part of a set of data that capture both supply and demand of residence buildings market. They could be very useful for National account department when aggregate primary data in Goods and services account, consumption and investments.

FTD have a good potential to show what is going on in the economy when considered within a broader set of statistics. Moreover, it is possible to detect a mismatch between supply and demand when looking at a set

of indicators. Suppose, there are too many building permits and loans extended to real estate developers. And there are not much mortgage loans. May be supply is above demand? Or this is a temporary phenomenon? The financial crisis in 2007 originated from huge mortgage loan extension and overheated real estate market. When supply greatly outstripped demand house prices collapsed, many homeowners found themselves under water, bad loans increased, and crisis erased quite a piece from banks, investors and households' balance sheet. May be a set of indicators could be helpful in detecting a mismatch between supply and demand on construction market and other markets.

Appendix 3. Germany: financial transactions data (FTD)

A3.1 What transaction data exist?

The *Deutsche Bundesbank* publishes annual time series on the number of charge and credit cards circulating in the economy and on the value of transaction. Since 2007, the number of those cards has increased by more than 60%, while the amount of transaction value paid with charge and credit cards has increased by more than 170%. The *Deutsche Bundesbank* publishes data about the number of card transactions and payments to POS terminals, while the Statistical data Warehouse by the *ECB* publishes data for the EU about payment card functions and accepting devices, and about payments per type of payment service. The available data are however only at the aggregate level and do not inform about the nature of purchase done per transaction or payment.

A3.2 Process for getting FTD

Like all euro area central banks, the *Deutsche Bundesbank* is part of the TARGET2 system. About half of the payments in terms of volume and over one-third in terms of value are submitted via the *Deutsche Bundesbank* through the Target2 payment system, with the German system called TARGET2-Bbk.

FTD relates to individual and retail payments and hence operate under TARGET2, under the TARGET Instant Payment Settlement (TIPS) and under the Retail Payment System (RPS). Under the TIPS, electronic retail payment solutions are processed almost immediately through interbank clearing of the transaction and the payee's account is credited. The settlement happens in secure central bank money.
For non-urgent payments, the *Deutsche Bundesbank* operates the Retail Payment System (RPS) platform. The RPS comprises of i) the cheque clearing service for the processing of euro-denominated cheque payments between credit institutions domiciled in Germany and ii) the SEPA-Clearer for the processing of euro-denominated national and cross-border transfers, direct debits and card payments.

Two Systemically Important Payment Systems (SIPS) are available through EBA Clearing: i) EURO1, a unique Real Time Gross Settlement (RTGS) equivalent net settlement system, held to the highest oversight requirements and overseen by the European Central Bank with the participation of National Central Banks of the Eurosystem and ii) STEP2, a central clearing house, provides banks across Europe the channel through which SEPA Credit Transfers and SEPA Direct Debits can be sent and received. The *Deutsche Bundesbank* participates in the Eurosystem's joint monitoring of the STEP2-T retail payment system operated by EBA CLEARING under the lead of the European Central Bank in its capacity as a competent authority. For instant payment activities, EBA clearing launched RT1, an infrastructure solution for the processing of instant SEPA credit transfers at pan-European level.

A3.3 Metadata for FTD

- Bundesbank data

In addition to the data passing through the payments platforms stated in the previous sections the *Deutsche Bundesbank* currently receives monthly data on credit and debit card turnover in relation to travel (balance of payments) pursuant to section 70 (1) 4a and 4b of the Foreign Trade and Payments Ordinance. This includes payments cards transactions of German residents travelling abroad as well as payments cards transactions of foreign travellers in Germany. However, more detailed information as for example the Merchant Category Code (MCC) is not known. The current data therefore does not allow for empirical analyses on estimating shared economy transactions.
However, the European Central Bank is currently revising the ECB regulation on payments statistics. Balance of Payments requirements may be imposed in the context of this update in particular with the intention of using payment card information to compile the travel item.

- Schufa data

The Federal Statistical Office also contacted the Schufa Holding AG, a German private credit bureau holding credit rating information about 67.7 million persons and 6 million companies for possible FTD data.
Schufa informed that they own some data, amongst others: the existence of a credit or debit card and its creation date, as well as the limit with revolving credit. However, their data does not inform about transaction data using credit and debit cards and can therefore not be considered for this exercise.

- Authorised payment service providers

A possible future data collection could emanate from authorised payment service providers. Those now fall within the second Payment Services Directive (PSD2) scope of applicability and are subject to supervision and monitoring by the National Federal Financial Supervisory Authority, the BaFin in Germany. A list of payment and electronic money institutions authorised or registered within the European Union (EU) and the European Economic Area countries (EEA) are available on the website of the European Banking Authority[55]. In future work it could be examined whether this may provide indications on representative payment service providers.

A3.4 Official statistics potentially benefitting from FTD

The collaboration between the Deutsche Bundesbank and the German Federal Statistics Office allowed evaluating potential future FTD statistics for official statistic production.

The type of statistics that would benefit from FTD depends on the level of detail, the population and coverage of FTD that would be available. Given that no German financial transactions data is currently available that would allow for empirical analyses on estimating shared economy transactions, it is not possible to give concrete examples of official statistics that could benefit from German Financial transactions data.

[55] https://eba.europa.eu/risk-analysis-and-data/register-of-payment-and-e-money-institutions-under-psd2

Appendix 4. Italy: financial transactions data (FTD)

A4.1 Types of FTD

The Bank of Italy directly manages the information on the interbank retail payment systems, through BI-COMP clearance and TARGET2 settlement systems. This information does not include some payment and settlement data directly managed by private entities (e.g., international credit and debit cards networks).

A4.1.1 FTD controlled by the Central bank

BI-COMP is a multilateral clearing system created in the late 1990s to guarantee the settlement in central bank money of euro-denominated retail interbank payments. The Bank of Italy provides the management of BI-COMP as a non-profit public service. BI-COMP handles the transactions and the bilateral balances between banks entered by the clearing systems and operates a multilateral clearing in six daily compensation cycles. At the end of each cycle, BI-COMP calculates a multilateral credit or debit balance for each participant that is sent for settlement in central bank money on the accounts held by intermediaries in TARGET2 (Trans-European Automated Real-Time Gross Settlement Express Transfer System). Preceding the multilateral clearing of electronic payments, there are activities (transmission, matching, confirmation of payments and computation of bilateral balances between banks) carried out by automated clearing house operating in a regime of open competition. Four private automatic clearinghouses (SIA, NEXI and 'NEXI Instant and ICCREA') operate in Italy, whereas Bank of Italy operates CABI.

A4.1.2 FTD controlled by private operators

The EBA-clearance system operated by SIA and NEXI (the main card processors in Italy) is necessary in addition to the national central bank system in order to cover a sufficient part of the population through the clearance process. According to the last financial statements[56], SIA managed the processing of over 70 million cards per year, with over 1.1 million of points of sales; NEXI managed about 40 million payment cards and about 0.9 million points of sales.

The card processors are an alternative data source and their data benefits from the high data coverage (over 85% of the market) and the central management of the database (in terms of data reporting, aggregation, privacy issues, etc.). The internal archives of the (private) payment data processing entities have granular and timely payment data, which manage on behalf of banks. Data are collected on a daily (and infra-daily) basis and contain information on the acquiring side (incoming payments by merchants) and the issuing side (outgoing payments by cardholders). On the issuing side information on the cardholder is not always available for the processor because it is usually only in the internal database of the originating banks.

However, data collection from card processors requires an intense experimental phase before proceeding with their exploitation for statistical production purposes, because problems related to either the harmonisation of different IT platforms or the set-up of ad-hoc data lakes might arise.

A4.2 Getting FTD

A4.2.1 The Bank of Italy - Istat collaboration agreement

The Italian Institute of Statistics has undertaken a collaboration agreement with the Bank of Italy for the exchange of aggregated statistical information. Istat has the possibility, through this agreement, to get information from electronic payment transactions available at the Bank of Italy, as well as other aggregated data deriving from the Anti-Money Laundering Aggregate Reports (SARA) by the Bank of Italy's Financial Intelligence Unit (FIU). A joint Bank of Italy-Istat working group was set up to produce new time series from

[56] See https://www.sia.eu/en/media-events/news-press-releases/sia-2018-financial-statement-a-record-year-great-progress-in-card-payments; https://www.nexi.it/investor-relations.html.

these new sources, to study the characteristics of such series and test their adoption in the nowcasting or forecasting of macroeconomic aggregates. Moreover, to foster the participation to the ESSnet Project "Big Data 2018-2020, Bank of Italy and Istat signed a Memorandum of understanding for mutual collaboration aimed at accessing relevant aggregated FTD held by the Bank of Italy or by other private data-holders and in line with the legal framework described in Section 2.2.1.

BI-COMP and TARGET2 retail database - The joint Bank of Italy-Istat working group focussed on 15 monthly time series of transactions (eight in amount and nine in volume) broken down by payment instruments and reconstructed backwards since January 2000. Microdata of the two settlement systems were aggregated to comply with the General Data Protection Regulation (GDPR). The series are timely but the information has little granularity. Furthermore, it was necessary a pre-treatment and inspection phase before to use the series in order to take into account the changes in the payments regulation that created some shocks during the considered period.

Anti-money laundering aggregate reports (SARA) database - banks and other financial intermediaries have to report[57] monthly to the FIU all transactions amounting to over 15,000 euros, after aggregating them by branch, customer sector and type of transaction. SARA reports do not contain reference to the customers' personal data, but they have information on the customer's sector of economic activity and place of residence. SARA data collection started in January 1993 and, over the years, intervening amendments to the regulatory frameworks affected the reporting scheme (e.g., the thresholds for transactions to be reported and the domains of some classification variables). As some series revealed perturbations, due to changes in the technology used for reporting the series of cash/debit amounts, even in this case a pre-treatment phase was needed to overcome the fluctuations.

A4.2.1.2 Private operators database

During the first quarter of 2019, SIA and NEXI were contacted by the Bank of Italy in order to:

1) identify a possible information set based on aggregations of data already prepared by the managers for internal purposes (at the monthly or daily frequency), grouped by type of technology, or on a socio-economic, sectoral, product or geographical basis;
2) evaluate privacy and confidentiality issues (e.g., consent by banks who are the real owner of card processors' data);
3) define technical methods for collecting, storing and exploiting the data (e.g., periodic data transmission vs. direct remote access to the card processors' data lake).

Bank of Italy is informed on the centralised data collection by the processors in compliance with any profiles of privacy and confidentiality. Bank of Italy should start the realization phase as soon as the pre-treatment and inspection phases are completed, conditional to the solution of a plethora of technical issues such as the analysis of the reporting burden for the card processors, and the construction of a data lake inside the card processors to harmonise different data platforms, etc.

A4.3 Metadata for FTD

Payment system data are timely available with a high frequency pattern. However, such data are not granular in terms of commercial and geographical breakdowns because they come from predefined interbank procedures at the settlement level. The monthly and daily aggregated data we could access are the BI-COMP transactions broken down by payment instruments, namely, credit transfer, direct debit (rav, mav, rid, sepa), direct debit payments cards (atm, pos), cheques, other; and the TARGET2 customer payments total and cross-border series.

Referring to the SARA UIF database system, about fifty monthly series, referring to the total value of the banking transactions, cash operations and domestic or foreign wire transfers are considered since the

[57] Pursuant to Article no. 33 of Italian Anti-Money Laundering Law (Legislative Decree no. 231/2007).

beginning of years 2000. Information on the bank branch where the transaction took place and the customer residence (region or foreign country) are also available since 2012. The classification of the sector of economic activity (Total Services, Retail Trade Wholesale Trade Manufacturing Industry Other) has undergone various amendments over the years.

Private Operators

The card processors are sharing with the Bank of Italy a small aggregated dataset for preliminary investigation and testing based on a short string of variables. Daily information allows to geo-localise the direction of the payment flows: the location of the merchant (or Point of Sales) and the location of the issuer (as a proxy of the location of the consumer, if such information is not available), both from the acquiring side (incoming transactions) and the issuing side (outgoing transactions). This kind of breakdown is useful especially for measuring the cross-border flows. Other information also allows the distinction between e-commerce and proximity payments, useful to better measure the retail trade turnover. A valuable piece of information we obtained is the Merchant Category Code (a four-digit code describing the merchant's primary business based on annual sales volume, measured in local currency). About 700 MCCs are provided from the card networks to track the business and the type of goods or services provided. Such codes are similar to the NACE codes, but there is no standard methodology to match the two classifications. However, in the experimental phase of the project, following the card processor need to reduce the computational burden, MCCs codes have been aggregated. Consequently, an accurate preliminary study of the correspondence of the sectors between the MCCs groups of codes and the NACE rev.2 classification groups is needed.

A4.4 Promising official statistics based on FTD

Istat carried out, in January 2019, an internal (across the Directorates) survey has been carried out to highlight the statistical production potentially interested in the use of private operators' aggregate data. Seven case studies emerged; four of them refer to Business statistics and National accounts Directorates, the others to Social statistics Directorate. The National accounts Directorate proposed the joint use of FTD and short-term indicators for the early estimates of macroeconomic aggregates and the Economic research Unit suggested the use of FTD to improve the forecasts of private consumption and investment. The Business Short-term statistics Directorate nominated a case study aimed at improving the quality of two monthly indicators (retail trade index and services turnover index) besides a case study on the analysis of the Sharing economy platform characteristics. The Social statistics Directorate proposed the study of the use of the sharing economy platform made by citizens (e.g., for sharing personal assets, for direct peer-to-peer exchange, on-demand services and work, peer to business in the sharing mobility, etc.). Other Social Statistics cases proposed are the study of the e-commerce expenditure and the expenditures for tourism. All pilots envisaged the use of monthly data (and at a lesser extent, yearly data or daily data) and both the value and the number of transactions. The breakdown by type, location where the payments is made, payment channel and industry are also largely requested.
Considering the available information in the FTD managed by the Bank of Italy and the degree of aggregation of information supplied by private providers we could have access, we expect that the statistics that could benefit the most from using FTD are the following:

- *household expenditure of residents for tourism and balance of payments estimates* - available information, from the acquirer side, on the location of the issuing bank of the card in addition to information on the location of the sellers could allow distinguishing the purchases of foreigners in Italy from the purchases of the Italians abroad. These data can be compared with those of the balance of payments and new indicators on the (inbound and outbound) tourism spending could be set on a national basis (and possibly on a regional basis).

- *e-commerce turnover by MCCs groups* - using private credit card operators data a pilot study to estimate the e-commerce turnover can be carried out in order to compare this information with the current e-commerce turnover index and to get some insights on the breakdown by MCCs groups.
- *forecasting activities* - BI-COMP and TARGET2 (monthly and daily) series can be used for forecasting purposes. The availability of private credit card operators FTD for a time horizon of at least 5 years (broken down by type of payment channel and MCCs groups) will allow refining the forecasting of private consumption and retail trade index at national level.
- *early estimates of turnover in Services sector* – monthly time series built with the private operators data (broken down by MCCs groups) will be included in the National accounts model for early estimates in Services sector, to verify their predictive performance.

Appendix 5. Norway: financial transactions data (FTD)

A5.1 The different FTD

Norway has the usual payment instruments, i.e. giro, debit cards, delayed debit cards, credit cards and prepaid cards. For system where data are generated or maintained, however, there are some differences from what we described in Section 2.1.2: Firstly, the National Central bank has a limited role in handling single transactions, c.f. Section A5.2. Secondly, in addition to clearing/settlement systems, card operators/issuers and tax/control authorities, there are the following additional sources in Norway:

- The national payment system for use of debit cards in POS terminals and ATMs.
- The national payment system for giro transfers
- Issuers and acquirers of international payment cards (that can be debit cards, delayed debit cards, credit cards and prepaid cards) and issuers of national credit cards.
- Mobile Payment systems
- DSOP kontroll: A comprehensive structure of FTD made in cooperation between governmental and financial institutions.
- The Register of Cross border Currency Transactions and Currency Exchange (The Norwegian Tax Administration)

Below we will for all the systems look at the data's relevance for official statistics.

In Norway, the clearance of financial transactions between two bank accounts involving two banks operating in Norway, goes through *NICS, the interbank clearing system of the Norwegian Krone*. This is a pure clearing system that has no role in the settlement. Further, no credit is involved by neither any of the parties nor by NICS.[58] The settlement process is depending on whether a bank is a "level 1" or a "level 2" bank. A "level 1" bank has a settlement account in Norges Bank to make settlements on behalf of themselves and other banks / "level 2" banks. "Level 2" banks have their settlement accounts in a "level 1" bank.

The gross clearance system of NICS involves all transactions that are either exceeding 25 million NOK, marked with REG or HASTE (priority/hurry), or being a SWIFT-transaction received after 11:30.[59]. These micro-transactions are cleared one by one immediately. Gross clearances can only be made by "level 1" banks. Five times each day, multilateral net clearings are calculated within NICS.

All banks can take part in the net clearing system which involves all transactions that are not a part of the gross clearance system above. First the net position of each bank is calculated. As an example, for bank A, in NICS it is calculated how much money bank A should send to or receive from each of the other banks based on the transaction between accounts in these banks and bank A. The transactions here considered, are those that took place during the hours since the previous clearing.

The net position of a level 2 bank is added to the net position of level 2 bank's private settlement bank at level 1. The net position of the private settlement bank, including the underlying level 2 banks' net positions, is sent to Norges Bank for settlement via the level 1 bank's settlement account in the Norwegian central bank settlement system – NBO. Afterwards, each level 1 bank settles the situation for each of their level 2 banks, i.e. moves money between these banks according to the clearance result. Finally, each bank performs a settlement for each of the single bank accounts in that bank.

[58] https://www.bits.no/en/nics/prinsippet-om-kreditering-etter-oppgjor/

[59] https://www.bits.no/document/arsrapport-for-nics-u-vedlegg/ (in Norwegian). SWIFT transactions are mostly (only?) transactions to and from foreign accounts.

Whereas the gross clearance system of NICS has FTD on micro-level between payer and receiver, the net clearance system has only the net transaction between each of the involved banks that will settle the aggregated total between the banks after all the transactions between the banks during the 3-4-hour period since the last clearance was performed.

The settlement system has been studied for the purpose of being a possible source to FTD. However, as we have seen, in Norway the settlement is partly done within the NBO, the National central bank system, and partly within the level 1 banks. For both these sources, the data are similar to that of the clearance system. To get the micro-data transactions hidden behind the net clearances, it would probably be necessary to go to each bank where the internal settlement between the bank accounts are done or to one or more of the bank data centrals handling the data on behalf of banks. However, given necessary historical information about transactions from the payer in bank A all the way to the payee in bank B might not be easily obtained.

The national card payment system of Norway is a scheme that can be used in ATMs for cash withdrawals and POS terminals for both payments and cash withdrawals ("cash-back"). In 2018, 79 per cent of all payments in POS-terminals in Norway were such payments. Other payment cards can be used in ATMs and POS-terminals as well. Micro-data exists in one database for use of all payment cards in all POS-terminals accepting the national card scheme (almost all terminals) in Norway. This database is a potential source for Official statistics.

The national giro payment system covers both paper-based and electronic credit transfers and direct debits. A major part of these transactions are registered in one database. A considerable part of giro payments is linked to payment orders and electronic invoices. B2C Payment orders exist in one database only, with information about the payer, payee, amount and date of maturity. Electronic Invoices with specifications of the purposes of giro payments are stored in invoice hotels (About 20 B2C invoice hotels) or (for B2B invoices) access points.[60]

DSOP Kontroll is a system that is being constructed as a cooperation project between the financial institutions and three governmental institutions (Tax Authority, the police and the National Insurance Agency (NAV)) for putting less response burden on the financial institutions when getting auditing requests from the governmental institutions. The acronym DSOP is an abbreviation from "Digital cooperation public private". Below we will look more closely into the DSOP Kontroll system.

Today, when the tax authorities perform an audit of a person or an organisation, it contacts the financial institutions. This is a time-consuming process for the financial institutions as well as for the governmental institutions since it is not automated as much as possible.

The system is also helpful for the financial institutions requests to each other, e.g. when a person is applying for a loan, a credit check is made where the banks and other financial institutions are contacted and finds relevant information that is provided to the credit checking company. Receiving such information will be much easier through this digitalised and integrated system.

DSOP Kontroll is a solution to the above challenge. Once operational in 2020, the system will work in the following way:

- First step: Through a simple API-interface, an institution A having legitimate authority to get information, will send a list of personal/organisational identificators to the system. Also, an institution having a consent from the persons/organisations in question, will be able to use this system eventually, i.e. accountants hired by a private firm to audit the firm's accounts.

[60] Registered at https://www.anskaffelser.no/verktoy/veiledere/aksesspunkter-ehf-og-bis-formater

- The list is sent to a register, *KAR,* maintained by the finance industry. This is a register that for all persons and organisations, contains have a list of all the 130[61] banks where each person/organisation has an affiliation.
- Automatically, a request is sent from institution A to the banks identified above for getting a list of the account numbers for this person/institution.
- Institution A asks for account summary or transaction history, and this information is then sent automatically.

Notice that the DSOP Kontroll system, upon authority to request information, returns potentially all financial transactions for a person/organisation on micro level, at least all bank account to bank account transactions whether being debit card transactions or giro transactions. Statistics Norway has authority based on the Statistical act since this is a register.

The register of transactions crossing the national border, is a register containing information about all cross-border transactions. Bank transactions (SWIFT and currency cheques) exceeding NOK 100 000 (approx. 10 000 euro) and card transactions exceeding 25 000 NOK (approx. 2500 euro) must be reported with detailed information about the payer, the payee, currency and the purpose of the payment. Smaller transactions are available aggregated to receiving country, i.e. for each person or institution sending money abroad.; All transactions from one card holder to e.g. UK are aggregated to one record giving the total amount sent to that country (during the reference time).

A5.2 Getting FTD (Process)

The process of getting FTD in Norway is similar to the general considerations described in Section 2.2, where also the legal situation in Norway was described. Below we will make some additional comments.

The National central bank is a natural source for obtaining information on the financial system as in other countries, but it seems not to be a natural source for getting FTD. It has no role in the clearance process and only a part of the role in the settlement process. The clearance system is preferred over the settlement system which is Norway is distributed among several actors, of which the National central bank is only one of these. Clearance data have been received by Statistics Norway for one day in 2016. More details on these data are given in Section A5.3, where we also will discuss the limitations more in detail.

Debit card data from the National payment system (Section A5.1) covers about 80% of POS payments and about 70% of all card payments in Norway, and is a natural most prioritised source for card data. Such data has been acquired by Statistics Norway for one day in 2016, more details are found in Section A5.3.

Debit and credit card data from international debit cards used in Norway are maintained by seven different acquirers. National credit card data are maintained by a few issuers. Thus, it is a somewhat larger process to obtain data from the use of such cards. However, these data are relevant for Statistics Norway since most payments over the internet as well as contactless payments against POS terminals are made by these cards.

Regarding the possibility the get more data than authorised by the Statistical act, this is impossible at least for microdata. However, the Statistical act enables access to all data needed for the development and production of official statistics. Statistics Norway cannot pay for data and neither for the workload put onto a data provider. However, as mentioned in Section 2.2.2.4, it could be useful both for the data provider as well as for Statistics Norway that an employee in Statistics Norway, for a period of time is working together with the data provider' staff in their office, both for getting insight into the data and to help the data provider preparing data for Statistics Norway.

[61] About 130 banks are already a part of the DSOP Kontroll system a year before it starts to operate.

A5.3 Metadata for FTD

Variables and coverage in the National debit card payment system:

First, we notice that the national debit card payment system data cover most of the debit cards payments and the POS terminal payment platform (c.f. the dimensions of Section 2.1.2) . The relevant variables are:

- total transaction amount (NOK)
- cash withdrawal amount (NOK)
- date of transaction
- time of transaction (HH:MM:SS)
- total amount of purchase (NOK) (this is equal to the total transaction amount in the case when no cash withdrawal has been made, otherwise the difference is reflecting the amount of cash withdrawal).
- account number of payer
- account number of receiver (usually not the final receiver but the interim account of the bank)

The last two variables are pseudonymised in the processing at Statistics Norway.

For each individual transaction, register-information at Statistics Norway on payer (person) and enterprise (receiver) can be linked, using the bank account register from the Tax authorities. Example: demographic variables, household variables, number of employed in enterprise etc. Notice that due to the pseudonymisation above, the data sets used in Statistics Norway has no identification of the person or household involved in the different transactions.

A variable describing the Industry classification code was created, but this is the industry of the enterprise, not of the local unit.

Notice that apart from the payment instrument, platform and total purchase, the variables above are only those needed to identify the pure financial transaction: the transaction amount, the payer, the receiver, and the time of the transaction.

Coverage: The National payment system covers most of the debit card payments in POS terminals. As such it covers almost all of groceries sales except the proportion (roughly 20%) made by cash and the proportion made by international payment cards, domestic credit cards and cards issued by retail chains and used in POS-terminals only (in total 17 per cent of the sales in 2018). For other retail sales types, the proportion will be smaller due to the internet purchases which is often done with international payment cards, and giro, and which is still rather smaller on groceries in Norway than for sports equipment, electronics, books etc.

As seen above, groceries are covered roughly well by the national payment system. In the future, also for groceries there will be more internet sales, which will probably make the national payment system less representative for our purpose.

Contents of the purchase: Payment transactions have little or no information on what has been purchased. However, the stores have purchase transactions (corresponding to the receipts) containing: POS terminal, total purchase amount, and for each item purchased the amount of money and description of item[62]. However, the receipt has no information on who made the purchase.

[62] A relevant tool here is the so-called COICOP codes, "Classification of Individual Consumption According to Purpose". As an example, some of the categories are: "01.1.1.3 Bread and bakery products", "01.1.1.4 Breakfast cereals", "01.1.4.1 Raw and whole milk", "01.1.4.2 Skimmed milk", c.f. https://unstats.un.org/unsd/classifications/business-trade/desc/COICOP_english/COICOP_2018_-_pre-edited_white_cover_version_-_2018-12-26.pdf

Variables and coverage in the NICS system:

The NICS system data has the pure financial transaction identification variables: payer (account), receiver (account), amount and time. These are the interesting variables for our purpose in this system. For giro transactions, there might in some cases be overall information on what the transaction is about, but this seems only to be the text field filled in for the giro, e.g. manually by the payer, and is only meant as notes for the payer and receiver within the context of their relation. Thus, it is difficult to classify the information into meaningful categories such as "payment for hiring an apartment/house". As an example, the text could just be the name of the city or country – information being precise for both parties.

Since the NICS system is the output from the clearance system which has both net clearing and gross clearing, the only relevant FTD for Statistics Norway, are from the gross clearing system, containing transactions exceeding 25 million NOK (about 2.5 million euros), in addition to lower value transactions marked "REG", "HASTE" (urgent/priority), or being SWIFT transactions received after 11:30, c.f. Section A5.1. Thus, for most of the transactions, a micro-level transaction is only the sum of all transactions between two certain banks made during the corresponding time interval (about 3-4 hours) of the day.

For the population of interest, i.e. debit cards, credit cards and giro transactions, we here have both debit cards and giro, but we only cover the part of the population cleared by the gross clearance system, which we can assume to be systematically different from the transactions hidden behind the net clearance system, at least when it comes to amount of money, but possibly also when it comes to purposes of the payments.

Variables and coverage in the DSOP system

We do not yet know which variables will be contained in the DSOP system, but it should at least be the pure financial transaction variables (payer, receiver, amount, time and type of account). Supposedly, the payment instrument or the type of payment should be identified. Type of payment at a specific payment platform must perhaps be identified by elimination process: those card transactions not being payments in POS terminals, are mostly made on web or as mobile payments

The DSOP kontroll system is covering information connected to the use of debit cards, cheque and giro except those for a few banks that are not yet included. A challenge will be lack of information connected to the use of credit cards and mobile payments and technical difficulties of requesting the whole population or larger parts of the population from the system. Fortunately, there is as solution to this challenge: a random sample of units (persons/organisations) can be drawn by Statistics Norway and a request can be sent to the system for only the sample. Then, estimates can be made in the usual way for statistics based on sample surveys.

Variables and coverage in the register of transactions crossing the national border

The register of transactions crossing the national border is as the name tells, not covering transactions within Norway, but it is covering all cross-border transactions by use of international payment cards, SWIFT bank to bank transfers (in Norway often initiated by use of the national giro system) as well as other cross border transactions, i.e. transactions effectuated by entities registered as payment institutions

Although covering all domestic persons and organizations for the cross-border transactions mentioned above, the register has partly aggregated data as we noticed in Section A5.1. In more detail: The information for bank transactions exceeding 100 000 NOK and card transactions exceeding 25000 NOK is about the person/organisation id of payer, payee, country, currency and the purpose of the transaction, and total amount. Bank transactions less than 100 000 NOK lack information about the purposes of the payments. When a card transaction is less than 25000 NOK, all transactions with a person/organisation's card are aggregated to monthly numbers for a certain country where the card has been used (for outgoing transactions or (for ingoing transactions) the country where the card is issued. For ingoing card transactions

less than 25 000 NOK, the banks also send a monthly report with aggregated numbers and amounts of payments for every single user location, with detailed information about the user location. Thus, from this data, it should be possible to construct a total overview of the use of card issued abroad on user locations in Norway.

A5.4 Official statistics potentially benefitting from FTD

A5.4.1 Suggested statistics reported from within the NSI

Internal surveys within Statistics Norway has been performed. In which official statistics is it foreseen a potential improvement by using Financial transactions data? Different usages have been pointed out:

- In foreign trade statistics: flows of payments across the border associated with Norwegian companies
- Statistics on payment instruments.
- In statistics in border trade: amount of purchases abroad for inhabitants of Norway and purchases in Norway made by foreigners/tourists.
- Consumer expenditure survey: Consumption (detailed)
- Consumer price index: price development of the purchases made by households
- Structural statistics on wholesale and retail trade: the activity in retail sales industry
- Retail sales index: value and volume of retail sales
- Index of monthly turnover: turnover in the construction industry and the service industry.
- Construction cost indexes: price of the assets used in road construction and construction of new dwellings.
- Energy statistics: energy consumption expenditure by household type.
- National accounts on retail sale.

A5.4.2 The promising statistics

Below is a list of promising statistics, either as an additional source for improving the statistics or quality evaluating the statistics, or as a source that can replace the existing source.

- Retails sales index is promising statistics, since potentially the current monthly data collection from individual stores could be replaced by aggregated data from the debit card national payment system, together with aggregated purchase transaction data from the grocery stores.
    - For the grocery store sector, it is then possible to see the total sales from the purchase data, and then calculate the proportion of sales performed with the debit card instrument.
    - Outside the grocery store sector, the debit cards total sales are available. Then by some modelling based on the grocery store sector, the total sales from all payment instruments can be estimated. The result will rely on strong modelling assumptions, and it might not be possible to replace the current source completely but perhaps partially. In any case, FTD can be used as an alternative source that might be used either in the estimation or in quality assessment of the current statistics.
- Statistics on economic relations between companies and industries**:** based on giro transactions, the transactions could be aggregated so that each record/line in a dataset would be the sum of transactions between two companies during e.g. a year. The (composite) unit is then the pair (payer, receiver). Then it would be possible to use the NICS data to quantify the amount of transactions between e.g. different industries at the chosen level of industry classification code. This would give a valuable insight into the business structure since it shows which industries is trading with each other. The clearance data could be used, but due to the aggregation property (Section A5.1), DSOP Kontroll is a source more likely to be successful.

Appendix 6. Portugal: financial transactions data (FTD)

A6.1 The different FTD

A6.1.1 Types of FTD

In Portugal FTD can be accessed in different ways, each of them covering slightly different domains. The universe of Financial transactions data includes electronic and non-electronic transactions. Among electronic transactions we may distinguish from four main types of FTD:

- MBWAY and MULTIBANCO (electronic payment and money transfer processes managed by SIBS, a payment services provider). SIBS is also responsible for automatic payment terminals (POS);
- Electronic transactions managed by Credit Card companies (we assume they are mostly electronic nowadays);
- peer–to-peer electronic transfers between banks.
- e-invoice

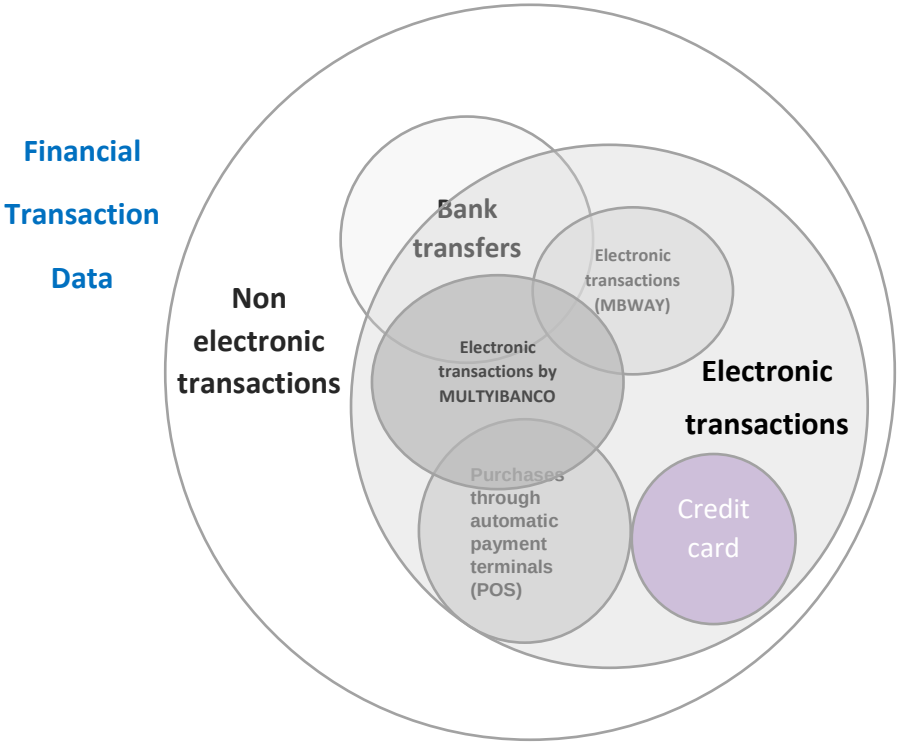


Figure A 6.1: Financial transactions data in Portugal

Electronic transactions (MBWAY) are mainly mobile phone transactions, MULTIBANCO are mainly desktop, notebook and ATM transactions. Purchases through automatic payment terminals (POS)

FTD are, then, all the universe of financial transactions being electronic or non-electronic transactions, and that may use an electronic system (POS terminals, ATM MULTIBANCO, or MBWAY) or not (simple bank transfer). In the latter case, we refer to peer-to-peer bank transfer where electronic system may be needed (or not) as intermediary. FTD from e-Invoice are related to all transactions (electronic and non-electronic) collected by the Tax Authority. This type of data covers all FTD corresponding to the whole universe depicted in Figure A6.1.

A6.1.2 Data owners and data providers

The data owners consist of the private companies, and banks, and the Bank of Portugal. The access to data from private companies and banks is difficult since these companies are not obligated by law to report microdata to Statistics Portugal. The Bank of Portugal collects this information for the transactions data in aggregated way, and SIBS (who manages MBWAY and MULTIBNCO), manages almost all electronic transactions (over 60% or all transactions are made electronically and almost all are monitored by SIBS).

SIBS is a leading payment services provider in various countries in Europe and Africa. The company is responsible for the ATM Express network and MULTIBANCO which include ATM machines, automatic payment terminals and mobile and online alternatives. Recently, a new system has been created, MBWAY, a framework developed for mobile phones.

Financial transactions or services covered by SIBS are the following:

- Card business

  SIBS is responsible for the emission and processing of debit, credit or pre-paid cards from national and international brands.

- Payment services

  Management of services of interbank transfers (credit transfers, direct debits) in SEPA space.

- ATM and POS terminals

  SIBS manages and monitors ATM terminal networks (MULTIBANCO or others). It also supervises POS network.

In Portugal, by law, all the purchases are associated with invoices. Financial transactions that correspond to purchases of goods or services are also monitored by the Tax Authority, that keeps track of all the FTD that correspond to the issue of an invoice. The E-invoice is a dataset of the Tax Authority to monitor all the electronic invoices.

Therefore, the data providers we may consider are:

Data provider	Data
SIBS, Interbank Settlement of the System of Payments	Electronic transactions (payments) made by ATM in MULTIBANCO and MBWAY and purchases through POS
Bank of Portugal	Direct debits and transfers, payments, withdrawals and purchases
Tax Authority	FTD from E-Invoice

A6.2 Getting FTD

A6.2.1 Legal requirements for FTD

Statistical Act

The Portuguese National Statistical System (NSS) includes the Statistical Council, the State body that superintends and coordinates the system; the National Statistical Institute, IP (Statistics Portugal), the central body responsible for the production and dissemination of official statistics, that ensures the supervision and the technical and scientific coordination of the NSS; the Bank of Portugal that, as part of its mission, is responsible for the collection and compilation of monetary, financial, foreign exchange and balance of

payments statistics; the Regional Services of Statistics of the Autonomous Regions of Açores and Madeira that act as delegations of the Statistics Portugal in relation to nationwide statistics as statistical authorities in what concerns regional statistics; and other entities producing official statistics by delegation of Statistics Portugal.

Statistics Portugal, Bank of Portugal, Regional Services of Statistics of the Autonomous Regions of Açores and Madeira and entities producing official statistics by delegation of Statistics Portugal are considered statistical authorities, having responsibility for the production of official statistics, and empowered to require (mandatory and gratuitously) to all departments or agencies, individuals and legal entities, information necessary for the production of official statistics.

The official statistics are produced with technical independence and considered as public good, observing the national and international quality standards, and meeting the users' needs in an efficient manner, preventing the providers of information to the statistical authorities from an excessive burden, by using increasingly the administrative data.

All personal data collected by the statistical authorities for statistical purposes, are considered confidential, and legally protected. All people connected with the production of official statistics are obliged to professional secrecy. To break confidentiality is considered a very serious administrative offence that implies criminal responsibility.

The entities responsible for the production of official statistics have the power to access any service or organism, single or plural person in order to gather any element necessary to the production of official statistics. However, there may be limitations to this access (article 4) and obtaining information can be dismissed if it appears impossible or of disproportional efforts.

A6.2.2 Process for getting FTD

The different transactions, B2B, B2C through bank cards, direct debits and transfers, payments, withdrawals and purchases are all monitored by the Bank of Portugal and the Tax Authority.
The Bank of Portugal publishes a variety of aggregated statistics concerning the Portuguese economy as public finances, financial national accounts, the banking system, the payment system, financial and monetary statistics amongst others.

The information provided by the Bank of Portugal is not detailed and there is no current agreement between Statistics Portugal and Bank of Portugal that facilitates the access to this data.

SIBS also detains many financial transactions data since it is responsible for processing transactions with various banking payment instruments. Although it is a private company, Statistics Portugal already has access to some data by SIBS in an aggregated way. Recently, contacts have been made between Statistics Portugal in order to explore new forms of interchanging data.

FTD registered in the Tax authority are currently the best data to be used in the ESSnet Big Data. A protocol between Statistics Portugal and the Tax Authority has been signed last year that allows the access of Statistics Portugal to data from E-Invoice data set.

Nonetheless, we will try to continue to contact SIBS and Bank of Portugal.

In case we are unable to make an agreement with any of these entities, as mentioned before, Statistics Portugal will use the aggregated data from SIBS, namely the withdrawals from ATM machines and the purchases through automatic payment terminals. We will try to draw results from this data, which is a proxy of financial transactions data for internal demand.

A6.3 Metadata for FTD

Relevant topics that should be considered for each FTD dataset:

E-invoice contains the following information for each register (FTD):

- VAT number (taxpayer number)
- VAT of the buyer
- Month of transaction
- Country of the buyer
- Taxable value (aggregated by VAT number)

Regarding the other data providers, we do not know what kind of metadata we would have access to.

A6.4 Official statistics potentially benefitting from FTD

A6.4.1 Suggested statistics reported from within the NSI

(This subsection is only relevant if experts on the different official statistics at the NSI have been asked what potential they see in FTD)

Depending on the access to the E-Invoice dataset, Statistics Portugal may use this data to produce flash estimates for macroeconomic indicators and private consumption estimation. Furthermore, this information is important to measure Collaborative Economy, namely to estimate the business volume in specific activity sectors. By having access to the identifiers of the companies in some domains (tourism, transports, etc.), we may estimate the aggregate turnover in these sectors.

A6.4.2 The promising statistics

Below is a list of promising statistics, either as an additional source for improving the statistics or quality evaluating the statistics, or as a source that can replace the existing source. For each statistics, it is indicated which FTD that should be used (and possibly which variable in the FTD dataset).

Statistics	**FTD**	**Source**	**Variable**
Turnover of collaborative Economy (total)	E-Invoice	Tax Authority	all
Turnover of collaborative Economy (specific sector)	E-Invoice	Tax Authority	all
Internal demand	Withdrawals and payments	SIBS	

Appendix 7. Slovenia: financial transactions data (FTD)

A7.1 The different FTD

Types of FTD that could possibly be obtained:

- Payment cards (debit cards and credit cards)
- Cash withdrawals and deposits
- Electronic banking
- Credit transfers
- Direct debits (giro payment)
- Fiscal cash registers data
- e-invoice data (In Slovenia e-invoice is only used by the government -- for direct budget users)

A7.1.1 FTD controlled by the Central bank

Since organizations that record electronic payments with payment or credit cards are subject to Slovenian national law, all financial transactions performed between consumers and businesses (B2C) and between businesses (B2B), both debit and credit card and giro transactions, are legally monitored and recorded by Bank of Slovenia.

There are currently 18 financial institutions in Slovenia, which according to the law are obliged to report the data to Bank of Slovenia:

- Addiko bank d.d.
- BKS BANK AG, Bančna podružnica
- Banka Sparkasse d.d.
- RCI Banque Societe Anonyme, podr
- Abanka d.d.
- SKB banka d.d.
- Banka Intesa Sanpaolo d.d.
- Gorenjska banka d.d., Kranj
- Primorska hranilnica d.d.
- DBS d.d. Ljubljana
- UniCredit Banka Slovenija d.d.
- Delavska hranilnica d.d.
- Sberbank banka d.d.
- Hranilnica LON d.d. Kranj
- SID banka d.d., Ljubljana
- Nova LB d.d. Ljubljana
- Nova Kreditna banka Maribor d.d.
- MBILLS d.o.o.

Bank of Slovenia publishes information on payment cards, withdrawals and deposits in cash, electronic banking, credit transfers and direct debits. Data are published quarterly, and are available from 2015 onwards. Below is a more detailed breakdown of data collected and published by the BS.

Payment cards

Data on payment cards are shown by type: credit cards, deferred payment cards, debit cards, and cards with an e-money function. The latter is similar to prepaid cards. Because only one bank issues cards with e-money function in Slovenia, these transactions are added to debit cards in order not to disclose individual data. The

number and value of payments is broken down according to whether payments were made at home or abroad. Payments with foreign cards in Slovenia are also shown.

Table A7.1: Number of payment cards issued by resident issuers

Payment cards - Total	Debit cards and cards with e-money function	Delayed debit cards	Credit cards

Table A7.2: Volume of card payments

Payment cards issued by resident issuers						Payment cards issued by non-resident issuers
Debit cards and cards with e-money function		Delayed debit cards		Credit cards		
in Slovenia	cross-border	in Slovenia	cross-border	in Slovenia	cross-border	in Slovenia

Cash withdrawals and deposits

Bank of Slovenia publishes data on withdrawals and deposits at ATMs in Slovenia, withdrawals with Slovenian cards at ATMs abroad and withdrawals and deposits at bank counters in Slovenia.

Table A7.3: ATMs in Slovenia - transactions with cards issued by resident and non-resident issuers

Number of ATMs	Withdrawals						Deposits	
	Volume			Value in mio EUR			Volume	Value in mio EUR
	with cards issued by resident issuers - on us	with cards issued by resident issuers - not on us	with cards issued by non-resident issuers	with cards issued by resident issuers - on us	with cards issued by resident issuers - not on us	with cards issued by non-resident issuers	with cards issued by resident issuers	with cards issued by resident issuers

Table A7.4: ATMs abroad - cash withdrawals with cards issued by resident issuers

Withdrawals	
Volume	Value in mio EUR

Table A7.5: Over The Counter cash withdrawals and cash deposits in Slovenia

Withdrawals		Deposits	
Volume	Value in mio EUR	Volume	Value in mio EUR

Electronic banking

Electronic banking data are published for online banking, telephone banking and mobile banking in the form of a breakdown in Table A7.6.

Table A7.6: Electronic banking

Number of users		Volume of payments		Value of payments in mio EUR	
Natural persons, Sole proprietors	Legal persons	in Slovenia	cross-border	in Slovenia	cross-border

Credit transfers

Credit transfers data for domestic transactions, cross-border transactions and all transactions together are published in the form of the breakdown below.

Table A7.5: Credit transfers by volume and value

Credit transfers - volume				Credit transfers - value in mio EUR			
initiated in a paper-based form	initiated electronically			initiated in a paper-based form	initiated electronically		
		initiated ina file/batch	initiated on a single payment basis			initiated in a file/batch	initiated on a single payment basis

Direct debits

The direct debit data for domestic transactions, cross-border transactions and all transactions together are published in the form of the breakdown below.

Table A7.6: Direct debits by volume and value

Direct debits - volume		Direct debits - value in mio EUR	
initiated in a file/batch	initiated on a single payment basis	initiated in a file/batch	initiated on a single payment basis

A7.1.2 FTD controlled by private operators

BANKART is a company founded in 1997 by the 22 Slovenian banks on the initiative of Nova Ljubljanska banka, SKB Banka and Abanka for the processing of modern payment instruments. The main mission of the company is to ensure reliable, safe and cost-effective processing of transactions with various banking payment instruments and is almost 100% owned by banks.

Business areas that Bankart covers:

• Processing of ATM operations

Bankart supervises and manages a network of ATMs for commercial banks and savings banks active in retail banking in Slovenia and covers a 93% market share.

• Processing card business

As at 31 December 2017, there were 1,273,461 credit cards and 2,628,499 debit cards in Slovenia - a total of 3,901,960 cards. Bankart processes 628,567 credit and 2,157,767 debit cards, representing 49% of the market share in the field of credit card processing and 82% of the market share in debit cards in Slovenia.

According to the Bank of Slovenia data, as at 31 December 2017, there were 35,842 POS terminals in Slovenia. Bankart processes for 26,674 POS terminals, which represents a 74% market share in Slovenia.

• Payment systems and services

The business area of payment systems and services comprises the management of payment systems and systems for processing SEPA transactions and the E-invoice system. Both SEPA and the SIMP-PS payment system enable the execution of credit payments and direct debits in accordance with SEPA schemes. Each system provides participants with access to all SEPA banks, which means that transactions are executed

between SEPA participants and other euro area banks. The E-invoice system enables the exchange of invoices and other documents in electronic format among the participants.

• ATM and POS terminals

Bankart manages and maintains ATMs and POS network both in terms of daily operational operations and in terms of ensuring compliance with MasterCard and Visa mandates.

A7.1.3 FTD controlled by Financial Administration of the Republic of Slovenia

Another possible source of financial transactions data are data on fiscal cash registers. With fiscal cash register Financial Administration of the Republic of Slovenia monitors all issued invoices, which are paid in money. This includes all payments except credit transfers, meaning also payments with credit or debit cards are included. For the payment cards, both usage in store and online are included.

Not every country uses fiscal money register so there is no uniform definition for it like for example with SEPA. These are systems and models that ensure that data on issued invoices and the turnover of taxable persons reflects the actual economic situation. It is one of the adopted measures against grey economy and tax evasion.

The taxpayers are connected via the internet connection to the information system of the Financial Administration of the Republic of Slovenia. The financial administration confirms and stores data on cash invoices in the process of issuing them in real time. Thus, upon each delivery of goods or services for payment, a taxable person must issue an invoice through an electronic device or using a bound book of accounts, approve it and deliver it to the buyer.

The purpose of carrying out the procedure for the tax invoicing is to obtain a unique identification code of the invoice from the tax authority for each account paid in cash representing a certificate that the turnover generated in cash operations has been reported to the tax authority in the actual scope and cannot therefore be retrofitted with the purpose evading tax.

The invoice (in addition to classic data) must include the time of issue of the invoice (hour and minute), the code of the person/company issuing the invoice using an electronic invoicing device (the name of the person issuing the invoice must be associated with the tax number of that person) , a unique invoice identifier and the security code of the issuer of the invoice.

The invoice number must consist of three parts:

- labels of the business premises of issuer;
- codes of an electronic device for issuing invoices and
- consecutive invoice numbers.

By the law a natural person who carries out activities does not have to pay for the goods supplied and the services rendered to the transaction account of a legal person or natural person performing an activity if the individual payment does not exceed 420 EUR. Likewise, the legal person of payment for the goods supplied and services rendered by him shall not be liable to transfer to the transaction account of the legal or natural person performing the activity if the individual payment does not exceed EUR 420.

Persons engaged in the activity of selling goods or providing services in the Republic of Slovenia may not receive payment in cash from the customer or from any third party in the sale of certain goods or the provision of a particular service, provided that this exceeds the value of EUR 5,000.

A7.2 Getting FTD

A7.2.1 Process for getting FTD

Since all financial transactions are legally monitored and recorded by Bank of Slovenia, data on financial transactions exist, only the access to them is questionable. At present, at national level, the number and value of financial transactions in the framework of the monitoring of payment statistics is monitored by the Bank of Slovenia.

Bank of Slovenia is willing to share the data with the Statistical Office of Slovenia, but according to the current legislation and demands from ECB, banks are obliged to report aggregated quarterly data, within ten days after the completed quarter. In such form, financial transactions data do not represent added value for the Statistical Office of Slovenia, and it would not be sensible to monitor them. At the same time Bank of Slovenia has not expressed willingness to cooperate any further within this project. Nevertheless, the published data can help us, as we can see different types of data that could be obtained.

One possibility of accessing financial transactions data, is also to try to get them from the banks themselves. At the end of year 2018 there were 18 financial institutions in Slovenia, which according to the law are obliged to report Bank of Slovenia data and as such represent a potential source of data.

Since this is a time-consuming process and, in order to increase market share coverage, it would be necessary to obtain data from a large number of banks, it would make sense to look for a third party that would have access to a wider network of banks.

Possible solution is already mentioned privately owned company Bankart that is in charge of processing of transactions with various banking payment instruments. Currently we are in talks with Bankart regarding the data, but at this stage, we have not yet agreed on obtaining the data and as such, we cannot predict if we will be able to access any data before the end on the project ESSnet BD II. The granularity has not been decided, but it will not be microdata.

Another potential source is data on Fiscal cash registers. Based on our talks with Financial Administration we will most likely be able to obtain the data on fiscal cash registers, but we do not know yet, when that will occur. At this stage, we were given an insight into structure of data, we could obtain.

A7.3 Metadata for FTD

As currently we are most likely to obtain the data on fiscal cash registers, this chapter will emphasise on those data.

- **Population**: all issued invoices, which are paid in cash, credit and debit cards, and do not exceed value of EUR 5,000 for consumers (B2C) and EUR 420 for businesses (B2B).
- **Base unit**: individual cash or card payment
- **Composite units** that could be formed: enterprise
- **Variables:**
    - the tax number of the taxpayer who issues the invoice
    - time of issue of the invoice;
    - date of issue of the invoice;
    - the invoice number and method of assigning an invoice number;
    - the value of the invoice, the total value of the base by type of tax, broken down by tax rates;
    - value for payment (EUR);
    - the tax number of the natural person issuing an invoice using an electronic device (actual cashier);
    - the security code of the issuer of the invoice;

- the tax number or identification number for the purpose of the value added tax value of the buyer or the contracting authority in cases where these data are indicated in the account in accordance with the tax regulations;
- number of the original invoice in the case of carrying out the process of confirming the subsequent change of data in the account.
- the code of the obligor's business premises;
- the identification code of the building or part of the building where the business premises of the taxpayer are located, as determined in the real estate register (number of the cadastral municipality, building number and part number of the building);
- the address of the business premises of the taxpayer;
- the type of business premises of the taxpayer when the debtor issues invoices in a movable business premises.

There is no information available about the payer.

All units have an identifier (tax number of enterprise) that can be used at the Statistics of Slovenia to link the data with existing internal data obtained from both surveys and administrative sources.

A7.4 Official statistics potentially benefitting from FTD

A7.4.1 Suggested statistics reported from within the NSI

To recognise areas with potential to use FTD we designed a questionnaire and sent it to experts on the different official statistics at our NSI. Based on the answers the possible need for FTD was detected within following areas:

- Business statistics: foreign trade, retail trade, wholesale, services, industry
- Demographic and social statistics: household consumption, income and living conditions, income and poverty of the population / households, satellite accounts for education, expenditure on formal education, culture statistics
- Environmental statistics: use of ICT, foreign travellers / tourists, travels of domestic population
- Macroeconomic statistics: quarterly GDP, investments, non-financial sector accounts, final consumption of households, PPP and ICES / HICP

As at the moment we do not have any data, we tried to recognise type of data that would be most needed. Based on the questionnaire the most useful data would be debit and credit card data, followed by electronic banking data (number of users, number of payments and value of payments in Slovenia and abroad) and direct debits (number and value of direct debits for domestic and cross-border transactions). Credit transfers and withdrawals and cash deposits would also be useful for some statistics.

All potential users are interested in the value of transactions and in the category of products and services to which the payment relates. More than half of them indicated that they would also need information about the recipient's or payer's activity sector and the location of the payee.

When asked about importance, information on the value of transactions was the most important data, followed by data on the category of products / services to which the payment relates, and the activity of the payer or the recipient.

All of potential users would need microdata more than aggregates, and most of them would use FTD as an additional data source for already existing statistics, or to increase the quality of existing statistics. Others would like to replace existing resources or develop new statistics.

A7.4.2 The promising statistics

Based on the data we are most likely to obtain (unfortunately not in the time scope of this project), the most promising statistics are statistics on retail trade.

Currently indices on retail trade are calculated based on data from survey and administrative data. Observation units that answer the questionnaires are determined on the basis of their previous turnover in two steps: at the level of the survey and at the level of the activity group. The units are sorted by descending turnover, and then we select a sufficient number of units from the begging of the list to exceed the defined share (approximately 60%) of turnover in the total turnover of units covered in the selected activity group. For the remaining observation units we use administrative data being reported by enterprises to the Financial Administration of the Republic of Slovenia for the value added tax purpose (DDV-O forms).

Administrative data on value added tax, that we are currently using, becomes available to us 45 days after reference period. As the data on cash registers would be available 10 days after reference period at the latest, we could get more accurate results at time of first publishing of provisional data, which is 30 days after reference period.

Based on analysis we could determine whether quality of data on cash registers is sufficient to be used in regular production of Index of turnover in retail trade.

Appendix 8. What is an operational definition

The purpose of literature review aims to define what sharing economy is in an operational way. Operational definition suggests quantitative measures of sharing economy: output, employees, investments, etc. Wikipedia provides the following example of operational definition: *weight is the numbers that appeared when that object in placed on a weighting scale.*

Operational definition differs from purely theoretical definitions, which suggested a way to think about the object. Measurement is an operationalization of theoretical insights. Let us look at two examples from National Accounts that demonstrate difference between operational and "scientific" definition.

- Production boundary definition is fundamental for measuring output of an economy. Production boundary, as defined in the ESA 2010, excludes services produced and consumed within the same household: cleaning, cooking, care of children, old people, sick people, etc. The same holds for volunteer services. ESA 2010 does not include volunteer services (e.g., cleaning with no payment) in the production boundary and hence services provided are not output.

Economic theory considers both the efforts of households and volunteer as production activity which produces output. The ESA 2010 definition about production boundary shrinks the economic concept of production in order to overcome difficulties when measure the output of households. It is hard to measure the services provided by the household's members and consumed within the same households. We could add it is hard to measure the volume of services provided by one household to another household for free, e.g., sharing households' assets with no middleman between.

When a household contracts someone to clean, cook, etc. and pays for it services provided by contractor fall into production boundary and are part of households' output. There is a contract, payments, social security number, and so on. Statisticians have enough sources of info to measure the output even using administrative data sources.

- ESA 2010 measures a good part of general government sector output as a sum of inputs. For example, the output of a hospital is a sum of expenditures on wages and salaries, medicines, water, electricity, and so on. This way of evaluation of output as a sum of inputs is at odds with economic theory. Production function, which is at the heart of the neoclassical economics links inputs with outputs and built around that linkage a lot of theory. Moreover, evaluation of output as a sum of inputs make comparisons across countries with different size of government sector highly suspicious.

The above example show that ESA definition of production boundary and output is narrow than what economic theory understands about production. ESA 2010 definitions are operational definition, which took into account both the economic logic of phenomena and the ability of national statistics to measure different forms of economic activity. Operational definition reflects limitation in ability to measure.

System of national accounts develop over time and there is a tendency to come closer to economic theory concept of production boundary, output, income, etc. 1953 edition of SNA under auspices of UNSC had about 50 pages. ESA 2010 is as thick is 651 pages. SNA refine concepts and measurement methods and number of pages increases accordingly.

Appendix 9. Questionnaire of the WPG mini-survey on SEP-related indicators

This questionnaire is aimed at assessing how much information available in WPG countries on FTD could be used also to compare the potential for SEP-related business at national, or sub-national, level.
It focuses only on those FTD which refer to credit card transactions and, among them, to those credit card transactions which are managed on-line, on the Web (whether or not genuine e-commerce transactions).
In this respect, questions 1.1, 1.2 and 1.3 ask about credit cards' FTD in general, while the following questions will be mainly about on-line FTD.

Essnet Big Data 2 WPG SEP **Questionnaire on the availability of credit card FTD data to estimate the role of digital platforms in EU countries**	
Country:	
Respondent:	
Date:	

Section 1	FTD data source availability		
1.1 Please, list the sources of FTD data (maintainers) which are actual/future data providers for data that will be received within this Essnet WPG and what is their rate of coverage of national FTD total			
Name (or just a code if name cannot be disclosed)	Status (public vs. private)	Coverage (national vs. regional)	Percentage of national total (approx.)
		National total	100%

1.2 As a result of the planned data collection, do you will be able to get any aggregated TOTAL with reference to the volume of FTD credit card data for any period of time (from 1 day to 1 year, either on issuing or acquiring side)?	
Yes ☐	No ☐

1.3 Please, describe below the broadest coverage of FTD credit card data which could be available in your country by two dimensions: time and territory (country/region(s)).	
Time:	
Territory:	
The above coverage can be achieved:	Using data from a single source ☐ Combining more sources ☐
Available aggregated data refer to:	Transactions ☐ Value ☐ Both ☐

1.4 With reference to the above-referred "broadest coverage", do you will be able to make a distinction between credit card FTD as a result of an on-site (proximity) purchases and credit card FTD from online transactions (e-commerce)?		
Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
If the answer is "No", please go to Section 3, otherwise go to question 1.5		

1.5 Please, describe below the broadest coverage of online credit card FTD data which could be available in your country by two dimensions: time and area (country/region(s)).	
Time:	
Area: (e.g. "whole country", "only the region ....")	
The above coverage can be achieved:	Using data from a single source ☐ Combining more sources ☐
Available aggregated data refer to:	Transactions ☐ Value ☐ Both ☐

Section 2	Data breakdown		
2.1 With reference to the broadest available coverage of online credit card FTD data (Q.1.5), please report whether a further breakdown could be made available by (either on issuing or acquiring side):			
2.1.1 Payer			
2.1.1.1 Nature (Individual/Firms/Other)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.1.2 Nationality 1 (domestic/foreign)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.1.3 Nationality 2 (country where the card has been issued	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.2 Receiver			
2.1.2.1 Nature (Individual/Firms/Other)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.2.2 Nationality 1 (domestic/foreign)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.2.3 Nationality 2 (country where the card has been issued	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.2.4 Sector of activity (if a firm)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.3 Transaction			
2.1.3.1 Value	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.3.2 Domain (at least making a distinction between goods and service selling)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:

2.2 Please, specify whether some of the backgrounds listed in Q 2.1 can be combined with reference to the national total of FTD from online transactions.				
2.2.1 Transactions by value				
	Receiver's nature	Receiver's nationality 1	Receiver's nationality 2	Receiver's activity
Payer's nature	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
Payer's nationality 1	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
Payer's nationality 1	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
2.2.2 Transactions by domain				
	Receiver's nature	Receiver's nationality 1	Receiver's nationality 2	Receiver's activity
Payer's nature	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
Payer's nationality 1	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
Payer's nationality 1	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐

2.3 As a result of the information provided in Q 2.1 and Q 2.2, please describe the main aggregated indicators (or tables) which could be made available.				
	Description	Estimated national coverage (%) of FTD from online transactions	Number of cells	Reference period
Indicator 1				
Indicator 2				
Indicator 3				
Indicator 4				
Indicator 5				
Indicator 6				

Section 3.	Complementary information	
3.1 In order to complement or to replace a distinction between "proximity" and online FTD, can other methods be implemented (e.g. by nature of the receiver/merchant)?	Yes ☐ please, specify:	No ☐
3.2 Is it any distinction between "desktop" and "mobile" online FTD available in your country?	Yes ☐ please, specify:	No ☐
3.3 Is there any way, in your country, to assess the relevance (such as a concentration rate, both in terms of number of transactions and value) of the receivers with the highest number of incoming online payments?	Yes ☐ please, specify:	No ☐
3.4 The same as Q 3.2, by country?	Yes ☐ please, specify:	No ☐
3.5 With reference to Q. 2.1, please describe, if any, the breakdown used to describe the nature of payers/receivers		
3.6 With reference to Q. 2.1, please describe, if any, the breakdown used to describe the nationality of payers/receivers		

3.7 With reference to Q. 2.1.2.4, please describe, if any, the breakdown used to describe the sector of activity of receivers			
3.8 With reference to Q. 2.1.3.2, please describe, if any, the breakdown used to describe the domain of transactions			
3.9 With reference to the indicators described in Q. 2.3, which frequency is expected for the release of new FTD data?	Yearly ☐	Monthly ☐	Other ☐
3.10 In addition to the standard use of FTD aggregated data, is it any use of FTD (anonymised) microdata envisaged (e.g. to calculate ratios, to identify clusters/patterns of use, etc.)?	Yes ☐ please, specify:	No ☐	
3.11 Is it any indicator on the "transaction time" (peak, average, etc.), that is the time needed to complete a financial transaction, available in your country?	Yes ☐ please, specify:	No ☐	
3.12 Which quality assessment methods are going to be implemented to check for accuracy and representativeness of the FTD' sample used). Please, describe if any.			

Appendix 10. Results of the WPG mini-survey

Table A10.1 Results of the WPG mini-survey

	Germany	Norway	Bulgaria	Italy
1.2	Y	Y	Y	Y
1.3	Annual National Single source T+V	Q/M National 2 sources T+V	Annual National Single source V	Annual National Single source T+V
1.4	N	Y, quarterly	Y	Y
1.5	-	Single source T+V	Single source V	Single source T+V
2.1.1.1	-	Only private cards vs. business cards	Not available for non-residents	No
2.1.1.2	-	No	Y	Y
2.1.1.3	-	Y	Not available for non-residents	Y
2.1.2.1	-	From a single source via MCCs	Not available for non-residents	No
2.1.2.2	-	Y	Y	Y
2.1.2.3	-	Y	Not available for non-residents	Only for selected countries
2.1.2.4	-	Y	No	Only via MCCs
2.1.3.1	-	Y	Y	Y
2.1.3.2	-	Only via MCCs	To be confirmed	Only via MCCs
2.2.1a1	-	No	Y	No
2.2.1a2	-	Y	Y	No
2.2.1a3	-	Y	No	No
2.2.1a4	-	Y	No	No
2.2.1b1	-	Y	-	Y
2.2.1b2	-	Y	-	Y
2.2.1b3	-	Y	-	Y
2.2.1b4	-	Y	-	Y
2.2.1c1	-	Y	-	Y
2.2.1c2	-	Y	-	Y
2.2.1c3	-	Y	-	Y
2.2.1c4	-	Y	-	Y
2.2.2a1	-	No	No	No
2.2.2a2	-	Y	No	-
2.2.2a3	-	Y	No	No
2.2.2a4	-	Y	No	No
2.2.2b1	-	Y	No	No
2.2.2b2	-	Y	No	Y
2.2.2b3	-	Y	No	Y
2.2.2b4	-	Y	No	Y
2.2.2c1	-	Y	No	No
2.2.2c2	-	Y	No	Y
2.2.2c3	-	Y	No	Y
2.2.2c4		Y	No	Y
2.3 Ind.1	-	Payments with Norwegian cards over the Internet in Norway and abroad, and payments by foreigners' cards over the Internet in Norway (100%)	Aggregated online payments across residents and across residents and non-residents (100%)	-

2.3 Ind.2	-	Country where the card has been issued, and country where the card has been used (100%)	-	-
2.3 Ind.3	-	Merchant Category Codes (100%)	-	-
2.3 Ind.4	-	Use of cards issued abroad on user locations in Norway broken down geographically by country, and for Norway also by postal code (100%)	-	-
3.1	No	Y. "Online FTD transactions" = "Card not present"- transactions in the VISA and MasterCard systems	No	No
3.2	No	Y. Mobile payments" = "Card not present" – combined with "tokenised transactions" in the VISA and MasterCard systems	No	No
3.3	No	Y. Source 1: Highest number of income online payments can be measured for receivers indirectly only -- by MCC-codes. Norwegian virtual user locations may be identified by source 2	No	No
3.4	No	Y	No	No
3.5	No MCC available	We have no information about the nature of payers over the internet today, but we have been informed from source 1 that it should be possible to report the internet use of Norwegian cards held by persons or entities. Receivers can be broken down by MCCs	-	MCC classification
3.6	Only domestic/foreign card or payment at domestic/foreign terminals	See answer of Q 2-3, indicator 2	-	The level of detail will depend on the number of transactions in order to protect confidential data.
3.7	-	See answer of Q 2-3, indicator 3 – MCC can be linked by NACE to sector. Notice: assuming Sector is the variable that can have the values "Public/Private"	-	MCC classification
3.8	-	MCC-codes are used, but they will not be exact to distinguish between goods and services	-	MCC classification
3.9	-	Quarterly	Yearly	Yearly
3.10	-	Y. Source 2: Individual transactions >=NOK 25 000 (Not yet, but from 2020: DSOP-data)	No	No
3.11	-	Y. Information in general about type of transaction and times of settlement /settlement frequency	No	Y. Data could be made available on a daily basis.
3.12	-	Procedures controlling the validity of data and metadata and completion of the reporting	For the time being we have no idea how to check FTD for accuracy and representativeness.	Data from various sources will be systematically compared to check for consistency.